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Alpha-amylase variants and polynucleotides encoding same

Abstract

The present invention relates to alpha-amylase variants. The present invention also relates to polynucleotides encoding the variants; nucleic acid constructs, vectors, and host cells comprising the polynucleotides; and methods of using the variants.

Inventors: **Andersen; Carsten (Vaerloese, DK), Ghadiyaram; Chakshusmathi (Bangalore, IN), Sainathan; Rajendra Kulothungan (Bangalore, IN), Balumuri; Padmavathi (Chennai, IN), Iyer; Padma Venkatachalam (Mumbai, IN), Damager; Iben (Valby, DK), Munch; Astrid (Frederiksberg, DK), Dalal; Sohel (Ahmedabad, IN)**

Applicant: **Novozymes A/S** (Bagsvaerd, DK)

Family ID: **1000008819353**

Assignee: **Novozymes A/S (Bagsvaerd, DK)**

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Primary Examiner: Rao; Manjunath N

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/702,431 filed on Mar. 23, 2022 which is a divisional of U.S. application Ser. No. 16/847,773 filed on Apr. 14, 2020 now U.S. Pat. No. 11,319,509 which is a divisional of U.S. application Ser. No. 15/571,219 filed May 9, 2016, now U.S. Pat. No. 10,647,946 which is a 35 U.S.C. 371 national application of PCT/EP2016/060266 filed May 9, 2016, which claims priority or the benefit under 35 U.S.C. 119 of Indian application no. 2335/CHE/2015 filed May 8, 2015, the contents of which are fully incorporated herein by reference.

REFERENCE TO A SEQUENCE LISTING
(1) This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference. This application contains a Sequence Listing in computer readable form The contents of the electronic sequence listing created on Mar. 5, 2024, named SQ.xml and 26,105 bytes in size, is hereby incorporated by reference in its entirety.
BACKGROUND OF THE INVENTION
Field of the Invention
(2) The present invention relates to alpha-amylase variants, polynucleotides encoding the variants, methods of producing the variants, and methods of using the variants.
Description of the Related Art
(3) Alpha-amylases have for many years been used in laundry where is it well-known that alpha-amylases have a beneficial effect in removal of starch containing, or starch-based, stains.
(4) WO95/26397 discloses alkaline *Bacillus* amylases having good wash performance measured at temperatures in the range of 30-60° C.
(5) WO00/60060 and WO00/60058 discloses further bacterial alpha-amylases having good wash performance.
(6) In recent years there has been a desire to reduce the temperature of the laundry in order to reduce the energy consumption. Lowering the temperature in laundry often means that the performance of the detergent composition and the enzyme is reduced and a lower wash performance is therefore obtained at low temperature. It is therefore desired to find new alpha-amylases having good wash performance at low temperature. Accordingly, it is an object of the present invention to provide alpha-amylases which have good wash performance at low temperature, such as at 15° C. Thus, the present invention provides such further improved alpha-amylse variants with improved properties compared to its parent.
SUMMARY OF THE INVENTION
(7) The present invention relates to an alpha-amylase variant of a parent polypeptide having alpha-amylase activity, wherein the variant is a fusion polypeptide which has an improved wash performance at low temperature, and wherein said variant has alpha-amylase activity.
(8) The present invention also relates to polynucleotides encoding the variants; nucleic acid constructs, vectors, and host cells comprising the polynucleotides; and methods of producing the variants.
(9) The present invention also relates to methods of improving wash performance of a parent alpha-amylase.
(10) Conventions for Designation of Variants
(11) For purposes of the present invention, the mature polypeptide disclosed in SEQ ID NO: 7 is used to determine the corresponding amino acid residue in another alpha-amylase. Alternatively, the mature polypeptide disclosed in SEQ ID NO: 13 and 14 may be used to determine the corresponding amino acid residue in another alpha-amylase. The amino acid sequence of another alpha-amylase is aligned with the mature polypeptide disclosed in SEQ ID NO: 7, and based on the alignment, the amino acid position number corresponding to any amino acid residue in the mature polypeptide disclosed in any of SEQ ID NOs: 13 and 14 is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48: 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16: 276-277), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix.

(12) Identification of the corresponding amino acid residue in another alpha-amylase can be determined by an alignment of multiple polypeptide sequences using several computer programs including, but not limited to, MUSCLE (multiple sequence comparison by log-expectation; version 3.5 or later; Edgar, 2004, *Nucleic Acids Research* 32: 1792-1797), MAFFT (version 6.857 or later; Katoh and Kuma, 2002, *Nucleic Acids Research* 30: 3059-3066; Katoh et al., 2005, *Nucleic Acids Research* 33: 511-518; Katoh and Toh, 2007, *Bioinformatics* 23: 372-374; Katoh et al., 2009, *Methods in Molecular Biology* 537:_39-64; Katoh and Toh, 2010, *Bioinformatics* 26:_1899-1900), and EMBOSS EMMA employing ClustalW (1.83 or later; Thompson et al., 1994, *Nucleic Acids Research* 22: 4673-4680), using their respective default parameters.
(13) When the other enzyme has diverged from the mature polypeptide of SEQ ID NO: 7 such that traditional sequence-based comparison fails to detect their relationship (Lindahl and Elofsson, 2000, *J. Mol. Biol.* 295: 613-615), other pairwise sequence comparison algorithms can be used. Greater sensitivity in sequence-based searching can be attained using search programs that utilize probabilistic representations of polypeptide families (profiles) to search databases. For example, the PSI-BLAST program generates profiles through an iterative database search process and is capable of detecting remote homologs (Atschul et al., 1997, *Nucleic Acids Res.* 25: 3389-3402). Even greater sensitivity can be achieved if the family or superfamily for the polypeptide has one or more representatives in the protein structure databases. Programs such as GenTHREADER (Jones, 1999, *J. Mol. Biol.* 287: 797-815; McGuffin and Jones, 2003, *Bioinformatics* 19: 874-881) utilize information from a variety of sources (PSI-BLAST, secondary structure prediction, structural alignment profiles, and solvation potentials) as input to a neural network that predicts the structural fold for a query sequence. Similarly, the method of Gough et al., 2000, *J. Mol. Biol.* 313: 903-919, can be used to align a sequence of unknown structure with the superfamily models present in the SCOP database. These alignments can in turn be used to generate homology models for the polypeptide, and such models can be assessed for accuracy using a variety of tools developed for that purpose.
(14) For proteins of known structure, several tools and resources are available for retrieving and generating structural alignments. For example the SCOP superfamilies of proteins have been structurally aligned, and those alignments are accessible and downloadable. Two or more protein structures can be aligned using a variety of algorithms such as the distance alignment matrix (Holm and Sander, 1998, *Proteins* 33: 88-96) or combinatorial extension (Shindyalov and Bourne, 1998, *Protein Engineering* 11: 739-747), and implementation of these algorithms can additionally be utilized to query structure databases with a structure of interest in order to discover possible structural homologs (e.g., Holm and Park, 2000, *Bioinformatics* 16: 566-567).
(15) In describing the variants of the present invention, the nomenclature described below is adapted for ease of reference. The accepted IUPAC single letter or three letter amino acid abbreviation is employed.
(16) Substitutions: For an amino acid substitution, the following nomenclature is used: Original amino acid, position, substituted amino acid. Accordingly, the substitution of threonine at position 226 with alanine is designated as “Thr226Ala” or “T226A”. Multiple mutations are separated by addition marks (“+”), e.g., “Gly205Arg+Ser411Phe” or “G205R+S411F”, representing substitutions at positions 205 and 411 of glycine (G) with arginine (R) and serine (S) with phenylalanine (F), respectively.
(17) Deletions: For an amino acid deletion, the following nomenclature is used: Original amino acid, position, *. Accordingly, the deletion of glycine at position 195 is designated as “Gly195*” or “G195*”. Multiple deletions are separated by addition marks (“+”), e.g., “Gly195*+Ser411*” or “G195*+S411*”.
(18) Multiple modifications: Variants comprising multiple modifications are separated by addition marks (“+”), e.g., “Arg170Tyr+Gly195Glu” or “R170Y+G195E” representing a substitution of arginine and glycine at positions 170 and 195 with tyrosine and glutamic acid, respectively.
(19) Different modifications: Where different modifications can be introduced at a position, the different alterations are separated by a comma, e.g., “Arg170Tyr,Glu” represents a substitution of arginine at position 170 with tyrosine or glutamic acid. Thus, “Tyr167Gly,Ala+Arg170Gly,Ala” designates the following variants: “Tyr167Gly+Arg170Gly”, “Tyr167Gly+Arg170Ala”, “Tyr167Ala+Arg170Gly”, and “Tyr167Ala+Arg170Ala”.

Description

DETAILED DESCRIPTION OF THE INVENTION
(1) The present invention relates to a set of sequences. For ease, these are listed below; SEQ ID NO: 1 is the nucleotide sequence of an alpha-amylase (AAI10) SEQ ID NO: 2 is the nucleotide sequence of an alpha-amylase originating from *Alicyclobacillus* sp. SEQ ID NO: 3 is the nucleotide sequence of an alpha-amylase originating from *Bacillus amyloliquefaciens.* SEQ ID NO: 4 is the amino acid sequence of AAI10 including the signal peptide. SEQ ID NO: 5 is the amino acid sequence of *Alicyclobacillus* sp. including the signal peptide. SEQ ID NO: 6 is the amino acid sequence of *Bacillus amyloliquefaciens* including the signal peptide. SEQ ID NO: 7 is the amino acid sequence of the mature AAI10 SEQ ID NO: 8 is the amino acid sequence of the mature *Alicyclobacillus* sp. SEQ ID NO: 9 is the amino acid sequence of the mature *Bacillus amyloliquefaciens* SEQ ID NO: 10 is the amino acid sequence corresponding the A and B domains of AAI10. SEQ ID NO: 11 is the amino acid sequence corresponding to the C domain of *Alicyclobacillus* sp. SEQ ID NO: 12 is the amino acid sequence corresponding to the C domain of *Bacillus amyloliquefaciens* SEQ ID NO: 13 is the amino acid sequence of the fusion polypeptide consisting of the A and B domain of AAI10 and the C domain of *Alicyclobacillus* sp. SEQ ID NO: 14 is the amino acid sequence of the fusion polypeptide consisting of the A and the B domains of AAI10 and the C domain of *Bacillus amyloliquefaciens* SEQ ID NO: 15 is the amino acid sequence of the mature AAI10 alpha-amylase comprising a double deletion at positions corresponding to positions 182 and 183 of SEQ ID NO: 7
Variants of the Invention
(2) In one aspect, the present invention relates to variants of a parent polypeptide having alpha-amylase activity. Thus, in particular aspect, the present invention relates to variant of a parent polypeptide having alpha-amylase activity, wherein said variant is a fusion polypeptide which has an improved wash performance at low temperature, and wherein said variant has alpha-amylase activity.
(3) It has been shown that variants according to the present invention have improved performance, such as wash performance, at a low temperature, in particular at 15° C., in liquid detergent compositions.
(4) The term “variant” as used herein, refers to a polypeptide having alpha-amylase activity comprising an alteration, i.e., a substitution, insertion, and/or deletion, at one or more (e.g., several) positions. A substitution means replacement of the amino acid occupying a position with a different amino acid; a deletion means removal of the amino acid occupying a position; and an insertion means adding an amino acid adjacent to and immediately following the amino acid occupying a position, all as defined above. The variants of the present invention have at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 100% of the alpha-amylase activity of the polypeptide of SEQ ID NOs: 7, 13, 14, 15, or the mature polypeptide of SEQ ID NO: 4. The variants of the present invention have at least 65%, such as at least 70%, such as at least 75%, such as at least 80%, such as at least 85%, such at least 90%, such as at least 95%, such as at least 97%, such as at least 99%, but less than 100% sequence identity to any one of SEQ ID NOs: 4, 7, 13, 14, or 15
(5) The term “parent polypeptide” as used herein, refers to a polypeptide to which an alteration is made to produce enzyme variants. The polypeptide having any of SEQ ID NOs: 4, 7, 13, 14, and 15 may e.g. be a parent for the variants of the present invention. Any polypeptide having at least 65%, such as at least 70%, such as at least 75%, such as at least 80%, such as at least 85%, such at least 90%, such as at least 95%, such as at least 97%, such as at least 99%, sequence identity to any one of SEQ ID NOs: 4, 7, 13, 14, or 15 may also be a parent polypeptide for the variants of the present invention.
(6) The parent polypeptide may be a fusion polypeptide or cleavable fusion polypeptide. Such fusion polypeptide may consist of a subsequence of one parent polypeptide and a subsequence of second parent polypeptide. In particular, a fusion polypeptide may consist of an A and B domain of one species of alpha-amylase and a C domain of another species of alpha-amylase, and thereby providing a parent polypeptide to generate a variant of the

present invention. Techniques for producing fusion polypeptides are known in the art, and include ligating the coding sequences encoding the polypeptides so that they are in frame and that expression of the fusion polypeptide is under control of the same promoter(s) and terminator. Fusion polypeptides may also be constructed using intein technology in which fusion polypeptides are created post-translationally (Cooper et al., 1993, *EMBO J.* 12: 2575-2583; Dawson et al., 1994, *Science* 266: 776-779).

(7) A fusion polypeptide may further comprise a cleavage site between the two polypeptides. Upon secretion of the fusion protein, the site is cleaved releasing the two polypeptides. Examples of cleavage sites include, but are not limited to, the sites disclosed in Martin et al., 2003, *J. Ind. Microbiol. Biotechnol.* 3: 568-576; Svetina et al., 2000, *J. Biotechnol.* 76: 245-251; Rasmussen-Wilson et al., 1997, *Appl. Environ. Microbiol.* 63: 3488-3493; Ward et al., 1995, *Biotechnology* 13: 498-503; and Contreras et al., 1991, *Biotechnology* 9: 378-381; Eaton et al., 1986, *Biochemistry* 25: 505-512; Collins-Racie et al., 1995, *Biotechnology* 13: 982-987; Carter et al., 1989, *Proteins: Structure, Function, and Genetics* 6: 240-248; and Stevens, 2003, *Drug Discovery World* 4: 35-48.

(8) The parent polypeptide may be obtained from microorganisms of any genus. For purposes of the present invention, the term “obtained from” as used herein in connection with a given source shall mean that the parent polypeptide encoded by a polynucleotide is produced by the source or by a strain in which the polynucleotide from the source has been inserted.

(9) In some aspects of the present invention, the parent polypeptide is *Bacillus* sp. alpha-amylase, e.g., the alpha-amylase of SEQ ID NOs: 4, the mature polypeptide thereof, i.e. SEQ ID NO: 7. In particular aspects, the parent polypeptide is a fusion protein, and originates from both a *Bacillus* sp and a *Alicyclobacillus* sp., e.g. the alpha-amylase of SEQ ID NO: 13. In other aspects, the parent polypeptide is a fusion protein and originates from two different *Bacillus* sps., e.g. resulting in the alpha-amylase of SEQ ID NO: 14.

(10) It will be understood that for the aforementioned species, the invention encompasses both the perfect and imperfect states, and other taxonomic equivalents, e.g., anamorphs, regardless of the species name by which they are known. Those skilled in the art will readily recognize the identity of appropriate equivalents.

(11) Strains of these species are readily accessible to the public in a number of culture collections, such as the American Type Culture Collection (ATCC), Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), Centraalbureau Voor Schimmelcultures (CBS), and Agricultural Research Service Patent Culture Collection, Northern Regional Research Center (NRRL).

(12) The parent polypeptide may be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) using the above-mentioned probes. Techniques for isolating microorganisms and DNA directly from natural habitats are well known in the art. A polynucleotide encoding a parent polypeptide may then be obtained by similarly screening a genomic DNA or cDNA library of another microorganism or mixed DNA sample. Once a polynucleotide encoding a parent polypeptide has been detected with the probe(s), the polynucleotide can be isolated or cloned by utilizing techniques that are known to those of ordinary skill in the art (see, e.g., Sambrook et al., 1989, supra).

(13) In one embodiment, the parent polypeptide comprises or consists of the amino acid sequence set forth in SEQ ID NO: 7, 13, or 15. In another embodiment, the parent polypeptide comprises or consists of the mature polypeptide of SEQ ID NO: 4. In particular, the parent polypeptide may comprise or consist of amino acids 1 to 485 of SEQ ID NO: 4.

(14) In another embodiment, the parent polypeptide is a fragment of the mature polypeptide of SEQ ID NO: 4 containing at least 350, such as at least 390 amino acid residues, e.g., at least 395 and at least 397 amino acid residues of SEQ ID NO: 4.

(15) In another embodiment, the parent polypeptide is an allelic variant of the mature polypeptide of SEQ ID NO: 4.

(16) The term “sequence identity” as used herein, refers to the relatedness between two amino acid sequences or between two nucleotide sequences. For purposes of the present invention, the sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48: 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16: 276-277), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix. The output of Needle labeled “longest identity” (obtained using the—nobrief option) is used as the percent identity and is calculated as follows:

(Identical Residues×100)/(Length of Alignment−Total Number of Gaps in Alignment)

(17) For purposes of the present invention, the sequence identity between two deoxyribonucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, supra) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, supra), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix. The output of Needle labeled “longest identity” (obtained using the—nobrief option) is used as the percent identity and is calculated as follows:

(Identical Deoxyribonucleotides×100)/(Length of Alignment−Total Number of Gaps in Alignment)

(18) The term “alpha-amylase activity” as used herein, refers to the activity of alpha-1,4-glucan-4-glucanohydrolases, E.C. 3.2.1.1, which constitute a group of enzymes, catalyzing hydrolysis of starch and other linear and branched 1,4-glucosidic oligo- and polysaccharides. Thus, the term “alpha-amylase” as used herein, refers to an enzyme that has alpha-amylase activity (Enzyme Class; EC 3.2.1.1) that hydrolyses alpha bonds of large, alpha-linked polysaccharides, such as starch and glycogen, yielding glucose and maltose. The terms “alpha-amylase” and “amylase” may be used interchangeably and constitute the same meaning and purpose within the scope of the present invention. For purposes of the present invention, alpha-amylase activity is determined according to the procedure described in the Examples. In one embodiment, the variants of the present invention have at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 100% of the alpha-amylase activity of the polypeptide of SEQ ID NOs: 7, 13, 14, or 15; or the mature polypeptide of SEQ ID NO: 4.

(19) The term “fusion polypeptide” as used herein, refers to a polypeptide which comprises amino acid sequences originating from more than one, such as two, three, or four, species. Such a fusion polypeptide may have been generated by molecular techniques well-known to the skilled person. A fusion polypeptide of the present invention has alpha-amylase activity. In particular, a fusion polypeptide of the present invention, comprises e.g. an A and B domain of one alpha-amylase and a C domain from another alpha-amylase.

(20) The term “wash performance” as used herein, refers to an enzyme's ability to remove starch or starch-containing stains present on the object to be cleaned during e.g. laundry or hard surface cleaning, such as dish wash. The term “wash performance” includes cleaning in general e.g. hard surface cleaning as in dish wash, but also wash performance on textiles such as laundry, and also industrial and institutional cleaning. The wash performance may be quantified by calculating the so-called Intensity value, and results may be displayed as “Improvement Factor” (IF). Wash performance may be determined as in described in the Examples herein.

(21) The term “Intensity value” as used herein, refers to the wash performance measured as the brightness expressed as the intensity of the light reflected from the sample when illuminated with white light. When the sample is stained the intensity of the reflected light is lower, than that of a clean sample. Therefore the intensity of the reflected light can be used to measure wash performance, where a higher intensity value correlates with higher wash performance.

(22) Color measurements are made with a professional flatbed scanner (Kodak iQsmart, Kodak) used to capture an image of the washed textile.

(23) To extract a value for the light intensity from the scanned images, 24-bit pixel values from the image are converted into values for red, green and blue (RGB). The intensity value (Int) is calculated by adding the RGB values together as vectors and then taking the length of the resulting vector:

(24) $\text{Int} = \sqrt{r^2 + g^2 + b^2}$

(25) The term “improved wash performance” as used herein, refers to an improvement of the wash performance of an alpha-amylase of the present

invention relative to the wash performance of the parent polypeptide. Improved wash performance may be measured by comparing of the so-called Intensity value and calculating the Improvement Factor (IF). The improved wash performance is determined according to the section "Wash performance of alpha-amylases using Automatic Mechanical Stress Assay" and using model detergent J at 15° C., 30° C. or 40° C. Other model detergents may be used, such as Model detergent A.
(26) Thus, in one embodiment, the improved wash performance is determined by a method comprising the steps of; a) washing a fabric stained with starch with an alpha-amylase variant and a parent polypeptide sample added, respectively, to a model detergent composition, such as Model A or Model J; b) measuring the intensity of light reflected from the sample when illuminated with white light; and c) optionally, calculating the improvement factor (IF) as the ratio of delta intensity of the alpha-amylase sample over the delta intensity of the parent polypeptide sample.
(27) In one embodiment, the improved wash performance is determined by a method comprising the steps of; a) washing a fabric stained with starch with an alpha-amylase variant and a parent polypeptide sample added, respectively, to a model detergent composition, such as Model A or Model J for 20 minutes at 15° C., 30° C., and 40° C.; b) measuring the intensity of light reflected from the sample when illuminated with white light; and c) optionally, calculating the improvement factor (IF) as the ration of delta intensity of the alpha-amylase sample over the delta intensity of the parent polypeptide sample.
(28) The term "low temperature" as used herein, refers to is a temperature of 5-40° C., such as 5-35° C., preferably 5-30° C., more preferably 5-25° C., more preferably 5-20° C., most preferably 5-15° C., and in particular 5-10° C. In a preferred embodiment, "Low temperature" is a temperature of 10-35° C., preferably 10-30° C., more preferably 10-25° C., most preferably 10-20° C., and in particular 10-15° C. Most preferred, low temperature means 15° C.
(29) Thus, in one embodiment, the low temperature is 5° C. to 40° C., preferably 10° C. to 30°, event more preferred 15° C. to 20° C.
(30) In one embodiment, the improved wash performance has an Improvement Factor (IF) of ≥1 when compared to said parent polypeptide having alpha-amylase activity.
(31) The term "Improvement Factor" as used herein, refers to a quantitative way of calculating the improvement of a particular property of a variant according to the present invention. Determination of the Improvement Factor may be according to the following formula:
(32) $\frac{\text{Intensityvalueofvariant - Intensityvalueofblank}}{\text{Intensityvalueofparent - Intensityvalueofblank}}$
(33) Other formulas may be used to determine the Improvement Factor. The skilled person knows the presently presented formula as well as alternative ways of calculating the Improvement Factor.
(34) According to the present invention, a value of 1.0 corresponds to the performance observed for the parent polypeptide. A value above 1.0 indicates an improvement of performance of the variant tested compared to the parent polypeptide. Accordingly, any value of ≥1.0 is indicative for improvement of property, such as performance, of the variant compared to the parent polypeptide.
(35) According to the present invention, a variant showing improvement of property under at least one condition tested, is considered a variant having improved property as compared to the parent polypeptide.
(36) In one embodiment, said parent polypeptide comprises an A and a B domain comprising the amino acid sequence of SEQ ID NO: 10, or an A and a B domain having at least 80%, such as 85%, such as 90%, such as 95%, such as 97% sequence identity to SEQ ID NO: 10, and a C domain of a low temperature alpha-amylase.
(37) The terms "A domain", "B domain" and "C domain" as used herein, refers to three distinct domains A, B and C, all part of the alpha-amylase structure, see, e.g., Machius et al., 1995, *J. Mol. Biol.* 246: 545-559. Thus, an alpha-amylase, such as a parent polypeptide and a variant according to the invention, may comprise both an A, B, and C domain. The term "domain" means a region of a polypeptide that in itself forms a distinct and independent substructure of the whole molecule. Alpha-amylases consist of a beta/alpha-8 barrel harboring the active site residues, which is denoted the A domain, a rather long loop between the beta-sheet 3 and alpha-helix 3, which is denoted the B domain (together; "A and B domain"), and a C-domain and in some cases also a carbohydrate binding domain (e.g., WO 2005/001064; Machius et al., supra).
(38) The domains of an alpha-amylase may be determined by structure analysis such as using crystallographically techniques. An alternative method for determining the domains of an alpha-amylase is by sequence alignment of the amino acid sequence of the alpha-amylase with another alpha-amylase for which the domains have been determined. The sequence that aligns with, e.g., the C-domain sequence in the alpha-amylase for which the C-domain has been determined can be considered the C domain for the given alpha-amylase.
(39) The term "A and B domain" as used herein, refers to two domains of an alpha-amylase taken as one unit, whereas the C domain is another unit of the alpha-amylases. Thus, the amino acid sequence of the "A and B domain" is understood as one sequence or one part of a sequence of an alpha-amylase comprising an "A and B domain" and other domains (such as the C domain). Thus, the term "the A and B domain domain has at least 65% sequence identity to SEQ ID NO: 10" means that the amino acid sequence that form the A and B domain has at least 65% sequence identity to SEQ ID NO: 10. As used herein, the "A and B domain" of a parent polypeptide corresponds to amino acids 1 to 399 of SEQ ID NO: 7, or to amino acids 1 to 397 of SEQ ID NO: 15. The A and B domain of a parent polypeptide may have the amino acid sequence of SEQ ID NO: 10.
(40) The term "C domain" as used herein, refers to a domain of an alpha-amylase as one unit. The "C domain" of an alpha-amylase corresponds to amino acids 400 to 485 of SEQ ID NO: 7, or to amino acid 398 to 483 of SEQ ID NO: 15. Thus, the C domain of an alpha amylase may be found by alignment of said alpha-amylase with the polypeptide of SEQ ID NO: 7 or 15. The part of said alpha-amylase that aligns with amino acids 400 to 485 of SEQ ID NO: 7 or to amino acids 398 to 483 of SEQ ID NO: 15 is according to the present invention "the C domain" of the alpha-amylase. The C domain of a parent polypeptide may have the amino acid sequence of SEQ ID NOs: 11 or 12.
(41) The term "C domain of a low temperature alpha-amylase" as used herein, refers to the C domain of an alpha-amylase showing particular properties at low temperatures. Non-limiting examples of such low temperature alpha-amylases are those originating from *Alicyclobacillus* sp. as well as from *Bacillus amyloliquefaciens* (BAN).
(42) In one embodiment, the parent polypeptide comprises a C domain comprising the amino acid sequence of SEQ ID NOs: 11 or 12, or a C domain having at 80%, such as 85%, such as 90%, such as 95%, such as 97%, sequence identity to SEQ ID NO: 11 or 12.
(43) In one embodiment, the parent polypeptide comprises the amino acid sequence of SEQ ID NOs: 13 or 14, or a sequence having at least 80%, such as 85%, such as 90%, such as 95%, such as 97% sequence identity to SEQ ID NOs: 13 or 14.
(44) As stated elsewhere herein, the parent polypeptide may be any polypeptide having alpha-amylase activity and at least 65% sequence identity to any one of the amino acid sequences as set forth in SEQ ID NOs: 13 and 14. In certain embodiments, the parent polypeptide comprises the amino acid sequences as set forth in SEQ ID NOs: 10, 11, and 12.
(45) In one embodiment, the parent polypeptide a sequence identity to the amino acid sequence set forth in SEQ ID NOs: 13 or 14 of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100%, and which parent polypeptide has alpha-amylase activity. In one embodiment, the amino acid sequence of the parent polypeptide differs by up to 10 amino acids, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, from the amino acid sequences set forth in SEQ ID NOs: 7, 13, and 14.
(46) The present invention provides variants of such parent polypeptides.
(47) The variants according to the present invention may advantageously comprise alterations in specific positions of the A and B domain. Thus, in a particular embodiment, the variant comprises amino acid alterations in amino acid positions corresponding the amino acid sequence set forth in SEQ ID NO: 10, or a sequence having at least 65% sequence identity to the amino acid sequence set forth in SEQ ID NO: 10.
(48) In particular, the present invention relates to an embodiment, the A and B domains comprise a modification in one or more of the following positions corresponding to positions; H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, A60, K72, R87, Q98, M105,

G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, G196, A204, V206, P211, I214, V215, L217, L219, A225, L228, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, N295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, and Q395 of SEQ ID NOs: 7 or 10.

(49) Such variants have shown to have improved wash performance in at least one assay wherein the temperature under which the variants have been tested was a low temperature, such as 15° C., 30° and 40° C. Specific conditions may be seen in the Examples herein.

(50) The term “modification” as used herein, refers to both substitutions and deletions of amino acid within the amino acid sequence of a polypeptide. The terms “alteration” and “modification” may be used interchangeably herein. This should not be understood as any limitation and thus, the terms constitute the same meaning and purpose unless explicitly stated otherwise.

(51) The term “corresponding to” as used herein, refers to way of determining the specific amino acid of a sequence wherein reference is made to a specific amino acid sequence. E.g. for the purposes of the present invention, when references are made to specific amino acid positions, the skilled person would be able to align another amino acid sequence to said amino acid sequence that reference has been made to, in order to determine which specific amino acid may be of interest in said another amino acid sequence. Alignment of another amino acid sequence with e.g. the sequence as set forth in SEQ ID NO: 4, 7, or any other sequence listed herein, has been described elsewhere herein. Alternative alignment methods may be used, and are well-known for the skilled person.

(52) In a particular embodiment, the variant comprises an alteration in two, three, four, five, six, seven, eight, nine, ten, eleven, twelfth, thirteen, fourteen, fifteen, or sixteen positions corresponding to the following positions; H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, A60, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, G196, A204, V206, P211, I214, V215, L217, L219, A225, L228, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, and Q395 of SEQ ID NOs: 7 or 10.

(53) In a particular embodiment, the variant comprises an alteration in the following positions;

(54) TABLE-US-00001 H1 + T5 T40 + L217 N54 + T285 Q125 + G255 G184 + P211 M248 + Q299 H1 + G7 T40 + L219 N54 + M286 Q125 + Q256 G184 + I214 M248 + S304 H1 + Q11 T40 + L228 N54 + F289 Q125 + A263 G184 + V215 M248 + R320 H1 + N16 T40 + L235 N54 + V291 Q125 + V264 G184 + L217 M248 + H321 H1 + V17 T40 + V238 N54 + Q299 Q125 + A265 G184 + L219 M248 + S323 H1 + Q32 T40 + M246 N54 + S304 Q125 + Y267 G184 + L228 M248 + H324 H1 + A37 T40 + M248 N54 + R320 Q125 + N270 G184 + L235 M248 + F328 H1 + T40 T40 + L250 N54 + H321 Q125 + G273 G184 + V238 M248 + T334 H1 + P45 T40 + G255 N54 + S323 Q125 + S280 G184 + M246 M248 + D337 H1 + W48 T40 + Q256 N54 + H324 Q125 + T285 G184 + M248 M248 + Q345 H1 + G50 T40 + A263 N54 + F328 Q125 + M286 G184 + L250 M248 + G346 H1 + T51 T40 + V264 N54 + T334 Q125 + F289 G184 + G255 M248 + G348 H1 + N54 T40 + A265 N54 + D337 Q125 + V291 G184 + Q256 M248 + T355 H1 + V56 T40 + Y267 N54 + Q345 Q125 + Q299 G184 + A263 M248 + S376 H1 + A60 T40 + N270 N54 + G346 Q125 + S304 G184 + V264 M248 + D377 H1 + K72 T40 + G273 N54 + G348 Q125 + R320 G184 + A265 M248 + D379 H1 + R87 T40 + S280 N54 + T355 Q125 + H321 G184 + Y267 M248 + Y382 H1 + Q98 T40 + T285 N54 + S376 Q125 + S323 G184 + N270 M248 + S383 H1 + M105 T40 + M286 N54 + D377 Q125 + H324 G184 + G273 M248 + Q385 H1 + G109 T40 + F289 N54 + D379 Q125 + F328 G184 + S280 M248 + K391 H1 + F113 T40 + V291 N54 + Y382 Q125 + T334 G184 + T285 M248 + K393 H1 + R116 T40 + Q299 N54 + S383 Q125 + D337 G184 + M286 M248 + Q395 H1 + Q118 T40 + S304 N54 + Q385 Q125 + Q345 G184 + F289 L250 + G255 H1 + Q125 T40 + R320 N54 + K391 Q125 + G346 G184 + V291 L250 + Q256 H1 + G133 T40 + H321 N54 + K393 Q125 + G348 G184 + Q299 L250 + A263 H1 + T134 T40 + S323 N54 + Q395 Q125 + T355 G184 + S304 L250 + V264 H1 + W140 T40 + H324 V56 + A60 Q125 + S376 G184 + R320 L250 + A265 H1 + G142 T40 + F328 V56 + K72 Q125 + D377 G184 + H321 L250 + Y267 H1 + G149 T40 + T334 V56 + R87 Q125 + D379 G184 + S323 L250 + N270 H1 + T165 T40 + D337 V56 + Q98 Q125 + Y382 G184 + H324 L250 + G273 H1 + W167 T40 + Q345 V56 + M105 Q125 + S383 G184 + F328 L250 + S280 H1 + R171 T40 + G346 V56 + G109 Q125 + Q385 G184 + T334 L250 + T285 H1 + Q172 T40 + G348 V56 + F113 Q125 + K391 G184 + D337 L250 + M286 H1 + L173 T40 + T355 V56 + R116 Q125 + K393 G184 + Q345 L250 + F289 H1 + A174 T40 + S376 V56 + Q118 Q125 + Q395 G184 + G346 L250 + V291 H1 + G184 T40 + D377 V56 + Q125 G133 + T134 G184 + G348 L250 + Q299 H1 + T193 T40 + D379 V56 + G133 G133 + W140 G184 + T355 L250 + S304 H1 + N195 T40 + Y382 V56 + T134 G133 + G142 G184 + S376 L250 + R320 H1 + G196 T40 + S383 V56 + W140 G133 + G149 G184 + D377 L250 + H321 H1 + A204 T40 + Q385 V56 + G142 G133 + T165 G184 + D379 L250 + S323 H1 + V206 T40 + K391 V56 + G149 G133 + W167 G184 + Y382 L250 + H324 H1 + P211 T40 + K393 V56 + T165 G133 + R171 G184 + S383 L250 + F328 H1 + I214 T40 + Q395 V56 + W167 G133 + Q172 G184 + Q385 L250 + T334 H1 + V215 P45 + W48 V56 + R171 G133 + L173 G184 + K391 L250 + D337 H1 + L217 P45 + G50 V56 + Q172 G133 + A174 G184 + K393 L250 + Q345 H1 + L219 P45 + T51 V56 + L173 G133 + G184 G184 + Q395 L250 + G346 H1 + L228 P45 + N54 V56 + A174 G133 + T193 T193 + N195 L250 + G348 H1 + L235 P45 + V56 V56 + G184 G133 + N195 T193 + G196 L250 + T355 H1 + V238 P45 + A60 V56 + T193 G133 + G196 T193 + A204 L250 + S376 H1 + M246 P45 + K72 V56 + N195 G133 + A204 T193 + V206 L250 + D377 H1 + M248 P45 + R87 V56 + G196 G133 + V206 T193 + P211 L250 + D379 H1 + L250 P45 + Q98 V56 + A204 G133 + P211 T193 + I214 L250 + Y382 H1 + G255 P45 + M105 V56 + V206 G133 + I214 T193 + V215 L250 + S383 H1 + Q256 P45 + G109 V56 + P211 G133 + V215 T193 + L217 L250 + Q385 H1 + A263 P45 + F113 V56 + I214 G133 + L217 T193 + L219 L250 + K391 H1 + V264 P45 + R116 V56 + V215 G133 + L219 T193 + L228 L250 + K393 H1 + A265 P45 + Q118 V56 + L217 G133 + L228 T193 + L235 L250 + Q395 H1 + Y267 P45 + Q125 V56 + L219 G133 + L235 T193 + V238 G255 + Q256 H1 + N270 P45 + G133 V56 + L228 G133 + V238 T193 + M246 G255 + A263 H1 + G273 P45 + T134 V56 + L235 G133 + M246 T193 + M248 G255 + V264 H1 + S280 P45 + W140 V56 + V238 G133 + M248 T193 + L250 G255 + A265 H1 + T285 P45 + G142 V56 + M246 G133 + L250 T193 + G255 G255 + Y267 H1 + M286 P45 + G149 V56 + M248 G133 + G255 T193 + Q256 G255 + N270 H1 + F289 P45 + T165 V56 + L250 G133 + Q256 T193 + A263 G255 + G273 H1 + V291 P45 + W167 V56 + G255 G133 + A263 T193 + V264 G255 + S280 H1 + Q299 P45 + R171 V56 + Q256 G133 + V264 T193 + A265 G255 + T285 H1 + S304 P45 + Q172 V56 + A263 G133 + A265 T193 + Y267 G255 + M286 H1 + R320 P45 + L173 V56 + V264 G133 + Y267 T193 + N270 G255 + F289 H1 + H321 P45 + A174 V56 + A265 G133 + N270 T193 + G273 G255 + V291 H1 + S323 P45 + G184 V56 + Y267 G133 + G273 T193 + S280 G255 + Q299 H1 + H324 P45 + T193 V56 + N270 G133 + S280 T193 + T285 G255 + S304 H1 + F328 P45 + N195 V56 + G273 G133 + T285 T193 + M286 G255 + R320 H1 + T334 P45 + G196 V56 + S280 G133 + M286 T193 + F289 G255 + H321 H1 + D337 P45 + A204 V56 + T285 G133 + F289 T193 + V291 G255 + S323 H1 + Q345 P45 + V206 V56 + M286 G133 + V291 T193 + Q299 G255 + H324 H1 + G346 P45 + P211 V56 + F289 G133 + Q299 T193 + S304 G255 + F328 H1 + G348 P45 + I214 V56 + V291 G133 + S304 T193 + R320 G255 + T334 H1 + T355 P45 + V215 V56 + Q299 G133 + R320 T193 + H321 G255 + D337 H1 + S376 P45 + L217 V56 + S304 G133 + H321 T193 + S323 G255 + Q345 H1 + D377 P45 + L219 V56 + R320 G133 + S323 T193 + H324 G255 + G346 H1 + D379 P45 + L228 V56 + H321 G133 + H324 T193 + F328 G255 + G348 H1 + Y382 P45 + L235 V56 + S323 G133 + F328 T193 + T334 G255 + T355 H1 + S383 P45 + V238 V56 + H324 G133 + T334 T193 + D337 G255 + S376 H1 + Q385 P45 + M246 V56 + F328 G133 + D337 T193 + Q345 G255 + D377 H1 + K391 P45 + M248 V56 + T334 G133 + Q345 T193 + G346 G255 + D379 H1 + K393 P45 + L250 V56 + D337 G133 + G346 T193 + G348 G255 + Y382 H1 + Q395 P45 + G255 V56 + Q345 G133 + G348 T193 + T355 G255 + S383 T5 + G7 P45 + Q256 V56 + G346 G133 + T355 T193 + S376 G255 + Q385 T5 + Q11 P45 + A263 V56 + G348 G133 + S376 T193 + D377 G255 + K391 T5 + N16 P45 + V264 V56 + T355 G133 + D377 T193 + D379 G255 + K393 T5 + V17 P45 + A265 V56 + S376 G133 + D379 T193 + Y382 G255 + Q395 T5 + Q32 P45 + Y267 V56 + D377 G133 + Y382 T193 + S383 Q256 + A263 T5 + A37 P45 + N270 V56 + D379 G133 + S383 T193 + Q385 Q256 + V264 T5 + T40 P45 + G273 V56 + Y382 G133 + Q385 T193 + K391 Q256 + A265 T5 + P45 P45 + S280 V56 + S383 G133 + K391 T193 + K393 Q256 + Y267 T5 + W48 P45 + T285 V56 + Q385 G133 + K393 T193 + Q395 Q256 +

N270 T5 + G50 P45 + M286 V56 + K391 G133 + Q395 N195 + G196 Q256 + G273 T5 + T51 P45 + F289 V56 + K393 T134 + W140 N195 + A204 Q256 + S280 T5 + N54 P45 + V291 V56 + Q395 T134 + G142 N195 + V206 Q256 + T285 T5 + V56 P45 + Q299 A60 + K72 T134 + G149 N195 + P211 Q256 + M286 T5 + A60 P45 + S304 A60 + R87 T134 + T165 N195 + I214 Q256 + F289 T5 + K72 P45 + R320 A60 + Q98 T134 + W167 N195 + V215 Q256 + V291 T5 + R87 P45 + H321 A60 + M105 T134 + R171 N195 + L217 Q256 + Q299 T5 + Q98 P45 + S323 A60 + G109 T134 + Q172 N195 + L219 Q256 + S304 T5 + M105 P45 + H324 A60 + F113 T134 + L173 N195 + L228 Q256 + R320 T5 + G109 P45 + F328 A60 + R116 T134 + A174 N195 + L235 Q256 + H321 T5 + F113 P45 + T334 A60 + Q118 T134 + G184 N195 + V238 Q256 + S323 T5 + R116 P45 + D337 A60 + Q125 T134 + T193 N195 + M246 Q256 + H324 T5 + Q118 P45 + Q345 A60 + G133 T134 + N195 N195 + M248 Q256 + F328 T5 + Q125 P45 + G346 A60 + T134 T134 + G196 N195 + L250 Q256 + T334 T5 + G133 P45 + G348 A60 + W140 T134 + A204 N195 + G255 Q256 + D337 T5 + T134 P45 + T355 A60 + G142 T134 + V206 N195 + Q256 Q256 + Q345 T5 + W140 P45 + S376 A60 + G149 T134 + P211 N195 + A263 Q256 + G346 T5 + G142 P45 + D377 A60 + T165 T134 + I214 N195 + V264 Q256 + G348 T5 + G149 P45 + D379 A60 + W167 T134 + V215 N195 + A265 Q256 + T355 T5 + T165 P45 + Y382 A60 + R171 T134 + L217 N195 + Y267 Q256 + S376 T5 + W167 P45 + S383 A60 + Q172 T134 + L219 N195 + N270 Q256 + D377 T5 + R171 P45 + Q385 A60 + L173 T134 + L228 N195 + G273 Q256 + D379 T5 + Q172 P45 + K391 A60 + A174 T134 + L235 N195 + S280 Q256 + Y382 T5 + L173 P45 + K393 A60 + G184 T134 + V238 N195 + T285 Q256 + S383 T5 + A174 P45 + Q395 A60 + T193 T134 + M246 N195 + M286 Q256 + Q385 T5 + G184 W48 + G50 A60 + N195 T134 + M248 N195 + F289 Q256 + K391 T5 + T193 W48 + T51 A60 + G196 T134 + L250 N195 + V291 Q256 + K393 T5 + N195 W48 + N54 A60 + A204 T134 + G255 N195 + Q299 Q256 + Q395 T5 + G196 W48 + V56 A60 + V206 T134 + Q256 N195 + S304 A263 + V264 T5 + A204 W48 + A60 A60 + P211 T134 + A263 N195 + R320 A263 + A265 T5 + V206 W48 + K72 A60 + I214 T134 + V264 N195 + H321 A263 + Y267 T5 + P211 W48 + R87 A60 + V215 T134 + A265 N195 + S323 A263 + N270 T5 + I214 W48 + Q98 A60 + L217 T134 + Y267 N195 + H324 A263 + G273 T5 + V215 W48 + M105 A60 + L219 T134 + N270 N195 + F328 A263 + S280 T5 + L217 W48 + G109 A60 + L228 T134 + G273 N195 + T334 A263 + T285 T5 + L219 W48 + F113 A60 + L235 T134 + S280 N195 + D337 A263 + M286 T5 + L228 W48 + R116 A60 + V238 T134 + T285 N195 + Q345 A263 + F289 T5 + L235 W48 + Q118 A60 + M246 T134 + M286 N195 + G346 A263 + V291 T5 + V238 W48 + Q125 A60 + M248 T134 + F289 N195 + G348 A263 + Q299 T5 + M246 W48 + G133 A60 + L250 T134 + V291 N195 + T355 A263 + S304 T5 + M248 W48 + T134 A60 + G255 T134 + Q299 N195 + S376 A263 + R320 T5 + L250 W48 + W140 A60 + Q256 T134 + S304 N195 + D377 A263 + H321 T5 + G255 W48 + G142 A60 + A263 T134 + R320 N195 + D379 A263 + S323 T5 + Q256 W48 + G149 A60 + V264 T134 + H321 N195 + Y382 A263 + H324 T5 + A263 W48 + T165 A60 + A265 T134 + S323 N195 + S383 A263 + F328 T5 + V264 W48 + W167 A60 + Y267 T134 + H324 N195 + Q385 A263 + T334 T5 + A265 W48 + R171 A60 + N270 T134 + F328 N195 + K391 A263 + D337 T5 + Y267 W48 + Q172 A60 + G273 T134 + T334 N195 + K393 A263 + Q345 T5 + N270 W48 + L173 A60 + S280 T134 + D337 N195 + Q395 A263 + G346 T5 + G273 W48 + A174 A60 + T285 T134 + Q345 G196 + A204 A263 + G348 T5 + S280 W48 + G184 A60 + M286 T134 + G346 G196 + V206 A263 + T355 T5 + T285 W48 + T193 A60 + F289 T134 + G348 G196 + P211 A263 + S376 T5 + M286 W48 + N195 A60 + V291 T134 + T355 G196 + I214 A263 + D377 T5 + F289 W48 + G196 A60 + Q299 T134 + S376 G196 + V215 A263 + D379 T5 + V291 W48 + A204 A60 + S304 T134 + D377 G196 + L217 A263 + Y382 T5 + Q299 W48 + V206 A60 + R320 T134 + D379 G196 + L219 A263 + S383 T5 + S304 W48 + P211 A60 + H321 T134 + Y382 G196 + L228 A263 + Q385 T5 + R320 W48 + I214 A60 + S323 T134 + S383 G196 + L235 A263 + K391 T5 + H321 W48 + V215 A60 + H324 T134 + Q385 G196 + V238 A263 + K393 T5 + S323 W48 + L217 A60 + F328 T134 + K391 G196 + M246 A263 + Q395 T5 + H324 W48 + L219 A60 + T334 T134 + K393 G196 + M248 V264 + A265 T5 + F328 W48 + L228 A60 + D337 T134 + Q395 G196 + L250 V264 + Y267 T5 + T334 W48 + L235 A60 + Q345 W140 + G142 G196 + G255 V264 + N270 T5 + D337 W48 + V238 A60 + G346 W140 + G149 G196 + Q256 V264 + G273 T5 + Q345 W48 + M246 A60 + G348 W140 + T165 G196 + A263 V264 + S280 T5 + G346 W48 + M248 A60 + T355 W140 + W167 G196 + V264 V264 + T285 T5 + G348 W48 + L250 A60 + S376 W140 + R171 G196 + A265 V264 + M286 T5 + T355 W48 + G255 A60 + D377 W140 + Q172 G196 + Y267 V264 + F289 T5 + S376 W48 + Q256 A60 + D379 W140 + L173 G196 + N270 V264 + V291 T5 + D377 W48 + A263 A60 + Y382 W140 + A174 G196 + G273 V264 + Q299 T5 + D379 W48 + V264 A60 + S383 W140 + G184 G196 + S280 V264 + S304 T5 + Y382 W48 + A265 A60 + Q385 W140 + T193 G196 + T285 V264 + R320 T5 + S383 W48 + Y267 A60 + K391 W140 + N195 G196 + M286 V264 + H321 T5 + Q385 W48 + N270 A60 + K393 W140 + G196 G196 + F289 V264 + S323 T5 + K391 W48 + G273 A60 + Q395 W140 + A204 G196 + V291 V264 + H324 T5 + K393 W48 + S280 K72 + R87 W140 + V206 G196 + Q299 V264 + F328 T5 + Q395 W48 + T285 K72 + Q98 W140 + P211 G196 + S304 V264 + T334 G7 + Q11 W48 + M286 K72 + M105 W140 + I214 G196 + R320 V264 + D337 G7 + N16 W48 + F289 K72 + G109 W140 + V215 G196 + H321 V264 + Q345 G7 + V17 W48 + V291 K72 + F113 W140 + L217 G196 + S323 V264 + G346 G7 + Q32 W48 + Q299 K72 + R116 W140 + L219 G196 + H324 V264 + G348 G7 + A37 W48 + S304 K72 + Q118 W140 + L228 G196 + F328 V264 + T355 G7 + T40 W48 + R320 K72 + Q125 W140 + L235 G196 + T334 V264 + S376 G7 + P45 W48 + H321 K72 + G133 W140 + V238 G196 + D337 V264 + D377 G7 + W48 W48 + S323 K72 + T134 W140 + M246 G196 + Q345 V264 + D379 G7 + G50 W48 + H324 K72 + W140 W140 + M248 G196 + G346 V264 + Y382 G7 + T51 W48 + F328 K72 + G142 W140 + L250 G196 + G348 V264 + S383 G7 + N54 W48 + T334 K72 + G149 W140 + G255 G196 + T355 V264 + Q385 G7 + V56 W48 + D337 K72 + T165 W140 + Q256 G196 + S376 V264 + K391 G7 + A60 W48 + Q345 K72 + W167 W140 + A263 G196 + D377 V264 + K393 G7 + K72 W48 + G346 K72 + R171 W140 + V264 G196 + D379 V264 + Q395 G7 + R87 W48 + G348 K72 + Q172 W140 + A265 G196 + Y382 A265 + Y267 G7 + Q98 W48 + T355 K72 + L173 W140 + Y267 G196 + S383 A265 + N270 G7 + M105 W48 + S376 K72 + A174 W140 + N270 G196 + Q385 A265 + G273 G7 + G109 W48 + D377 K72 + G184 W140 + G273 G196 + K391 A265 + S280 G7 + F113 W48 + D379 K72 + T193 W140 + S280 G196 + K393 A265 + T285 G7 + R116 W48 + Y382 K72 + N195 W140 + T285 G196 + Q395 A265 + M286 G7 + Q118 W48 + S383 K72 + G196 W140 + M286 A204 + V206 A265 + F289 G7 + Q125 W48 + Q385 K72 + A204 W140 + F289 A204 + P211 A265 + V291 G7 + G133 W48 + K391 K72 + V206 W140 + V291 A204 + I214 A265 + Q299 G7 + T134 W48 + K393 K72 + P211 W140 + Q299 A204 + V215 A265 + S304 G7 + W140 W48 + Q395 K72 + I214 W140 + S304 A204 + L217 A265 + R320 G7 + G142 G50 + T51 K72 + V215 W140 + R320 A204 + L219 A265 + H321 G7 + G149 G50 + N54 K72 + 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G50 + G142 K72 + G273 W140 + Q385 A204 + T285 A265 + K391 G7 + L217 G50 + G149 K72 + S280 W140 + K391 A204 + M286 A265 + K393 G7 + L219 G50 + T165 K72 + T285 W140 + K393 A204 + F289 A265 + Q395 G7 + L228 G50 + W167 K72 + M286 W140 + Q395 A204 + V291 Y267 + N270 G7 + L235 G50 + R171 K72 + F289 G142 + G149 A204 + Q299 Y267 + G273 G7 + V238 G50 + Q172 K72 + V291 G142 + T165 A204 + S304 Y267 + S280 G7 + M246 G50 + L173 K72 + Q299 G142 + W167 A204 + R320 Y267 + T285 G7 + M248 G50 + A174 K72 + S304 G142 + R171 A204 + H321 Y267 + M286 G7 + L250 G50 + G184 K72 + R320 G142 + Q172 A204 + S323 Y267 + F289 G7 + G255 G50 + T193 K72 + H321 G142 + L173 A204 + H324 Y267 + V291 G7 + Q256 G50 + N195 K72 + S323 G142 + A174 A204 + F328 Y267 + Q299 G7 + A263 G50 + G196 K72 + H324 G142 + G184 A204 + T334 Y267 + S304 G7 + V264 G50 + A204 K72 + F328 G142 + T193 A204 + D337 Y267 + R320 G7 + A265 G50 + V206 K72 + T334 G142 + N195 A204 + Q345 Y267 + H321 G7 + Y267 G50 + P211 K72 + D337 G142 + G196 A204 + G346 Y267 + S323 G7 + N270 G50 + I214 K72 + Q345 G142 + A204 A204 + G348 Y267 + H324 G7 + G273 G50 + V215 K72 + G346 G142 + V206 A204 + T355 Y267 + F328 G7 + S280 G50 + L217 K72 + G348 G142 + P211 A204 + S376 Y267 + T334 G7 + T285 G50 + L219 K72 + T355 G142 + I214 A204 + D377 Y267 + D337 G7 + M286 G50 + L228 K72 + S376 G142 + V215 A204 + D379 Y267 +

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K391 G142 + M246 A204 + Q395 Y267 + D379 G7 + S323 G50 + Q256 K72 + K393 G142 + M248 V206 + P211 Y267 + Y382 G7 + H324 G50 +
A263 K72 + Q395 G142 + L250 V206 + I214 Y267 + S383 G7 + F328 G50 + V264 R87 + Q98 G142 + G255 V206 + V215 Y267 + Q385 G7 +
T334 G50 + A265 R87 + M105 G142 + Q256 V206 + L217 Y267 + K391 G7 + D337 G50 + Y267 R87 + G109 G142 + A263 V206 + L219 Y267 +
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Q385 G50 + R320 R87 + T165 G142 + V291 V206 + V264 N270 + R320 G7 + K391 G50 + H321 R87 + W167 G142 + Q299 V206 + A265 N270 +
H321 G7 + K393 G50 + S323 R87 + R171 G142 + S304 V206 + Y267 N270 + S323 G7 + Q395 G50 + H324 R87 + Q172 G142 + R320 V206 +
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G50 G50 + S376 R87 + V206 G142 + G346 V206 + S304 N270 + S376 Q11 + T51 G50 + D377 R87 + P211 G142 + G348 V206 + R320 N270 +
D377 Q11 + N54 G50 + D379 R87 + I214 G142 + T355 V206 + H321 N270 + D379 Q11 + V56 G50 + Y382 R87 + V215 G142 + S376 V206 +
S323 N270 + Y382 Q11 + A60 G50 + S383 R87 + L217 G142 + D377 V206 + H324 N270 + S383 Q11 + K72 G50 + Q385 R87 + L219 G142 +
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L235 G142 + S383 V206 + D337 N270 + K393 Q11 + M105 G50 + Q395 R87 + V238 G142 + Q385 V206 + Q345 N270 + Q395 Q11 + G109 T51
+ N54 R87 + M246 G142 + K391 V206 + G346 G273 + S280 Q11 + F113 T51 + V56 R87 + M248 G142 + K393 V206 + G348 G273 + T285 Q11 +
R116 T51 + A60 R87 + L250 G142 + Q395 V206 + T355 G273 + M286 Q11 + Q118 T51 + K72 R87 + G255 G149 + T165 V206 + S376 G273 +
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+ Q125 R87 + S280 G149 + N195 V206 + Q395 G273 + F328 Q11 + R171 T51 + G133 R87 + T285 G149 + G196 P211 + I214 G273 + T334 Q11 +
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L228 G273 + G348 Q11 + T193 T51 + T165 R87 + S304 G149 + V215 P211 + L235 G273 + T355 Q11 + N195 T51 + W167 R87 + R320 G149 +
L217 P211 + V238 G273 + S376 Q11 + G196 T51 + R171 R87 + H321 G149 + L219 P211 + M246 G273 + D377 Q11 + A204 T51 + Q172 R87 +
S323 G149 + L228 P211 + M248 G273 + D379 Q11 + V206 T51 + L173 R87 + H324 G149 + L235 P211 + L250 G273 + Y382 Q11 + P211 T51 +
A174 R87 + F328 G149 + V238 P211 + G255 G273 + S383 Q11 + I214 T51 + G184 R87 + T334 G149 + M246 P211 + Q256 G273 + Q385 Q11 +
V215 T51 + T193 R87 + D337 G149 + M248 P211 + A263 G273 + K391 Q11 + L217 T51 + N195 R87 + Q345 G149 + L250 P211 + V264 G273 +
K393 Q11 + L219 T51 + G196 R87 + G346 G149 + G255 P211 + A265 G273 + Q395 Q11 + L228 T51 + A204 R87 + G348 G149 + Q256 P211 +
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V264 P211 + G273 S280 + F289 Q11 + M246 T51 + I214 R87 + D377 G149 + A265 P211 + S280 S280 + V291 Q11 + M248 T51 + V215 R87 +
D379 G149 + Y267 P211 + T285 S280 + Q299 Q11 + L250 T51 + L217 R87 + Y382 G149 + N270 P211 + M286 S280 + S304 Q11 + G255 T51 +
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S376 Q98 + L228 T165 + G184 I214 + G255 T285 + G346 Q11 + Q395 T51 + D377 Q98 + L235 T165 + T193 I214 + Q256 T285 + G348 N16 +
V17 T51 + D379 Q98 + V238 T165 + N195 I214 + A263 T285 + T355 N16 + Q32 T51 + Y382 Q98 + M246 T165 + G196 I214 + V264 T285 +
S376 N16 + A37 T51 + S383 Q98 + M248 T165 + A204 I214 + A265 T285 + D377 N16 + T40 T51 + Q385 Q98 + L250 T165 + V206 I214 + Y267
T285 + D379 N16 + P45 T51 + K391 Q98 + G255 T165 + P211 I214 + N270 T285 + Y382 N16 + W48 T51 + K393 Q98 + Q256 T165 + I214 I214
+ G273 T285 + S383 N16 + G50 T51 + Q395 Q98 + A263 T165 + V215 I214 + S280 T285 + Q385 N16 + T51 N54 + V56 Q98 + V264 T165 +
L217 I214 + T285 T285 + K391 N16 + N54 N54 + A60 Q98 + A265 T165 + L219 I214 + M286 T285 + K393 N16 + V56 N54 + K72 Q98 + Y267
T165 + L228 I214 + F289 T285 + Q395 N16 + A60 N54 + R87 Q98 + N270 T165 + L235 I214 + V291 M286 + F289 N16 + K72 N54 + Q98 Q98 +
G273 T165 + V238 I214 + Q299 M286 + V291 N16 + R87 N54 + M105 Q98 + S280 T165 + M246 I214 + S304 M286 + Q299 N16 + Q98 N54 +
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G109 N54 + R116 Q98 + F289 T165 + G255 I214 + S323 M286 + H321 N16 + F113 N54 + Q118 Q98 + V291 T165 + Q256 I214 + H324 M286 +
S323 N16 + R116 N54 + Q125 Q98 + Q299 T165 + A263 I214 + F328 M286 + H324 N16 + Q118 N54 + G133 Q98 + S304 T165 + V264 I214 +
T334 M286 + F328 N16 + Q125 N54 + T134 Q98 + R320 T165 + A265 I214 + D337 M286 + T334 N16 + G133 N54 + W140 Q98 + H321 T165 +
Y267 I214 + Q345 M286 + D337 N16 + T134 N54 + G142 Q98 + S323 T165 + N270 I214 + G346 M286 + Q345 N16 + W140 N54 + G149 Q98 +
H324 T165 + G273 I214 + G348 M286 + G346 N16 + G142 N54 + T165 Q98 + F328 T165 + S280 I214 + T355 M286 + G348 N16 + G149 N54 +
W167 Q98 + T334 T165 + T285 I214 + S376 M286 + T355 N16 + T165 N54 + R171 Q98 + D337 T165 + M286 I214 + D377 M286 + S376 N16 +
W167 N54 + Q172 Q98 + Q345 T165 + F289 I214 + D379 M286 + D377 N16 + R171 N54 + L173 Q98 + G346 T165 + V291 I214 + Y382 M286 +
D379 N16 + Q172 N54 + A174 Q98 + G348 T165 + Q299 I214 + S383 M286 + Y382 N16 + L173 N54 + G184 Q98 + T355 T165 + S304 I214 +

Q385 M286 + S383 N16 + A174 N54 + T193 Q98 + S376 T165 + R320 I214 + K391 M286 + Q385 N16 + G184 N54 + N195 Q98 + D377 T165 +
H321 I214 + K393 M286 + K391 N16 + T193 N54 + G196 Q98 + D379 T165 + S323 I214 + Q395 M286 + K393 N16 + N195 N54 + A204 Q98 +
Y382 T165 + H324 V215 + L217 M286 + Q395 N16 + G196 N54 + V206 Q98 + S383 T165 + F328 V215 + L219 F289 + V291 N16 + A204 N54 +
P211 Q98 + Q385 T165 + T334 V215 + L228 F289 + Q299 N16 + V206 N54 + I214 Q98 + K391 T165 + D337 V215 + L235 F289 + S304 N16 +
P211 N54 + V215 Q98 + K393 T165 + Q345 V215 + V238 F289 + R320 N16 + I214 N54 + L217 Q98 + Q395 T165 + G346 V215 + M246 F289 +
H321 N16 + V215 N54 + L219 M105 + G109 T165 + G348 V215 + M248 F289 + S323 N16 + L217 N54 + L228 M105 + F113 T165 + T355 V215
+ L250 F289 + H324 N16 + L219 N54 + L235 M105 + R116 T165 + S376 V215 + G255 F289 + F328 N16 + L228 N54 + V238 M105 + Q118 T165
+ D377 V215 + Q256 F289 + T334 N16 + L235 N54 + M246 M105 + Q125 T165 + D379 V215 + A263 F289 + D337 N16 + V238 N54 + M248
M105 + G133 T165 + Y382 V215 + V264 F289 + Q345 N16 + M246 N54 + L250 M105 + T134 T165 + S383 V215 + A265 F289 + G346 N16 +
M248 N54 + G255 M105 + W140 T165 + Q385 V215 + Y267 F289 + G348 N16 + L250 N54 + Q256 M105 + G142 T165 + K391 V215 + N270
F289 + T355 N16 + G255 N54 + A263 M105 + G149 T165 + K393 V215 + G273 F289 + S376 N16 + Q256 N54 + V264 M105 + T165 T165 +
Q395 V215 + S280 F289 + D377 N16 + A263 N54 + A265 M105 + W167 W167 + R171 V215 + T285 F289 + D379 N16 + V264 N54 + Y267
M105 + R171 W167 + Q172 V215 + M286 F289 + Y382 N16 + A265 N54 + N270 M105 + Q172 W167 + L173 V215 + F289 F289 + S383 N16 +
Y267 N54 + G273 M105 + L173 W167 + A174 V215 + V291 F289 + Q385 N16 + N270 N54 + S280 M105 + A174 W167 + G184 V215 + Q299
F289 + K391 N16 + G273 Q118 + V215 M105 + G184 W167 + T193 V215 + S304 F289 + K393 N16 + S280 Q118 + L217 M105 + T193 W167 +
N195 V215 + R320 F289 + Q395 N16 + T285 Q118 + L219 M105 + N195 W167 + G196 V215 + H321 V291 + Q299 N16 + M286 Q118 + L228
M105 + G196 W167 + A204 V215 + S323 V291 + S304 N16 + F289 Q118 + L235 M105 + A204 W167 + V206 V215 + H324 V291 + R320 N16 +
V291 Q118 + V238 M105 + V206 W167 + P211 V215 + F328 V291 + H321 N16 + Q299 Q118 + M246 M105 + P211 W167 + I214 V215 + T334
V291 + S323 N16 + S304 Q118 + M248 M105 + I214 W167 + V215 V215 + D337 V291 + H324 N16 + R320 Q118 + L250 M105 + V215 W167 +
L217 V215 + Q345 V291 + F328 N16 + H321 Q118 + G255 M105 + L217 W167 + L219 V215 + G346 V291 + T334 N16 + S323 Q118 + Q256
M105 + L219 W167 + L228 V215 + G348 V291 + D337 N16 + H324 Q118 + A263 M105 + L228 W167 + L235 V215 + T355 V291 + Q345 N16 +
F328 Q118 + V264 M105 + L235 W167 + V238 V215 + S376 V291 + G346 N16 + T334 Q118 + A265 M105 + V238 W167 + M246 V215 + D377
V291 + G348 N16 + D337 Q118 + Y267 M105 + M246 W167 + M248 V215 + D379 V291 + T355 N16 + Q345 Q118 + N270 M105 + M248 W167
+ L250 V215 + Y382 V291 + S376 N16 + G346 Q118 + G273 M105 + L250 W167 + G255 V215 + S383 V291 + D377 N16 + G348 Q118 + S280
M105 + G255 W167 + Q256 V215 + Q385 V291 + D379 N16 + T355 Q118 + T285 M105 + Q256 W167 + A263 V215 + K391 V291 + Y382 N16 +
S376 Q118 + M286 M105 + A263 W167 + V264 V215 + K393 V291 + S383 N16 + D377 Q118 + F289 M105 + V264 W167 + A265 V215 + Q395
V291 + Q385 N16 + D379 Q118 + V291 M105 + A265 W167 + Y267 L217 + L219 V291 + K391 N16 + Y382 Q118 + Q299 M105 + Y267 W167 +
N270 L217 + L228 V291 + K393 N16 + S383 Q118 + S304 M105 + N270 W167 + G273 L217 + L235 V291 + Q395 N16 + Q385 Q118 + R320
M105 + G273 W167 + S280 L217 + V238 Q299 + S304 N16 + K391 Q118 + H321 M105 + S280 W167 + T285 L217 + M246 Q299 + R320 N16 +
K393 Q118 + S323 M105 + T285 W167 + M286 L217 + M248 Q299 + H321 N16 + Q395 Q118 + H324 M105 + M286 W167 + F289 L217 + L250
Q299 + S323 V17 + Q32 Q118 + F328 M105 + F289 W167 + V291 L217 + G255 Q299 + H324 V17 + A37 Q118 + T334 M105 + V291 W167 +
Q299 L217 + Q256 Q299 + F328 V17 + T40 Q118 + D337 M105 + Q299 W167 + S304 L217 + A263 Q299 + T334 V17 + P45 Q118 + Q345 M105
+ S304 W167 + R320 L217 + V264 Q299 + D337 V17 + W48 Q118 + G346 M105 + R320 W167 + H321 L217 + A265 Q299 + Q345 V17 + G50
Q118 + G348 M105 + H321 W167 + S323 L217 + Y267 Q299 + G346 V17 + T51 Q118 + T355 M105 + S323 W167 + H324 L217 + N270 Q299 +
G348 V17 + N54 Q118 + S376 M105 + H324 W167 + F328 L217 + G273 Q299 + T355 V17 + V56 Q118 + D377 M105 + F328 W167 + T334
L217 + S280 Q299 + S376 V17 + A60 Q118 + D379 M105 + T334 W167 + D337 L217 + T285 Q299 + D377 V17 + K72 Q118 + Y382 M105 +
D337 W167 + Q345 L217 + M286 Q299 + D379 V17 + R87 Q118 + S383 M105 + Q345 W167 + G346 L217 + F289 Q299 + Y382 V17 + Q98
Q118 + Q385 M105 + G346 W167 + G348 L217 + V291 Q299 + S383 V17 + M105 Q118 + K391 M105 + G348 W167 + T355 L217 + Q299 Q299
+ Q385 V17 + G109 Q118 + K393 M105 + T355 W167 + S376 L217 + S304 Q299 + K391 V17 + F113 Q118 + Q395 M105 + S376 W167 + D377
L217 + R320 Q299 + K393 V17 + R116 Q125 + G133 M105 + D377 W167 + D379 L217 + H321 Q299 + Q395 V17 + Q118 Q125 + T134 M105 +
D379 W167 + Y382 L217 + S323 S304 + R320 V17 + Q125 Q125 + W140 M105 + Y382 W167 + S383 L217 + H324 S304 + H321 V17 + G133
Q125 + G142 M105 + S383 W167 + Q385 L217 + F328 S304 + S323 V17 + T134 Q125 + G149 M105 + Q385 W167 + K391 L217 + T334 S304 +
H324 V17 + W140 Q125 + T165 M105 + K391 W167 + K393 L217 + D337 S304 + F328 V17 + G142 Q125 + W167 M105 + K393 W167 + Q395
L217 + Q345 S304 + T334 V17 + G149 Q125 + R171 M105 + Q395 R171 + Q172 L217 + G346 S304 + D337 V17 + T165 Q125 + Q172 G109 +
F113 R171 + L173 L217 + G348 S304 + Q345 V17 + W167 Q125 + L173 G109 + R116 R171 + A174 L217 + T355 S304 + G346 V17 + R171
Q125 + A174 G109 + Q118 R171 + G184 L217 + S376 S304 + G348 V17 + Q172 Q125 + G184 G109 + Q125 R171 + T193 L217 + D377 S304 +
T355 V17 + L173 Q125 + T193 G109 + G133 R171 + N195 L217 + D379 S304 + S376 V17 + A174 Q125 + N195 G109 + T134 R171 + G196
L217 + Y382 S304 + D377 V17 + G184 Q125 + G196 G109 + W140 R171 + A204 L217 + S383 S304 + D379 V17 + T193 Q125 + A204 G109 +
G142 R171 + V206 L217 + Q385 S304 + Y382 V17 + N195 Q125 + V206 G109 + G149 R171 + P211 L217 + K391 S304 + S383 V17 + G196
Q125 + P211 G109 + T165 R171 + I214 L217 + K393 S304 + Q385 V17 + A204 Q125 + I214 G109 + W167 R171 + V215 L217 + Q395 S304 +
K391 V17 + V206 Q125 + V215 G109 + R171 R171 + L217 L219 + L228 S304 + K393 V17 + P211 Q125 + L217 G109 + Q172 R171 + L219
L219 + L235 S304 + Q395 V17 + I214 Q125 + L219 G109 + L173 R171 + L228 L219 + V238 R320 + H321 V17 + V215 Q125 + L228 G109 +
A174 R171 + L235 L219 + M246 R320 + S323 V17 + L217 Q125 + L235 G109 + G184 R171 + V238 L219 + M248 R320 + H324 V17 + L219
Q125 + V238 G109 + T193 R171 + M246 L219 + L250 R320 + F328 V17 + L228 Q125 + M246 G109 + N195 R171 + M248 L219 + G255 R320 +
T334 V17 + L235 Q125 + M248 G109 + G196 R171 + L250 L219 + Q256 R320 + D337 V17 + V238 Q125 + L250 G109 + A204 R171 + G255
L219 + A263 R320 + Q345 V17 + M246 A37 + Q395 G109 + V206 R171 + Q256 L219 + V264 R320 + G346 V17 + M248 T40 + P45 G109 + P211
R171 + A263 L219 + A265 R320 + G348 V17 + L250 T40 + W48 G109 + I214 R171 + V264 L219 + Y267 R320 + T355 V17 + G255 T40 + G50
G109 + V215 R171 + A265 L219 + N270 R320 + S376 V17 + Q256 T40 + T51 G109 + L217 R171 + Y267 L219 + G273 R320 + D377 V17 +
A263 T40 + N54 G109 + L219 R171 + N270 L219 + S280 R320 + D379 V17 + V264 T40 + V56 G109 + L228 R171 + G273 L219 + T285 R320 +
Y382 V17 + A265 T40 + A60 G109 + L235 R171 + S280 L219 + M286 R320 + S383 V17 + Y267 T40 + K72 G109 + V238 R171 + T285 L219 +
F289 R320 + Q385 V17 + N270 T40 + R87 G109 + M246 R171 + M286 L219 + V291 R320 + K391 V17 + G273 T40 + Q98 G109 + M248 R171 +
F289 L219 + Q299 R320 + K393 V17 + S280 T40 + M105 G109 + L250 R171 + V291 L219 + S304 R320 + Q395 V17 + T285 T40 + G109 G109 +
G255 R171 + Q299 L219 + R320 H321 + S323 V17 + M286 T40 + F113 G109 + Q256 R171 + S304 L219 + H321 H321 + H324 V17 + F289 T40 +
R116 G109 + A263 R171 + R320 L219 + S323 H321 + F328 V17 + V291 T40 + Q118 G109 + V264 R171 + H321 L219 + H324 H321 + T334 V17
+ Q299 T40 + Q125 G109 + A265 R171 + S323 L219 + F328 H321 + D337 V17 + S304 T40 + G133 G109 + Y267 R171 + H324 L219 + T334
H321 + Q345 V17 + R320 T40 + T134 G109 + N270 R171 + F328 L219 + D337 H321 + G346 V17 + H321 T40 + W140 G109 + G273 R171 +
T334 L219 + Q345 H321 + G348 V17 + S323 T40 + G142 G109 + S280 R171 + D337 L219 + G346 H321 + T355 V17 + H324 T40 + G149 G109
+ T285 R171 + Q345 L219 + G348 H321 + S376 V17 + F328 T40 + T165 G109 + M286 R171 + G346 L219 + T355 H321 + D377 V17 + T334 T40
+ W167 G109 + F289 R171 + G348 L219 + S376 H321 + D379 V17 + D337 T40 + R171 G109 + V291 R171 + T355 L219 + D377 H321 + Y382
V17 + Q345 T40 + Q172 G109 + Q299 R171 + S376 L219 + D379 H321 + S383 V17 + G346 T40 + L173 G109 + S304 R171 + D377 L219 + Y382
H321 + Q385 V17 + G348 T40 + A174 G109 + R320 R171 + D379 L219 + S383 H321 + K391 V17 + T355 T40 + G184 G109 + H321 R171 +
Y382 L219 + Q385 H321 + K393 V17 + S376 T40 + T193 G109 + S323 R171 + S383 L219 + K391 H321 + Q395 V17 + D377 T40 + N195 G109 +
H324 R171 + Q385 L219 + K393 S323 + H324 V17 + D379 T40 + G196 G109 + F328 R171 + K391 L219 + Q395 S323 + F328 V17 + Y382 T40 +
A204 G109 + T334 R171 + K393 L228 + L235 S323 + T334 V17 + S383 T40 + V206 G109 + D337 R171 + Q395 L228 + V238 S323 + D337 V17
+ Q385 T40 + P211 G109 + Q345 Q172 + L173 L228 + M246 S323 + Q345 V17 + K391 T40 + I214 G109 + G346 Q172 + A174 L228 + M248

S323 + G346 V17 + K393 T40 + V215 G109 + G348 Q172 + G184 L228 + L250 S323 + G348 V17 + Q395 H1 + Q169 G109 + T355 Q172 + T193 L228 + G255 S323 + T355 Q32 + A37 H1 + A186 G109 + S376 Q172 + N195 L228 + Q256 S323 + S376 Q32 + T40 H1 + E190 G109 + D377 Q172 + G196 L228 + A263 S323 + D377 Q32 + P45 H1 + A225 G109 + D379 Q172 + A204 L228 + V264 S323 + D379 Q32 + W48 H1 + K242 G109 + Y382 Q172 + V206 L228 + A265 S323 + Y382 Q32 + G50 H1 + S244 G109 + S383 Q172 + P211 L228 + Y267 S323 + S383 Q32 + T51 H1 + N260 G109 + Q385 Q172 + I214 L228 + N270 S323 + Q385 Q32 + N54 H1 + K269 G109 + K391 Q172 + V215 L228 + G273 S323 + K391 Q32 + V56 H1 + W284 G109 + K393 Q172 + L217 L228 + S280 S323 + K393 Q32 + A60 H1 + Y295 G109 + Q395 Q172 + L219 L228 + T285 S323 + Q395 Q32 + K72 H1 + V326 F113 + R116 Q172 + L228 L228 + M286 H324 + F328 Q32 + R87 T5 + Q169 F113 + Q118 Q172 + L235 L228 + F289 H324 + T334 Q32 + Q98 T5 + A186 F113 + Q125 Q172 + V238 L228 + V291 H324 + D337 Q32 + M105 T5 + E190 F113 + G133 Q172 + M246 L228 + Q299 H324 + Q345 Q32 + G109 T5 + A225 F113 + T134 Q172 + M248 L228 + S304 H324 + G346 Q32 + F113 T5 + K242 F113 + W140 Q172 + L250 L228 + R320 H324 + G348 Q32 + R116 T5 + S244 F113 + G142 Q172 + G255 L228 + H321 H324 + T355 Q32 + Q118 T5 + N260 F113 + G149 Q172 + Q256 L228 + S323 H324 + S376 Q32 + Q125 T5 + K269 F113 + T165 Q172 + A263 L228 + H324 H324 + D377 Q32 + G133 T5 + W284 F113 + W167 Q172 + V264 L228 + F328 H324 + D379 Q32 + T134 T5 + Y295 F113 + R171 Q172 + A265 L228 + T334 H324 + Y382 Q32 + W140 T5 + V326 F113 + Q172 Q172 + Y267 L228 + D337 H324 + S383 Q32 + G142 G7 + Q169 F113 + L173 Q172 + N270 L228 + Q345 H324 + Q385 Q32 + G149 G7 + A186 F113 + A174 Q172 + G273 L228 + G346 H324 + K391 Q32 + T165 G7 + E190 F113 + G184 Q172 + S280 L228 + G348 H324 + K393 Q32 + W167 G7 + A225 F113 + T193 Q172 + T285 L228 + T355 H324 + Q395 Q32 + R171 G7 + K242 F113 + N195 Q172 + M286 L228 + S376 F328 + T334 Q32 + Q172 G7 + S244 F113 + G196 Q172 + F289 L228 + D377 F328 + D337 Q32 + L173 G7 + N260 F113 + A204 Q172 + V291 L228 + D379 F328 + Q345 Q32 + A174 G7 + K269 F113 + V206 Q172 + Q299 L228 + Y382 F328 + G346 Q32 + G184 G7 + W284 F113 + P211 Q172 + S304 L228 + S383 F328 + G348 Q32 + T193 G7 + Y295 F113 + I214 Q172 + R320 L228 + Q385 F328 + T355 Q32 + N195 G7 + V326 F113 + V215 Q172 + H321 L228 + K391 F328 + S376 Q32 + G196 Q11 + Q169 F113 + L217 Q172 + S323 L228 + K393 F328 + D377 Q32 + A204 Q11 + A186 F113 + L219 Q172 + H324 L228 + Q395 F328 + D379 Q32 + V206 Q11 + E190 F113 + L228 Q172 + F328 L235 + V238 F328 + Y382 Q32 + P211 Q11 + A225 F113 + L235 Q172 + T334 L235 + M246 F328 + S383 Q32 + I214 Q11 + K242 F113 + V238 Q172 + D337 L235 + M248 F328 + Q385 Q32 + V215 Q11 + S244 F113 + M246 Q172 + Q345 L235 + L250 F328 + K391 Q32 + L217 Q11 + N260 F113 + M248 Q172 + G346 L235 + G255 F328 + K393 Q32 + L219 Q11 + K269 F113 + L250 Q172 + G348 L235 + Q256 F328 + Q395 Q32 + L228 Q11 + W284 F113 + G255 Q172 + T355 L235 + A263 T334 + D337 Q32 + L235 Q11 + Y295 F113 + Q256 Q172 + S376 L235 + V264 T334 + Q345 Q32 + V238 Q11 + V326 F113 + A263 Q172 + D377 L235 + A265 T334 + G346 Q32 + M246 N16 + Q169 F113 + V264 Q172 + D379 L235 + Y267 T334 + G348 Q32 + M248 N16 + A186 F113 + A265 Q172 + Y382 L235 + N270 T334 + T355 Q32 + L250 N16 + E190 F113 + Y267 Q172 + S383 L235 + G273 T334 + S376 Q32 + G255 N16 + A225 F113 + N270 Q172 + Q385 L235 + S280 T334 + D377 Q32 + Q256 N16 + K242 F113 + G273 Q172 + K391 L235 + T285 T334 + D379 Q32 + A263 N16 + S244 F113 + S280 Q172 + K393 L235 + M286 T334 + Y382 Q32 + V264 N16 + N260 F113 + T285 Q172 + Q395 L235 + F289 T334 + S383 Q32 + A265 N16 + K269 F113 + M286 L173 + A174 L235 + V291 T334 + Q385 Q32 + Y267 N16 + W284 F113 + F289 L173 + G184 L235 + Q299 T334 + K391 Q32 + N270 N16 + Y295 F113 + V291 L173 + T193 L235 + S304 T334 + K393 Q32 + G273 N16 + V326 F113 + Q299 L173 + N195 L235 + R320 T334 + Q395 Q32 + S280 V17 + Q169 F113 + S304 L173 + G196 L235 + H321 D337 + Q345 Q32 + T285 V17 + A186 F113 + R320 L173 + A204 L235 + S323 D337 + G346 Q32 + M286 V17 + E190 F113 + H321 L173 + V206 L235 + H324 D337 + G348 Q32 + F289 V17 + A225 F113 + S323 L173 + P211 L235 + F328 D337 + T355 Q32 + V291 V17 + K242 F113 + H324 L173 + I214 L235 + T334 D337 + S376 Q32 + Q299 V17 + S244 F113 + F328 L173 + V215 L235 + D337 D337 + D377 Q32 + S304 V17 + N260 F113 + T334 L173 + L217 L235 + Q345 D337 + D379 Q32 + R320 V17 + K269 F113 + D337 L173 + L219 L235 + G346 D337 + Y382 Q32 + H321 V17 + W284 F113 + Q345 L173 + L228 L235 + G348 D337 + S383 Q32 + S323 V17 + Y295 F113 + G346 L173 + L235 L235 + T355 D337 + Q385 Q32 + H324 V17 + V326 F113 + G348 L173 + V238 L235 + S376 D337 + K391 Q32 + F328 Q32 + Q169 F113 + T355 L173 + M246 L235 + D377 D337 + K393 Q32 + T334 Q32 + A186 F113 + S376 L173 + M248 L235 + D379 D337 + Q395 Q32 + D337 Q32 + E190 F113 + D377 L173 + L250 L235 + Y382 Q345 + G346 Q32 + Q345 Q32 + A225 F113 + D379 L173 + G255 L235 + S383 Q345 + G348 Q32 + G346 Q32 + K242 F113 + Y382 L173 + Q256 L235 + Q385 Q345 + T355 Q32 + G348 Q32 + S244 F113 + S383 L173 + A263 L235 + K391 Q345 + S376 Q32 + T355 Q32 + N260 F113 + Q385 L173 + V264 L235 + K393 Q345 + D377 Q32 + S376 Q32 + K269 F113 + K391 L173 + A265 L235 + Q395 Q345 + D379 Q32 + D377 Q32 + W284 F113 + K393 L173 + Y267 V238 + M246 Q345 + Y382 Q32 + D379 Q32 + Y295 F113 + Q395 L173 + N270 V238 + M248 Q345 + S383 Q32 + Y382 Q32 + V326 R116 + Q118 L173 + G273 V238 + L250 Q345 + Q385 Q32 + S383 A37 + Q169 R116 + Q125 L173 + S280 V238 + G255 Q345 + K391 Q32 + Q385 A37 + A186 R116 + G133 L173 + T285 V238 + Q256 Q345 + K393 Q32 + K391 A37 + E190 R116 + T134 L173 + M286 V238 + A263 Q345 + Q395 Q32 + K393 A37 + A225 R116 + W140 L173 + F289 V238 + V264 G346 + G348 Q32 + Q395 A37 + K242 R116 + G142 L173 + V291 V238 + A265 G346 + T355 A37 + T40 A37 + S244 R116 + G149 L173 + Q299 V238 + Y267 G346 + S376 A37 + P45 A37 + N260 R116 + T165 L173 + S304 V238 + N270 G346 + D377 A37 + W48 A37 + K269 R116 + W167 L173 + R320 V238 + G273 G346 + D379 A37 + G50 A37 + W284 R116 + R171 L173 + H321 V238 + S280 G346 + Y382 A37 + T51 A37 + Y295 R116 + Q172 L173 + S323 V238 + T285 G346 + S383 A37 + N54 A37 + V326 R116 + L173 L173 + H324 V238 + M286 G346 + Q385 A37 + V56 T40 + Q169 R116 + A174 L173 + F328 V238 + F289 G346 + K391 A37 + A60 T40 + A186 R116 + G184 L173 + T334 V238 + V291 G346 + K393 A37 + K72 T40 + E190 R116 + T193 L173 + D337 V238 + Q299 G346 + Q395 A37 + R87 T40 + A225 R116 + N195 L173 + Q345 V238 + S304 G348 + T355 A37 + Q98 T40 + K242 R116 + G196 L173 + G346 V238 + R320 G348 + S376 A37 + M105 T40 + S244 R116 + A204 L173 + G348 V238 + H321 G348 + D377 A37 + G109 T40 + N260 R116 + V206 L173 + T355 V238 + S323 G348 + D379 A37 + F113 T40 + K269 R116 + P211 L173 + S376 V238 + H324 G348 + Y382 A37 + R116 T40 + W284 R116 + I214 L173 + D377 V238 + F328 G348 + S383 A37 + Q118 T40 + Y295 R116 + V215 L173 + D379 V238 + T334 G348 + Q385 A37 + Q125 T40 + V326 R116 + L217 L173 + Y382 V238 + D337 G348 + K391 A37 + G133 P45 + Q169 R116 + L219 L173 + S383 V238 + Q345 G348 + K393 A37 + T134 P45 + A186 R116 + L228 L173 + Q385 V238 + G346 G348 + Q395 A37 + W140 P45 + E190 R116 + L235 L173 + K391 V238 + G348 T355 + S376 A37 + G142 P45 + A225 R116 + V238 L173 + K393 V238 + T355 T355 + D377 A37 + G149 P45 + K242 R116 + M246 L173 + Q395 V238 + S376 T355 + D379 A37 + T165 P45 + S244 R116 + M248 A174 + G184 V238 + D377 T355 + Y382 A37 + W167 P45 + N260 R116 + L250 A174 + T193 V238 + D379 T355 + S383 A37 + R171 P45 + K269 R116 + G255 A174 + N195 V238 + Y382 T355 + Q385 A37 + Q172 P45 + W284 R116 + Q256 A174 + G196 V238 + S383 T355 + K391 A37 + L173 P45 + Y295 R116 + A263 A174 + A204 V238 + Q385 T355 + K393 A37 + A174 P45 + V326 R116 + V264 A174 + V206 V238 + K391 T355 + Q395 A37 + G184 W48 + Q169 R116 + A265 A174 + P211 V238 + K393 S376 + D377 A37 + T193 W48 + A186 R116 + Y267 A174 + I214 V238 + Q395 S376 + D379 A37 + N195 W48 + E190 R116 + N270 A174 + V215 M246 + M248 S376 + Y382 A37 + G196 W48 + A225 R116 + G273 A174 + L217 M246 + L250 S376 + S383 A37 + A204 W48 + K242 R116 + S280 A174 + L219 M246 + G255 S376 + Q385 A37 + V206 W48 + S244 R116 + T285 A174 + L228 M246 + Q256 S376 + K391 A37 + P211 W48 + N260 R116 + M286 A174 + L235 M246 + A263 S376 + K393 A37 + I214 W48 + K269 R116 + F289 A174 + V238 M246 + V264 S376 + Q395 A37 + V215 W48 + W284 R116 + V291 A174 + M246 M246 + A265 D377 + D379 A37 + L217 W48 + Y295 R116 + Q299 A174 + M248 M246 + Y267 D377 + Y382 A37 + L219 W48 + V326 R116 + S304 A174 + L250 M246 + N270 D377 + S383 A37 + L228 G50 + Q169 R116 + R320 A174 + G255 M246 + G273 D377 + Q385 A37 + L235 G50 + A186 R116 + H321 A174 + Q256 M246 + S280 D377 + K391 A37 + V238 G50 + E190 R116 + S323 A174 + A263 M246 + T285 D377 + K393 A37 + M246 G50 + A225 R116 + H324 A174 + V264 M246 + M286 D377 + Q395 A37 + M248 G50 + K242 R116 + F328 A174 + A265 M246 + F289 D379 + Y382 A37 + L250 G50 + S244 R116 + T334 A174 + Y267 M246 + V291 D379 + S383 A37 + G255 G50 + N260 R116 + D337 A174 + N270 M246 + Q299 D379 + Q385 A37 + Q256 G50 + K269 R116 + Q345 A174 + G273 M246 + S304 D379 + K391 A37 + A263 G50 + W284 R116 + G346 A174 + S280 M246 + R320 D379 + K393 A37 + V264 G50 + Y295 R116 + G348 A174 + T285 M246 + H321 D379 + Q395 A37 + A265 G50 + V326 R116 + T355 A174 + M286 M246 + S323 Y382 + S383 A37 + Y267 T51 + Q169 R116 + S376 A174 + F289 M246 + H324 Y382 + Q385 A37 + N270 T51 + A186 R116 + D377 A174 + V291 M246 + F328 Y382 + K391 A37 + G273

T51 + E190 R116 + D379 A174 + Q299 M246 + T334 Y382 + K393 A37 + S280 T51 + A225 R116 + Y382 A174 + S304 M246 + D337 Y382 + Q395 A37 + T285 T51 + K242 R116 + S383 A174 + R320 M246 + Q345 S383 + Q385 A37 + M286 T51 + S244 R116 + Q385 A174 + H321 M246 + G346 S383 + K391 A37 + F289 T51 + N260 R116 + K391 A174 + S323 M246 + G348 S383 + K393 A37 + V291 T51 + K269 R116 + K393 A174 + H324 M246 + T355 S383 + Q395 A37 + Q299 T51 + W284 R116 + Q395 A174 + F328 M246 + S376 Q385 + K391 A37 + S304 T51 + Y295 Q118 + Q125 A174 + T334 M246 + D377 Q385 + K393 A37 + R320 T51 + V326 Q118 + G133 A174 + D337 M246 + D379 Q385 + Q395 A37 + H321 N54 + Q169 Q118 + T134 A174 + Q345 M246 + Y382 K391 + K393 A37 + S323 N54 + A186 Q118 + W140 A174 + G346 M246 + S383 K391 + Q395 A37 + H324 N54 + E190 Q118 + G142 A174 + G348 M246 + Q385 K393 + Q395 A37 + F328 N54 + A225 Q118 + G149 A174 + T355 M246 + K391 M248 + F289 A37 + T334 N54 + K242 Q118 + T165 A174 + S376 M246 + K393 M248 + V291 A37 + D337 N54 + S244 Q118 + W167 A174 + D377 M246 + Q395 G109 + Q169 A37 + Q345 N54 + N260 Q118 + R171 A174 + D379 M248 + L250 G109 + A186 A37 + G346 N54 + K269 Q118 + Q172 A174 + Y382 M248 + G255 G109 + E190 A37 + G348 N54 + W284 Q118 + L173 A174 + S383 M248 + Q256 G109 + A225 A37 + T355 N54 + Y295 Q118 + A174 A174 + Q385 M248 + A263 G109 + K242 A37 + S376 N54 + V326 Q118 + G184 A174 + K391 M248 + V264 G109 + S244 A37 + D377 V56 + Q169 Q118 + T193 A174 + K393 M248 + A265 G109 + N260 A37 + D379 V56 + A186 Q118 + N195 A174 + Q395 M248 + Y267 G109 + K269 A37 + Y382 V56 + E190 Q118 + G196 G184 + T193 M248 + N270 G109 + W284 A37 + S383 V56 + A225 Q118 + A204 G184 + N195 M248 + G273 G109 + Y295 A37 + Q385 V56 + K242 Q118 + V206 G184 + G196 M248 + S280 G109 + V326 A37 + K391 V56 + S244 Q118 + P211 G184 + A204 M248 + T285 F113 + Q169 A37 + K393 V56 + N260 Q118 + I214 G184 + V206 M248 + M286 F113 + A186 A60 + Q169 V56 + K269 G133 + Q169 W167 + Q169 R171 + A186 F113 + A225 A60 + A186 V56 + W284 G133 + A186 W167 + A186 R171 + E190 F113 + E190 A60 + E190 V56 + Y295 G133 + E190 W167 + E190 R171 + A225 F113 + K242 A60 + A225 V56 + V326 G133 + A225 W167 + A225 R171 + K242 F113 + S244 A60 + K242 G184 + A186 G133 + K242 W167 + K242 R171 + S244 F113 + N260 A60 + S244 G184 + E190 G133 + S244 W167 + S244 R171 + N260 F113 + K269 A60 + N260 G184 + A225 G133 + N260 W167 + N260 R171 + K269 F113 + W284 A60 + K269 G184 + K242 G133 + K269 W167 + K269 R171 + W284 F113 + Y295 A60 + W284 G184 + S244 G133 + W284 W167 + W284 R171 + Y295 F113 + V326 A60 + Y295 G184 + N260 G133 + Y295 W167 + Y295 R171 + V326 R116 + Q169 A60 + V326 G184 + K269 G133 + V326 W167 + V326 Q172 + A186 R116 + A186 K72 + Q169 G184 + W284 T134 + Q169 Q169 + A186 Q172 + E190 R116 + E190 K72 + A186 G184 + Y295 T134 + A186 Q169 + E190 Q172 + A225 R116 + A225 K72 + E190 G184 + V326 T134 + E190 Q169 + L173 Q172 + K242 R116 + K242 K72 + A225 A186 + E190 T134 + A225 Q169 + A174 Q172 + S244 R116 + S244 K72 + K242 A186 + T193 T134 + K242 Q169 + G184 Q172 + N260 R116 + N260 K72 + S244 A186 + N195 T134 + S244 Q169 + T193 Q172 + K269 R116 + K269 K72 + N260 A186 + G196 T134 + N260 Q169 + N195 Q172 + W284 R116 + W284 K72 + K269 A186 + A204 T134 + K269 Q169 + G196 Q172 + Y295 R116 + Y295 K72 + W284 A186 + V206 T134 + W284 Q169 + A204 Q172 + V326 R116 + V326 K72 + Y295 A186 + P211 T134 + Y295 Q169 + V206 L173 + A186 Q118 + Q169 K72 + V326 A186 + I214 T134 + V326 Q169 + P211 L173 + E190 Q118 + A186 R87 + Q169 A186 + V215 W140 + Q169 Q169 + I214 L173 + A225 Q118 + A225 R87 + A186 A186 + L217 W140 + A186 Q169 + V215 L173 + K242 Q118 + K242 R87 + E190 A186 + L219 W140 + E190 Q169 + L217 L173 + S244 Q118 + S244 R87 + A225 A186 + A225 W140 + A225 Q169 + L219 L173 + N260 Q118 + N260 R87 + K242 A186 + L228 W140 + K242 Q169 + A225 L173 + K269 Q118 + K269 R87 + S244 A186 + L235 W140 + S244 Q169 + L228 L173 + W284 Q118 + W284 R87 + N260 A186 + V238 W140 + N260 Q169 + L235 L173 + Y295 Q118 + Y295 R87 + K269 A186 + K242 W140 + K269 Q169 + V238 L173 + V326 Q118 + V326 R87 + W284 A186 + S244 W140 + W284 Q169 + K242 A174 + A186 Q125 + Q169 R87 + Y295 A186 + M246 W140 + Y295 Q169 + S244 A174 + E190 Q125 + A186 R87 + V326 A186 + M248 W140 + V326 Q169 + M246 A174 + A225 Q125 + E190 Q98 + Q169 A186 + L250 G142 + Q169 Q169 + M248 A174 + K242 Q125 + A225 Q98 + A186 A186 + G255 G142 + A186 Q169 + L250 A174 + S244 Q125 + K242 Q98 + E190 A186 + Q256 G142 + E190 Q169 + G255 A174 + N260 Q125 + S244 Q98 + A225 A186 + N260 G142 + A225 Q169 + Q256 A174 + K269 Q125 + N260 Q98 + K242 A186 + A263 G142 + K242 Q169 + N260 A174 + W284 Q125 + K269 Q98 + S244 A186 + V264 G142 + S244 Q169 + A263 A174 + Y295 Q125 + W284 Q98 + N260 A186 + A265 G142 + N260 Q169 + V264 A174 + V326 Q125 + Y295 Q98 + K269 A186 + Y267 G142 + K269 Q169 + A265 T193 + A225 Q125 + V326 Q98 + W284 A186 + K269 G142 + W284 Q169 + Y267 T193 + K242 V215 + A225 Q98 + Y295 A186 + N270 G142 + Y295 Q169 + K269 T193 + S244 V215 + K242 Q98 + V326 A186 + G273 G142 + V326 Q169 + N270 T193 + N260 V215 + S244 M105 + Q169 A186 + S280 G149 + Q169 Q169 + G273 T193 + K269 V215 + N260 A186 + W284 Q169 + S280 T193 + W284 V215 + K269 M105 + A186 A186 + T285 G149 + A186 Q169 + W284 T193 + Y295 V215 + W284 M105 + E190 A186 + M286 G149 + E190 Q169 + T285 T193 + V326 V215 + Y295 M105 + A225 A186 + F289 G149 + A225 Q169 + M286 N195 + A225 V215 + V326 M105 + K242 A186 + V291 G149 + K242 Q169 + F289 N195 + K242 L217 + A225 M105 + S244 A186 + Y295 G149 + S244 Q169 + V291 N195 + S244 L217 + K242 M105 + N260 A186 + Q299 G149 + N260 Q169 + Y295 N195 + N260 L217 + S244 M105 + K269 A186 + S304 G149 + K269 Q169 + Q299 N195 + K269 L217 + N260 M105 + W284 A186 + R320 G149 + W284 Q169 + S304 N195 + W284 L217 + K269 M105 + Y295 A186 + H321 G149 + Y295 Q169 + R320 N195 + Y295 L217 + W284 M105 + V326 A186 + S323 G149 + V326 Q169 + H321 N195 + V326 L217 + Y295 A225 + L228 A186 + H324 T165 + Q169 Q169 + S323 G196 + A225 L217 + V326 A225 + L235 A186 + V326 T165 + A186 Q169 + H324 G196 + K242 L219 + A225 A225 + V238 A186 + F328 T165 + E190 Q169 + V326 G196 + K242 L219 + K242 A225 + K242 A186 + T334 T165 + A225 Q169 + F328 G196 + S244 L219 + S244 A225 + S244 A186 + D337 T165 + K242 Q169 + T334 G196 + N260 L219 + N260 A225 + M246 A186 + Q345 T165 + S244 Q169 + D337 G196 + K269 L219 + K269 A225 + M248 A186 + G346 T165 + N260 Q169 + Q345 G196 + W284 L219 + W284 A225 + L250 A186 + G348 T165 + K269 Q169 + G346 G196 + Y295 L219 + Y295 A225 + G255 A186 + T355 T165 + W284 Q169 + G348 G196 + V326 L219 + V326 A225 + Q256 A186 + S376 T165 + Y295 Q169 + T355 A204 + A225 M246 + N260 A225 + N260 A186 + D377 T165 + V326 Q169 + S376 A204 + K242 M246 + K269 A225 + A263 A186 + D379 L228 + K242 Q169 + D377 A204 + S244 M246 + W284 A225 + V264 A186 + Y382 L228 + S244 Q169 + D379 A204 + N260 M246 + Y295 A225 + A265 A186 + S383 L228 + N260 Q169 + Y382 A204 + K269 M246 + V326 A225 + Y267 A186 + Q385 L228 + K269 Q169 + S383 A204 + W284 M248 + N260 A225 + K269 A186 + K391 L228 + W284 Q169 + Q385 A204 + Y295 M248 + K269 A225 + N270 A186 + K393 L228 + Y295 Q169 + K391 A204 + V326 M248 + W284 A225 + G273 A186 + Q395 L228 + V326 Q169 + K393 V206 + A225 M248 + Y295 A225 + S280 E190 + T193 L235 + K242 Q169 + Q395 A206 + K242 M248 + V326 A225 + W284 E190 + N195 L235 + S244 Q256 + N260 A206 + S244 L250 + N260 A225 + T285 E190 + G196 L235 + N260 G256 + K269 A206 + N260 L250 + K269 A225 + M286 E190 + A204 L235 + K269 Q256 + W284 A206 + K269 L250 + W284 A225 + F289 E190 + V206 L235 + W284 Q256 + Y295 A206 + W284 L250 + Y295 A225 + V291 E190 + P211 L235 + Y295 Q256 + V326 A206 + Y295 L250 + V326 A225 + Y295 E190 + I214 L235 + V326 N260 + A263 A206 + V326 G255 + N260 A225 + Q299 E190 + V215 V238 + K242 N260 + V264 P211 + A225 G255 + K269 A225 + S304 E190 + L217 V238 + S244 N260 + A265 P211 + K242 G255 + W284 A225 + R320 E190 + L219 V238 + N260 N260 + Y267 P211 + S244 G255 + Y295 A225 + H321 E190 + A225 V238 + K269 N260 + K269 P211 + N206 G255 + V326 A225 + S323 E190 + L228 V238 + W284 N260 + N270 P211 + K269 N270 + W284 A225 + H324 E190 + L235 V238 + Y295 N260 + G273 P211 + W284 N270 + Y295 A225 + V326 E190 + V238 V238 + V326 N260 + S280 P211 + Y295 N270 + V326 A225 + F328 E190 + K242 K242 + S244 N260 + W284 P211 + V326 G273 + W284 A225 + T334 E190 + S244 K242 + M246 N260 + T285 I214 + A225 G273 + Y295 A225 + D337 E190 + M246 K242 + M248 N260 + M286 I214 + K242 G273 + V326 A225 + Q345 E190 + M248 K242 + L250 N260 + F289 I214 + S244 S280 + W284 A225 + G346 E190 + L250 K242 + G255 N260 + V291 I214 + N260 S280 + Y295 A225 + G348 E190 + G255 K242 + Q256 N260 + Y295 I214 + K269 S280 + V326 A225 + T355 E190 + Q256 K242 + N260 N260 + Q299 I214 + W284 W284 + T285 A225 + S376 E190 + N260 K242 + A263 N260 + S304 I214 + Y295 W284 + M286 A225 + D377 E190 + A263 K242 + V264 N260 + R320 I214 + V326 W284 + F289 A225 + D379 E190 + V264 K242 + A265 N260 + H321 K269 + N270 W284 + V291 A225 + Y382 E190 + A265 K242 + Y267 N260 + S323 K269 + G273 W284 + Y295 A225 + S383 E190 + Y267 K242 + K269 N260 + H324 K269 + S280 W284 + Q299 A225 + Q385 E190 + K269 K242 + N270 N260 + V326 K269 + W284 W284 + S304 A225 + K391 E190 + N270 K242 + G273 N260 + F328 K269 + T285 W284 + R320 A225 + K393 E190 + G273 K242 + S280 N260 + T334 K269 + M286 W284 + H321 A225 + Q395 E190 + S280 K242 + W284 N260 + D337 K269 + F289 W284 + S323 A263 + K269 E190 + W284 K242 + T285 N260 + Q345

K269 + V291 W284 + H324 A263 + W284 E190 + T285 K242 + M286 N260 + G346 K269 + Y295 W284 + V326 A263 + Y295 E190 + M286 K242 + F289 N260 + G348 K269 + Q299 W284 + F328 A263 + V326 E190 + F289 K242 + V291 N260 + T355 K269 + S304 W284 + T334 V264 + K269 E190 + V291 K242 + Y295 N260 + S376 K269 + R320 W284 + D337 V264 + W284 E190 + Y295 K242 + Q299 N260 + D377 K269 + H321 W284 + Q345 V264 + Y295 E190 + Q299 K242 + S304 N260 + D379 K269 + S323 W284 + G346 V264 + V326 E190 + S304 K242 + R320 N260 + Y382 K269 + H324 W284 + G348 A265 + K269 E190 + R320 K242 + H321 N260 + S383 K269 + V326 W284 + T355 A265 + W284 E190 + H321 K242 + S323 N260 + Q385 K269 + F328 W284 + S376 A265 + Y295 E190 + S323 K242 + H324 N260 + K391 K269 + T334 W284 + D377 A265 + V326 E190 + H324 K242 + V326 N260 + K393 K269 + D337 W284 + D379 A265 + K269 E190 + V326 K242 + F328 N260 + Q395 K269 + Q345 W284 + Y382 A265 + W284 E190 + F328 K242 + T334 Y295 + S376 K269 + G346 W284 + S383 A265 + Y295 E190 + T334 K242 + D337 V326 + S376 K269 + G348 W284 + Q385 A265 + V326 E190 + D337 K242 + Q345 V326 + D377 K269 + T355 W284 + K391 Y295 + D377 E190 + Q345 K242 + G346 V326 + D379 K269 + S376 W284 + K393 Y295 + D379 E190 + G346 K242 + G348 V326 + Y382 K269 + D377 W284 + Q395 Y295 + Y382 E190 + G348 K242 + T355 V326 + S383 K269 + D379 T285 + Y295 Y295 + S383 E190 + T355 K242 + S376 V326 + Q385 K269 + Y382 T285 + V326 Y295 + Q385 E190 + S376 K242 + D377 V326 + K391 K269 + S383 M286 + Y295 Y295 + K391 E190 + D377 K242 + D379 V326 + K393 K269 + Q385 M286 + V326 Y295 + K393 E190 + D379 K242 + Y382 V326 + Q395 K269 + K391 F289 + Y295 Y295 + Q395 E190 + Y382 K242 + S383 S244 + K393 K269 + K393 F289 + V326 Q299 + V326 E190 + S383 K242 + Q385 S244 + Q395 K269 + Q395 V291 + Y295 S304 + V326 E190 + Q385 K242 + K391 S244 + Y267 S244 + R320 V294 + V326 R320 + V326 E190 + K391 K242 + K393 S244 + K269 S244 + H321 Y295 + Q299 H321 + V326 E190 + K393 K242 + Q395 S244 + N270 S244 + S323 Y295 + S304 S323 + V326 E190 + Q395 S244 + M246 S244 + G273 S244 + H324 Y295 + R320 H324 + V326 S244 + F289 S244 + M248 S244 + S280 S244 + V326 Y295 + H321 V326 + F328 S244 + V291 S244 + L250 S244 + W284 S244 + F328 Y295 + S323 V326 + T334 S244 + Y295 S244 + G255 S244 + T285 S244 + T334 Y295 + H324 V326 + D337 S244 + Q299 S244 + Q256 S244 + S376 S244 + D337 Y295 + V326 V326 + Q345 S244 + S304 S244 + N260 S244 + D377 S244 + Q345 Y295 + F328 V326 + G346 S244 + K391 S244 + A263 S244 + D379 S244 + G346 Y295 + T334 V326 + G348 S244 + M286 S244 + V264 S244 + Y382 S244 + G348 Y295 + F328 V326 + T355 Y295 + Q345 S244 + A265 S244 + S383 S244 + T355 Y295 + T334 Y295 + G348 Y295 + G346 S244 + Q385 Y295 + D337 Y295 + T355

(55) In one particular embodiment, the modifications in one or more positions correspond to the following; H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40D, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51 G, T51A, T51 S, T51 G, T51 D, N54S, V56T, A60V, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, N125D, G133S, F133Q, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, 1L173T, L173F, 1L173M, A174S, A740, A174P, A174M, A174T, A174H, A174K, A174G, A174Q, A174N, A174V, A174L, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, N195F, G196R, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L228I, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255A, Q256A, N260G, A263G, V264I, V264T, A265G, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S304N, S304R, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, and Q395P, wherein said positions correspond to positions of SEQ ID NOs: 7 or 10.

(56) The inventors of the present invention have identified that these specific alterations within the A and B domain (as defined by SEQ ID NO: 10) of the amino acid sequence as set forth in SEQ ID NOs: 7 and 10, are particularly relevant for improving the performance of a variant alpha-amylase having at least 65% sequence identity to the parent polypeptide.

(57) In one embodiment, the variant comprises an alteration in two, three, four, five, six, seven, eight, nine, ten, eleven, twelfth, thirteen, fourteen, fifteen, sixteen, or seventeen positions corresponding to the following positions; H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40D, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, A60V, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, N125D, G133S, F133Q, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, A174S, A174D, A174P, A174M, A174T, A174H, A174K, A174G, A174Q, A174N, A174V, A174L, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, N195F, G196R, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, 1214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L228I, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, Q256A, N260G, A263G, V264I, V264T, A265G, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S304N, S304R, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, and Q395P, wherein said positions correspond to positions of SEQ ID NOs: 7 or 10.

(58) The C domain of a variant according to the present invention may also comprise alterations. Thus, in one embodiment, the variant comprises an alteration in one or more amino acid positions corresponding to the amino acid positions of the amino acid sequence set forth in SEQ ID NOs: 11 or 12, or an amino acid sequence having at least 65% sequence identity to an amino acid sequence as set forth in SEQ ID NOs: 11 or 12.

(59) In one particular embodiment, the C domain comprises an alteration in one or more of the following positions corresponding to positions; A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13 or I405, A421, A422, A428, R439 G448, W467, D476, and G477 of SEQ ID NO: 14.

(60) In one embodiment, the variant comprises an alteration in two, three, four, five, six, seven, or eight positions corresponding to the following positions; A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13 or I405, A421, A422, A428, R439 G448, W467, D476, and G477 of SEQ ID NO: 14.

(61) The inventors of the present invention have identified that specific amino acid substitutions within the C domain of parent polypeptide have an improved effect on performance of the variants. Thus, in a particular embodiment, the alteration in one or more positions correspond to the following T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S, wherein said positions correspond to positions of SEQ ID NO: 13; or I405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q, wherein said positions correspond to positions of SEQ ID NO: 14.

(62) In a particular embodiment, the variant comprises a modification in the following positions;

(63) TABLE-US-00002 H1 + T5 T40 + A174 R87 + N270 W140 + K393 V206 + V215 Y267 + C474 H1 + G7 T40 + G184 R87 + G273 W140 + Q395 V206 + L217 Y267 + G476 H1 + Q11 T40 + T193 R87 + S280 W140 + A420 V206 + L219 Y267 + G477 H1 + N16 T40 + N195 R87 + T285 W140 + G423 V206 + L235 Y267 + K484 H1 + V17 T40 + A204 R87 + M286 W140 + T444 V206 + V238 N270 + G273 H1 + Q32 T40 + V206 R87 + F289 W140 + A445 V206 + M246 N270 + S280 H1 + A37 T40 + P211 R87 + V291 W140 + Q449 V206 + M248 N270 + T285 H1 + T40 T40 + I214 R87 + Q299 W140 + T459 V206 + L250 N270 + M286 H1 + P45 T40 + V215 R87 + R320 W140 + P473 V206 + G255 N270 + F289 H1 + W48 T40 + L217 R87 + H321 W140 + C474 V206 + Q256 N270 + V291 H1 + G50 T40 + L219 R87 + S323 W140 + G476 V206 + A263 N270 + Q299 H1 + T51 T40 + L235 R87 + H324 W140 + G477 V206 + V264 N270 + R320 H1 + N54 T40 + V238 R87 + F328 W140 + K484 V206 + Y267 N270 + H321 H1 + V56 T40 + M246 R87 + T334 G142 + G149 V206 + N270 N270 + S323 H1 + K72 T40 + M248 R87 + D337 G142 + T165 V206 + G273 N270 + H324 H1 + R87 T40 + L250 R87 + Q345 G142 + W167 V206 + S280 N270 + F328 H1 + Q98 T40 + G255 R87 + G346 G142 + R171 V206 + T285 N270 + T334 H1 + M105 T40 + Q256 R87 + G348 G142 + Q172 V206 + M286 N270 + D337 H1 + G109 T40 + A263 R87 + T355 G142 + L173 V206 + F289 N270 + Q345 H1 + F113 T40 + V264 R87 + S376 G142 + A174 V206 + V291 N270 + G346 H1 + R116 T40 + Y267 R87 + D377 G142 + G184 V206 + Q299 N270 + G348 H1 + Q118 T40 + N270 R87 + D379 G142 + T193 V206 + R320 N270 + T355 H1 + Q125 T40 + G273 R87 + Y382 G142 + N195 V206 + H321 N270 + S376 H1 + G133 T40 + S280 R87 + S383 G142 + A204 V206 + S323 N270 + D377 H1 + T134 T40 + T285 R87 + Q385 G142 + V206 V206 + H324 N270 + D379 H1 + W140 T40 + M286 R87 + K391 G142 + P211 V206 + F328 N270 + Y382 H1 + G142 T40 + F289 R87 + K393 G142 + I214 V206 + T334 N270 + S383 H1 + G149 T40 + V291 R87 + Q395 G142 + V215 V206 + D337 N270 + Q385 H1 + T165 T40 + Q299 R87 + A420 G142 + L217 V206 + Q345 N270 + K391 H1 + W167 T40 + R320 R87 + G423 G142 + L219 V206 + G346 N270 + K393 H1 + R171 T40 + H321 R87 + T444 G142 + L235 V206 + G348 N270 + Q395 H1 + Q172 T40 + S323 R87 + A445 G142 + V238 V206 + T355 N270 + A420 H1 + L173 T40 + H324 R87 + Q449 G142 + M246 V206 + S376 N270 + G423 H1 + A174 T40 + F328 R87 + T459 G142 + M248 V206 + D377 N270 + T444 H1 + G184 T40 + T334 R87 + P473 G142 + L250 V206 + D379 N270 + A445 H1 + T193 T40 + D337 R87 + C474 G142 + G255 V206 + Y382 N270 + Q449 H1 + N195 T40 + Q345 R87 + G476 G142 + Q256 V206 + S383 N270 + T459 H1 + A204 T40 + G346 R87 + G477 G142 + A263 V206 + Q385 N270 + P473 H1 + V206 T40 + G348 R87 + K484 G142 + V264 V206 + K391 N270 + C474 H1 + P211 T40 + T355 Q98 + M105 G142 + Y267 V206 + K393 N270 + G476 H1 + I214 T40 + S376 Q98 + G109 G142 + N270 V206 + Q395 N270 + G477 H1 + V215 T40 + D377 Q98 + F113 G142 + G273 V206 + A420 N270 + K484 H1 + L217 T40 + D379 Q98 + R116 G142 + S280 V206 + G423 G273 + S280 H1 + L219 T40 + Y382 Q98 + Q118 G142 + T285 V206 + T444 G273 + T285 H1 + L235 T40 + S383 Q98 + Q125 G142 + M286 V206 + A445 G273 + M286 H1 + V238 T40 + Q385 Q98 + G133 G142 + F289 V206 + Q449 G273 + F289 H1 + M246 T40 + K391 Q98 + T134 G142 + V291 V206 + T459 G273 + V291 H1 + M248 T40 + K393 Q98 + W140 G142 + Q299 V206 + P473 G273 + Q299 H1 + L250 T40 + Q395 Q98 + G142 G142 + R320 V206 + C474 G273 + R320 H1 + G255 T40 + A420 Q98 + G149 G142 + H321 V206 + G476 G273 + H321 H1 + Q256 T40 + G423 Q98 + T165 G142 + S323 V206 + G477 G273 + S323 H1 + A263 T40 + T444 Q98 + W167 G142 + H324 V206 + K484 G273 + H324 H1 + V264 T40 + A445 Q98 + R171 G142 + F328 P211 + I214 G273 + F328 H1 + Y267 T40 + Q449 Q98 + Q172 G142 + T334 P211 + V215 G273 + T334 H1 + N270 T40 + T459 Q98 + L173 G142 + D337 P211 + L217 G273 + D337 H1 + G273 T40 + P473 Q98 + A174 G142 + Q345 P211 + L219 G273 + Q345 H1 + S280 T40 + C474 Q98 + G184 G142 + G346 P211 + L235 G273 + G346 H1 + T285 T40 + G476 Q98 + T193 G142 + G348 P211 + V238 G273 + G348 H1 + M286 T40 + G477 Q98 + N195 G142 + T355 P211 + M246 G273 + T355 H1 + F289 T40 + K484 Q98 + A204 G142 + S376 P211 + M248 G273 + S376 H1 + V291 P45 + W48 Q98 + V206 G142 + D377 P211 + L250 G273 + D377 H1 + Q299 P45 + G50 Q98 + P211 G142 + D379 P211 + G255 G273 + D379 H1 + R320 P45 + T51 Q98 + I214 G142 + Y382 P211 + Q256 G273 + Y382 H1 + H321 P45 + N54 Q98 + V215 G142 + S383 P211 + A263 G273 + S383 H1 + S323 P45 + V56 Q98 + L217 G142 + Q385 P211 + V264 G273 + Q385 H1 + H324 P45 + K72 Q98 + L219 G142 + K391 P211 + Y267 G273 + K391 H1 + F328 P45 + R87 Q98 + L235 G142 + K393 P211 + N270 G273 + K393 H1 + T334 P45 + Q98 Q98 + V238 G142 + Q395 P211 + G273 G273 + Q395 H1 + D337 P45 + M105 Q98 + M246 G142 + A420 P211 + S280 G273 + A420 H1 + Q345 P45 + G109 Q98 + M248 G142 + G423 P211 + T285 G273 + G423 H1 + G346 P45 + F113 Q98 + L250 G142 + T444 P211 + M286 G273 + T444 H1 + G348 P45 + R116 Q98 + G255 G142 + A445 P211 + F289 G273 + A445 H1 + T355 P45 + Q118 Q98 + Q256 G142 + Q449 P211 + V291 G273 + Q449 H1 + S376 P45 + Q125 Q98 + A263 G142 + T459 P211 + Q299 G273 + T459 H1 + D377 P45 + G133 Q98 + V264 G142 + P473 P211 + R320 G273 + P473 H1 + D379 P45 + T134 Q98 + Y267 G142 + C474 P211 + H321 G273 + C474 H1 + Y382 P45 + W140 Q98 + N270 G142 + G476 P211 + S323 G273 + G476 H1 + S383 P45 + G142 Q98 + G273 G142 + G477 P211 + H324 G273 + G477 H1 + Q385 P45 + G149 Q98 + S280 G142 + K484 P211 + F328 G273 + K484 H1 + K391 P45 + T165 Q98 + T285 G149 + T165 P211 + T334 S280 + T285 H1 + K393 P45 + W167 Q98 + M286 G149 + W167 P211 + D337 S280 + M286 H1 + Q395 P45 + R171 Q98 + F289 G149 + R171 P211 + Q345 S280 + F289 H1 + A420 P45 + Q172 Q98 + V291 G149 + Q172 P211 + G346 S280 + V291 H1 + G423 P45 + L173 Q98 + Q299 G149 + L173 P211 + G348 S280 + Q299 H1 + T444 P45 + A174 Q98 + R320 G149 + A174 P211 + T355 S280 + R320 H1 + A445 P45 + G184 Q98 + H321 G149 + G184 P211 + S376 S280 + H321 H1 + Q449 P45 + T193 Q98 + S323 G149 + T193 P211 + D377 S280 + S323 H1 + T459 P45 + N195 Q98 + H324 G149 + N195 P211 + D379 S280 + H324 H1 + P473 P45 + A204 Q98 + F328 G149 + A204 P211 + Y382 S280 + F328 H1 + C474 P45 + V206 Q98 + T334 G149 + V206 P211 + S383 S280 + T334 H1 + G476 P45 + P211 Q98 + D337 G149 + P211 P211 + Q385 S280 + D337 H1 + G477 P45 + I214 Q98 + Q345 G149 + I214 P211 + K391 S280 + Q345 H1 + K484 P45 + V215 Q98 + G346 G149 + V215 P211 + K393 S280 + G346 T5 + G7 P45 + L217 Q98 + G348 G149 + L217 P211 + Q395 S280 + G348 T5 + Q11 P45 + L219 Q98 + T355 G149 + L219 P211 + A420 S280 + T355 T5 + N16 P45 + L235 Q98 + S376 G149 + L235 P211 + G423 S280 + S376 T5 + V17 P45 + V238 Q98 + D377 G149 + V238 P211 + T444 S280 + D377 T5 + Q32 P45 + M246 Q98 + D379 G149 + M246 P211 + A445 S280 + D379 T5 + A37 P45 + M248 Q98 + Y382 G149 + M248 P211 + Q449 S280 + Y382 T5 + T40 P45 + L250 Q98 + S383 G149 + L250 P211 + T459 S280 + S383 T5 + P45 P45 + G255 Q98 + Q385 G149 + G255 P211 + P473 S280 + Q385 T5 + W48 P45 + Q256 Q98 + K391 G149 + Q256 P211 + C474 S280 + K391 T5 + G50 P45 + A263 Q98 + K393 G149 + A263 P211 + G476 S280 + K393 T5 + T51 P45 + V264 Q98 + Q395 G149 + V264 P211 + G477 S280 + Q395 T5 + N54 P45 + Y267 Q98 + A420 G149 + Y267 P211 + K484 S280 + A420 T5 + V56 P45 + N270 Q98 + G423 G149 + N270 I214 + V215 S280 + G423 T5 + K72 P45 + G273 Q98 + T444 G149 + G273 I214 + L217 S280 + T444 T5 + R87 P45 + S280 Q98 + A445 G149 + S280 I214 + L219 S280 + A445 T5 + Q98 P45 + T285 Q98 + Q449 G149 + T285 I214 + L235 S280 + Q449 T5 + M105 P45 + M286 Q98 + T459 G149 + M286 I214 + V238 S280 + T459 T5 + G109 P45 + F289 Q98 + P473 G149 + F289 I214 + M246 S280 + P473 T5 + F113 P45 + V291 Q98 + C474 G149 + V291 I214 + M248 S280 + C474 T5 + R116 P45 + Q299 Q98 + G476 G149 + Q299 I214 + L250 S280 + G476 T5 + Q118 P45 + R320 Q98 + G477 G149 + R320 I214 + G255 S280 + G477 T5 + Q125 P45 + H321 Q98 + K484 G149 + H321 I214 + Q256 S280 + K484 T5 + G133 P45 + S323 M105 + G109 G149 + S323 I214 + A263 T285 + M286 T5 + T134 P45 + H324 M105 + F113 G149 + H324 I214 + V264 T285 + F289 T5 + W140 P45 + F328 M105 + R116 G149 + F328 I214 + Y267 T285 + V291 T5 + G142 P45 + T334 M105 + Q118 G149 + T334 I214 + N270 T285 + Q299 T5 + G149 P45 + D337 M105 + Q125 G149 + D337 I214 + G273 T285 + R320 T5 + T165 P45 + Q345 M105 + G133 G149 + Q345 I214 + S280 T285 + H321 T5 + W167 P45 + G346 M105 + T134 G149 + G346 I214 + T285 T285 + S323 T5 + R171 P45 + G348 M105 + W140 G149 + G348 I214 + M286 T285 + H324 T5 + Q172 P45 + T355 M105 + G142 G149 + T355 I214 + F289 T285 + F328 T5 + L173 P45 + S376 M105 + G149 G149 + S376 I214 + V291 T285 + T334 T5 + A174 P45 + D377 M105 + T165 G149 + D377 I214 + Q299 T285 + D337 T5 + G184 P45 + D379 M105 + W167 G149 + D379 I214 + R320 T285 + Q345 T5 + T193 P45 + Y382 M105 + R171 G149 + Y382 I214 + H321 T285 + G346 T5 + N195 P45 + S383 M105 + Q172 G149 + S383 I214 + S323 T285 + G348 T5 + A204 P45 + Q385 M105 + L173 G149 + Q385 I214 + H324 T285 + T355 T5 + V206 P45 + K391 M105 + A174 G149 + K391 I214 + F328 T285 + S376 T5 + P211 P45 + K393 M105 + G184 G149 + K393 I214 + T334 T285 + D377 T5 + I214 P45 + Q395 M105 + T193 G149 + Q395 I214 + D337 T285 + D379 T5 + V215 P45 + A420 M105 + N195 G149 + A420 I214 + Q345 T285 + Y382 T5 + L217 P45 + G423 M105 + A204 G149 + G423 I214 + G346 T285 + S383 T5 + L219 P45 + T444 M105 + V206 G149 + T444 I214 + G348 T285 + Q385 T5 + L235 P45 + A445 M105 + P211 G149 + A445 I214 +

T355 T285 + K391 T5 + V238 P45 + Q449 M105 + I214 G149 + Q449 I214 + S376 T285 + K393 T5 + M246 P45 + T459 M105 + V215 G149 + T459 I214 + D377 T285 + Q395 T5 + M248 P45 + P473 M105 + L217 G149 + P473 I214 + D379 T285 + A420 T5 + L250 P45 + C474 M105 + L219 G149 + C474 I214 + Y382 T285 + G423 T5 + G255 P45 + G476 M105 + L235 G149 + G476 I214 + S383 T285 + T444 T5 + Q256 P45 + G477 M105 + V238 G149 + G477 I214 + Q385 T285 + A445 T5 + A263 P45 + K484 M105 + M246 G149 + K484 I214 + K391 T285 + Q449 T5 + V264 W48 + G50 M105 + M248 T165 + W167 I214 + K393 T285 + T459 T5 + Y267 W48 + T51 M105 + L250 T165 + R171 I214 + Q395 T285 + P473 T5 + N270 W48 + N54 M105 + G255 T165 + Q172 I214 + A420 T285 + C474 T5 + G273 W48 + V56 M105 + Q256 T165 + L173 I214 + G423 T285 + G476 T5 + S280 W48 + K72 M105 + A263 T165 + A174 I214 + T444 T285 + G477 T5 + T285 W48 + R87 M105 + V264 T165 + G184 I214 + A445 T285 + K484 T5 + M286 W48 + Q98 M105 + Y267 T165 + T193 I214 + Q449 M286 + F289 T5 + F289 W48 + M105 M105 + N270 T165 + N195 I214 + T459 M286 + V291 T5 + V291 W48 + G109 M105 + G273 T165 + A204 I214 + P473 M286 + Q299 T5 + Q299 W48 + F113 M105 + S280 T165 + V206 I214 + C474 M286 + R320 T5 + R320 W48 + R116 M105 + T285 T165 + P211 I214 + G476 M286 + H321 T5 + H321 W48 + Q118 M105 + M286 T165 + I214 I214 + G477 M286 + S323 T5 + S323 W48 + Q125 M105 + F289 T165 + V215 I214 + K484 M286 + H324 T5 + H324 W48 + G133 M105 + V291 T165 + L217 V215 + L217 M286 + F328 T5 + F328 W48 + T134 M105 + Q299 T165 + L219 V215 + L219 M286 + T334 T5 + T334 W48 + W140 M105 + R320 T165 + L235 V215 + L235 M286 + D337 T5 + D337 W48 + G142 M105 + H321 T165 + V238 V215 + V238 M286 + Q345 T5 + Q345 W48 + G149 M105 + S323 T165 + M246 V215 + M246 M286 + G346 T5 + G346 W48 + T165 M105 + H324 T165 + M248 V215 + M248 M286 + G348 T5 + G348 W48 + W167 M105 + F328 T165 + L250 V215 + L250 M286 + T355 T5 + T355 W48 + R171 M105 + T334 T165 + G255 V215 + G255 M286 + S376 T5 + S376 W48 + Q172 M105 + D337 T165 + Q256 V215 + Q256 M286 + D377 T5 + D377 W48 + L173 M105 + Q345 T165 + A263 V215 + A263 M286 + D379 T5 + D379 W48 + A174 M105 + G346 T165 + V264 V215 + V264 M286 + Y382 T5 + Y382 W48 + G184 M105 + G348 T165 + Y267 V215 + Y267 M286 + S383 T5 + S383 W48 + T193 M105 + T355 T165 + N270 V215 + N270 M286 + Q385 T5 + Q385 W48 + N195 M105 + S376 T165 + G273 V215 + G273 M286 + K391 T5 + K391 W48 + A204 M105 + D377 T165 + S280 V215 + S280 M286 + K393 T5 + K393 W48 + V206 M105 + D379 T165 + T285 V215 + T285 M286 + Q395 T5 + Q395 W48 + P211 M105 + Y382 T165 + M286 V215 + M286 M286 + A420 T5 + A420 W48 + I214 M105 + S383 T165 + F289 V215 + F289 M286 + G423 T5 + G423 W48 + V215 M105 + Q385 T165 + V291 V215 + V291 M286 + T444 T5 + T444 W48 + L217 M105 + K391 T165 + Q299 V215 + Q299 M286 + A445 T5 + A445 W48 + L219 M105 + K393 T165 + R320 V215 + R320 M286 + Q449 T5 + Q449 W48 + L235 M105 + Q395 T165 + H321 V215 + H321 M286 + T459 T5 + T459 W48 + V238 M105 + A420 T165 + S323 V215 + S323 M286 + P473 T5 + P473 W48 + 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F289 + G346 G7 + N54 W48 + Q299 G109 + T134 T165 + K393 V215 + K393 F289 + G348 G7 + V56 W48 + R320 G109 + W140 T165 + Q395 V215 + Q395 F289 + T355 G7 + K72 W48 + H321 G109 + G142 T165 + A420 V215 + A420 F289 + S376 G7 + R87 W48 + S323 G109 + G149 T165 + G423 V215 + G423 F289 + D377 G7 + Q98 W48 + H324 G109 + T165 T165 + T444 V215 + T444 F289 + D379 G7 + M105 W48 + F328 G109 + W167 T165 + A445 V215 + A445 F289 + Y382 G7 + G109 W48 + T334 G109 + R171 T165 + Q449 V215 + Q449 F289 + S383 G7 + F113 W48 + D337 G109 + Q172 T165 + T459 V215 + T459 F289 + Q385 G7 + R116 W48 + Q345 G109 + L173 T165 + P473 V215 + P473 F289 + K391 G7 + Q118 W48 + G346 G109 + A174 T165 + C474 V215 + C474 F289 + K393 G7 + Q125 W48 + G348 G109 + G184 T165 + G476 V215 + G476 F289 + Q395 G7 + G133 W48 + T355 G109 + T193 T165 + G477 V215 + G477 F289 + A420 G7 + T134 W48 + S376 G109 + N195 T165 + K484 V215 + K484 F289 + G423 G7 + W140 W48 + D377 G109 + A204 W167 + R171 L217 + L219 F289 + T444 G7 + G142 W48 + D379 G109 + 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+ T334 G7 + V215 W48 + G476 G109 + Y267 W167 + M246 L217 + F289 V291 + D337 G7 + L217 W48 + G477 G109 + N270 W167 + M248 L217 + V291 V291 + Q345 G7 + L219 W48 + K484 G109 + G273 W167 + L250 L217 + Q299 V291 + G346 G7 + L235 G50 + T51 G109 + S280 W167 + G255 L217 + R320 V291 + G348 G7 + V238 G50 + N54 G109 + T285 W167 + Q256 L217 + H321 V291 + T355 G7 + M246 G50 + V56 G109 + M286 W167 + A263 L217 + S323 V291 + S376 G7 + M248 G50 + K72 G109 + F289 W167 + V264 L217 + H324 V291 + D377 G7 + L250 G50 + R87 G109 + V291 W167 + Y267 L217 + F328 V291 + D379 G7 + G255 G50 + Q98 G109 + Q299 W167 + N270 L217 + T334 V291 + Y382 G7 + Q256 G50 + M105 G109 + R320 W167 + G273 L217 + D337 V291 + S383 G7 + A263 G50 + G109 G109 + H321 W167 + S280 L217 + Q345 V291 + Q385 G7 + V264 G50 + F113 G109 + S323 W167 + T285 L217 + G346 V291 + K391 G7 + Y267 G50 + R116 G109 + H324 W167 + M286 L217 + G348 V291 + K393 G7 + N270 G50 + Q118 G109 + F328 W167 + F289 L217 + T355 V291 + Q395 G7 + G273 G50 + Q125 G109 + T334 W167 + V291 L217 + S376 V291 + A420 G7 + S280 G50 + G133 G109 + D337 W167 + Q299 L217 + D377 V291 + G423 G7 + T285 G50 + T134 G109 + Q345 W167 + R320 L217 + D379 V291 + T444 G7 + M286 G50 + W140 G109 + G346 W167 + H321 L217 + Y382 V291 + A445 G7 + F289 G50 + G142 G109 + G348 W167 + S323 L217 + S383 V291 + Q449 G7 + V291 G50 + G149 G109 + T355 W167 + H324 L217 + Q385 V291 + T459 G7 + Q299 G50 + T165 G109 + S376 W167 + F328 L217 + K391 V291 + P473 G7 + R320 G50 + W167 G109 + D377 W167 + T334 L217 + K393 V291 + C474 G7 + H321 G50 + R171 G109 + D379 W167 + D337 L217 + Q395 V291 + G476 G7 + S323 G50 + Q172 G109 + Y382 W167 + Q345 L217 + A420 V291 + G477 G7 + H324 G50 + L173 G109 + S383 W167 + G346 L217 + G423 V291 + K484 G7 + F328 G50 + A174 G109 + Q385 W167 + G348 L217 + T444 Q299 + R320 G7 + T334 G50 + G184 G109 + K391 W167 + T355 L217 + A445 Q299 + H321 G7 + D337 G50 + T193 G109 + K393 W167 + S376 L217 + Q449 Q299 + S323 G7 + Q345 G50 + N195 G109 + Q395 W167 + D377 L217 + T459 Q299 + H324 G7 + G346 G50 + A204 G109 + A420 W167 + D379 L217 + P473 Q299 + F328 G7 + G348 G50 + V206 G109 + G423 W167 + Y382 L217 + C474 Q299 + T334 G7 + T355 G50 + P211 G109 + T444 W167 + S383 L217 + G476 Q299 + D337 G7 + S376 G50 + I214 G109 + A445 W167 + Q385 L217 + G477 Q299 + Q345 G7 + D377 G50 + V215 G109 + Q449 W167 + K391 L217 + K484 Q299 + G346 G7 + D379 G50 + L217 G109 + T459 W167 + K393 L219 + L235 Q299 + G348 G7 + Y382 G50 + L219 G109 + P473 W167 + Q395 L219 + V238 Q299 + T355 G7 + S383 G50 + L235 G109 + C474 W167 + A420 L219 + M246 Q299 + S376 G7 + Q385 G50 + V238 G109 + G476 W167 + G423 L219 + M248 Q299 + D377 G7 + K391 G50 + M246 G109 + G477 W167 + T444 L219 + L250 Q299 + D379 G7 + K393 G50 + M248 G109 + K484 W167 + A445 L219 + G255 Q299 + Y382 G7 + Q395 G50 + L250 F113 + R116 W167 + Q449 L219 + Q256 Q299 + S383 G7 + A420 G50 + G255 F113 + Q118 W167 + T459 L219 + A263 Q299 + Q385 G7 + G423 G50 + Q256 F113 + Q125 W167 + P473 L219 + V264 Q299 + K391 G7 + T444 G50 + A263 F113 + 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S280 F113 + T165 R171 + L173 L219 + M286 Q299 + A445 G7 + G476 G50 + T285 F113 + W167 R171 + A174 L219 + F289 Q299 + Q449 G7 + G477 G50 + M286 F113 + R171 R171 + G184 L219 + V291 Q299 + T459 G7 + K484 G50 + F289 F113 + Q172 R171 + T193 L219 + Q299 Q299 + P473 Q11 + N16 G50 + V291 F113 + L173 R171 + N195 L219 + R320 Q299 + C474 Q11 + V17 G50 + Q299 F113 + A174 R171 + A204 L219 + H321 Q299 + G476 Q11 + Q32 G50 + R320 F113 + G184 R171 + V206 L219 + S323 Q299 + G477 Q11 + A37 G50 + H321 F113 + T193 R171 + P211 L219 + H324 Q299 + K484 Q11 + T40 G50 + S323 F113 + N195 R171 + I214 L219 + F328 R320 + H321 Q11 + P45 G50 + H324 F113 + A204 R171 + V215 L219 + T334 R320 + S323 Q11 + W48 G50 + F328 F113 + V206 R171 + L217 L219 + D337 R320 + H324 Q11 + G50 G50 + T334 F113 + P211 R171 + L219 L219 + Q345 R320 + F328 Q11 + T51 G50 + D337 F113 + I214 R171 + L235 L219 + G346 R320 + T334 Q11 + N54 G50 + Q345 F113 + V215 R171 + V238 L219 + G348 R320 + D337 Q11 + V56 G50 + G346 F113 + L217 R171 + M246 L219 + T355 R320 + Q345 Q11 + K72 G50 + G348 F113 + L219 R171 + M248 L219 + S376 R320 + G346 Q11 + R87 G50 + T355 F113 + L235 R171 + L250 L219 + D377 R320 + G348 Q11 + Q98 G50 + S376 F113 + V238 R171 + G255 L219 + D379 R320 + T355 Q11 + M105 G50 + D377 F113 + M246 R171 + Q256 L219 + Y382 R320 + S376 Q11 + G109 G50 + D379 F113 + M248 R171 + A263 L219 + S383 R320 + D377 Q11 + F113 G50 + Y382 F113 + L250 R171 + V264 L219 + Q385 R320 + D379 Q11 + R116 G50 + S383 F113 + G255 R171 + Y267 L219 + K391 R320 + Y382 Q11 + Q118 G50 + Q385 F113 + Q256 R171 + N270 L219 + K393 R320 + S383 Q11 + Q125 G50 + K391 F113 + A263 R171 + G273 L219 + Q395 R320 + Q385 Q11 + G133 G50 + K393 F113 + V264 R171 + S280 L219 + A420 R320 + K391 Q11 + T134 G50 + Q395 F113 + Y267 R171 + T285 L219 + G423 R320 + K393 Q11 + W140 G50 + A420 F113 + N270 R171 + M286 L219 + T444 R320 + Q395 Q11 + G142 G50 + G423 F113 + G273 R171 + F289 L219 + A445 R320 + A420 Q11 + G149 G50 + T444 F113 + S280 R171 + V291 L219 + Q449 R320 + G423 Q11 + T165 G50 + A445 F113 + T285 R171 + Q299 L219 + T459 R320 + T444 Q11 + W167 G50 + Q449 F113 + M286 R171 + R320 L219 + P473 R320 + A445 Q11 + R171 G50 + T459 F113 + F289 R171 + H321 L219 + C474 R320 + Q449 Q11 + Q172 G50 + P473 F113 + V291 R171 + S323 L219 + G476 R320 + T459 Q11 + L173 G50 + C474 F113 + Q299 R171 + H324 L219 + G477 R320 + P473 Q11 + A174 G50 + G476 F113 + R320 R171 + F328 L219 + K484 R320 + C474 Q11 + G184 G50 + G477 F113 + H321 R171 + T334 L235 + V238 R320 + G476 Q11 + T193 G50 + K484 F113 + S323 R171 + D337 L235 + M246 R320 + G477 Q11 + N195 T51 + N54 F113 + H324 R171 + Q345 L235 + M248 R320 + K484 Q11 + A204 T51 + V56 F113 + F328 R171 + G346 L235 + L250 H321 + S323 Q11 + V206 T51 + K72 F113 + T334 R171 + G348 L235 + G255 H321 + H324 Q11 + P211 T51 + R87 F113 + D337 R171 + T355 L235 + Q256 H321 + F328 Q11 + I214 T51 + Q98 F113 + Q345 R171 + S376 L235 + A263 H321 + T334 Q11 + V215 T51 + M105 F113 + G346 R171 + D377 L235 + V264 H321 + D337 Q11 + L217 T51 + G109 F113 + G348 R171 + D379 L235 + Y267 H321 + Q345 Q11 + L219 T51 + F113 F113 + T355 R171 + Y382 L235 + N270 H321 + G346 Q11 + L235 T51 + R116 F113 + S376 R171 + S383 L235 + G273 H321 + G348 Q11 + V238 T51 + Q118 F113 + D377 R171 + Q385 L235 + S280 H321 + T355 Q11 + M246 T51 + Q125 F113 + D379 R171 + K391 L235 + T285 H321 + S376 Q11 + M248 T51 + G133 F113 + Y382 R171 + K393 L235 + M286 H321 + D377 Q11 + L250 T51 + T134 F113 + S383 R171 + Q395 L235 + F289 H321 + D379 Q11 + G255 T51 + W140 F113 + Q385 R171 + A420 L235 + V291 H321 + Y382 Q11 + Q256 T51 + G142 F113 + K391 R171 + G423 L235 + Q299 H321 + S383 Q11 + A263 T51 + G149 F113 + K393 R171 + T444 L235 + R320 H321 + Q385 Q11 + V264 T51 + T165 F113 + Q395 R171 + A445 L235 + H321 H321 + K391 Q11 + Y267 T51 + W167 F113 + A420 R171 + Q449 L235 + S323 H321 + K393 Q11 + N270 T51 + R171 F113 + G423 R171 + T459 L235 + H324 H321 + Q395 Q11 + G273 T51 + Q172 F113 + T444 R171 + P473 L235 + F328 H321 + A420 Q11 + S280 T51 + L173 F113 + A445 R171 + C474 L235 + T334 H321 + G423 Q11 + T285 T51 + A174 F113 + Q449 R171 + G476 L235 + D337 H321 + T444 Q11 + M286 T51 + G184 F113 + T459 R171 + G477 L235 + Q345 H321 + A445 Q11 + F289 T51 + T193 F113 + P473 R171 + K484 L235 + G346 H321 + Q449 Q11 + V291 T51 + N195 F113 + C474 Q172 + L173 L235 + G348 H321 + T459 Q11 + Q299 T51 + A204 F113 + G476 Q172 + A174 L235 + T355 H321 + P473 Q11 + R320 T51 + V206 F113 + G477 Q172 + G184 L235 + S376 H321 + C474 Q11 + H321 T51 + P211 F113 + K484 Q172 + T193 L235 + D377 H321 + G476 Q11 + S323 T51 + I214 R116 + Q118 Q172 + N195 L235 + D379 H321 + G477 Q11 + H324 T51 + V215 R116 + Q125 Q172 + A204 L235 + Y382 H321 + K484 Q11 + F328 T51 + L217 R116 + G133 Q172 + V206 L235 + S383 S323 + H324 Q11 + T334 T51 + L219 R116 + T134 Q172 + P211 L235 + Q385 S323 + F328 Q11 + D337 T51 + L235 R116 + W140 Q172 + I214 L235 + K391 S323 + T334 Q11 + Q345 T51 + V238 R116 + G142 Q172 + V215 L235 + K393 S323 + D337 Q11 + G346 T51 + M246 R116 + G149 Q172 + L217 L235 + Q395 S323 + Q345 Q11 + G348 T51 + M248 R116 + T165 Q172 + L219 L235 + A420 S323 + G346 Q11 + T355 T51 + L250 R116 + W167 Q172 + L235 L235 + G423 S323 + G348 Q11 + S376 T51 + G255 R116 + R171 Q172 + V238 L235 + T444 S323 + T355 Q11 + D377 T51 + Q256 R116 + Q172 Q172 + M246 L235 + A445 S323 + S376 Q11 + D379 T51 + A263 R116 + L173 Q172 + M248 L235 + Q449 S323 + D377 Q11 + Y382 T51 + V264 R116 + A174 Q172 + L250 L235 + T459 S323 + D379 Q11 + S383 T51 + Y267 R116 + G184 Q172 + G255 L235 + P473 S323 + Y382 Q11 + Q385 T51 + N270 R116 + T193 Q172 + Q256 L235 + C474 S323 + S383 Q11 + K391 T51 + G273 R116 + N195 Q172 + A263 L235 + G476 S323 + Q385 Q11 + K393 T51 + S280 R116 + A204 Q172 + V264 L235 + G477 S323 + K391 Q11 + Q395 T51 + T285 R116 + V206 Q172 + Y267 L235 + K484 S323 + K393 Q11 + A420 T51 + M286 R116 + P211 Q172 + N270 V238 + M246 S323 + Q395 Q11 + G423 T51 + F289 R116 + I214 Q172 + G273 V238 + M248 S323 + A420 Q11 + T444 T51 + V291 R116 + V215 Q172 + S280 V238 + L250 S323 + G423 Q11 + A445 T51 + Q299 R116 + L217 Q172 + T285 V238 + G255 S323 + T444 Q11 + Q449 T51 + R320 R116 + L219 Q172 + M286 V238 + Q256 S323 + A445 Q11 + T459 T51 + H321 R116 + L235 Q172 + F289 V238 + A263 S323 + Q449 Q11 + P473 T51 + S323 R116 + V238 Q172 + V291 V238 + V264 S323 + T459 Q11 + C474 T51 + H324 R116 + M246 Q172 + Q299 V238 + Y267 S323 + P473 Q11 + G476 T51 + F328 R116 + M248 Q172 + R320 V238 + N270 S323 + C474 Q11 + G477 T51 + T334 R116 + L250 Q172 + H321 V238 + G273 S323 + G476 Q11 + K484 T51 + D337 R116 + G255 Q172 + S323 V238 + S280 S323 + G477 N16 + V17 T51 + Q345 R116 + Q256 Q172 + H324 V238 + T285 S323 + K484 N16 + Q32 T51 + G346 R116 + A263 Q172 + F328 V238 + M286 H324 + F328 N16 + A37 T51 + G348 R116 + V264 Q172 + T334 V238 + F289 H324 + T334 N16 + T40 T51 + T355 R116 + Y267 Q172 + D337 V238 + V291 H324 + D337 N16 + P45 T51 + S376 R116 + N270 Q172 + Q345 V238 + Q299 H324 + Q345 N16 + W48 T51 + D377 R116 + G273 Q172 + G346 V238 + R320 H324 + G346 N16 + G50 T51 + D379 R116 + S280 Q172 + 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A445 N16 + W140 T51 + G476 R116 + G348 Q172 + Q449 V238 + Q385 H324 + Q449 N16 + G142 T51 + G477 R116 + T355 Q172 + T459 V238 + K391 H324 + T459 N16 + G149 T51 + K484 R116 + S376 Q172 + P473 V238 + K393 H324 + P473 N16 + T165 N54 + V56 R116 + D377 Q172 + C474 V238 + Q395 H324 + C474 N16 + W167 N54 + K72 R116 + D379 Q172 + G476 V238 + A420 H324 + G476 N16 + R171 N54 + R87 R116 + Y382 Q172 + G477 V238 + G423 H324 + G477 N16 + Q172 N54 + Q98 R116 + S383 Q172 + K484 V238 + T444 H324 + K484 N16 + L173 N54 + M105 R116 + Q385 L173 + A174 V238 + A445 F328 + T334 N16 + A174 N54 + G109 R116 + K391 L173 + G184 V238 + Q449 F328 + D337 N16 + G184 N54 + F113 R116 + K393 L173 + T193 V238 + T459 F328 + Q345 N16 + T193 N54 + R116 R116 + Q395 L173 + N195 V238 + P473 F328 + G346 N16 + N195 N54 + Q118 R116 + A420 L173 + A204 V238 + C474 F328 + G348 N16 + A204 N54 + Q125 R116 + G423 L173 + V206 V238 + G476 F328 + T355 N16 + V206 N54 + G133 R116 + T444 L173 + P211 V238 + G477 F328 + S376 N16 + P211 N54 + T134 R116 + A445 L173 + I214 V238 + K484 F328 + D377 N16 + I214 N54 + W140 R116 + Q449 L173 + V215 M246 + M248 F328 + D379 N16 + V215 N54 + G142 R116 + T459 L173 + L217 M246 + L250 F328 + Y382 N16 + L217 N54 + G149 R116 + P473 L173 + L219 M246 + G255 F328 + S383 N16 + L219 N54 + T165 R116 + C474 L173 + L235 M246 + Q256 F328 + Q385 N16 + L235 N54 + W167 R116 + G476 L173 + V238 M246 + A263 F328 + K391 N16 + V238 N54 + R171 R116 + G477 L173 + M246 M246 + V264 F328 + K393

N16 + M246 N54 + Q172 R116 + K484 L173 + M248 M246 + Y267 F328 + Q395 N16 + M248 N54 + L173 Q118 + Q125 L173 + L250 M246 + N270 F328 + A420 N16 + L250 N54 + A174 Q118 + G133 L173 + G255 M246 + G273 F328 + G423 N16 + G255 N54 + G184 Q118 + T134 L173 + Q256 M246 + S280 F328 + T444 N16 + Q256 N54 + T193 Q118 + W140 L173 + A263 M246 + T285 F328 + A445 N16 + A263 N54 + N195 Q118 + G142 L173 + V264 M246 + M286 F328 + Q449 N16 + V264 N54 + A204 Q118 + G149 L173 + Y267 M246 + F289 F328 + T459 N16 + Y267 N54 + V206 Q118 + T165 L173 + N270 M246 + V291 F328 + P473 N16 + N270 N54 + P211 Q118 + W167 L173 + G273 M246 + Q299 F328 + C474 N16 + G273 N54 + I214 Q118 + R171 L173 + S280 M246 + R320 F328 + G476 N16 + S280 N54 + V215 Q118 + Q172 L173 + T285 M246 + H321 F328 + G477 N16 + T285 N54 + L217 Q118 + L173 L173 + M286 M246 + S323 F328 + K484 N16 + M286 N54 + L219 Q118 + A174 L173 + F289 M246 + H324 T334 + D337 N16 + F289 N54 + L235 Q118 + G184 L173 + V291 M246 + F328 T334 + Q345 N16 + V291 N54 + V238 Q118 + T193 L173 + Q299 M246 + T334 T334 + G346 N16 + Q299 N54 + M246 Q118 + N195 L173 + R320 M246 + D337 T334 + G348 N16 + R320 N54 + M248 Q118 + A204 L173 + H321 M246 + Q345 T334 + T355 N16 + H321 N54 + L250 Q118 + V206 L173 + S323 M246 + G346 T334 + S376 N16 + S323 N54 + G255 Q118 + P211 L173 + H324 M246 + G348 T334 + D377 N16 + H324 N54 + Q256 Q118 + I214 L173 + F328 M246 + T355 T334 + D379 N16 + F328 N54 + A263 Q118 + V215 L173 + T334 M246 + S376 T334 + Y382 N16 + T334 N54 + V264 Q118 + L217 L173 + D337 M246 + D377 T334 + S383 N16 + D337 N54 + Y267 Q118 + L219 L173 + Q345 M246 + D379 T334 + Q385 N16 + Q345 N54 + N270 Q118 + L235 L173 + G346 M246 + Y382 T334 + K391 N16 + G346 N54 + G273 Q118 + V238 L173 + G348 M246 + S383 T334 + K393 N16 + G348 N54 + S280 Q118 + M246 L173 + T355 M246 + Q385 T334 + Q395 N16 + T355 N54 + T285 Q118 + M248 L173 + S376 M246 + K391 T334 + A420 N16 + S376 N54 + M286 Q118 + L250 L173 + D377 M246 + K393 T334 + G423 N16 + D377 N54 + F289 Q118 + G255 L173 + D379 M246 + Q395 T334 + T444 N16 + D379 N54 + V291 Q118 + Q256 L173 + Y382 M246 + A420 T334 + A445 N16 + Y382 N54 + Q299 Q118 + A263 L173 + S383 M246 + G423 T334 + Q449 N16 + S383 N54 + R320 Q118 + V264 L173 + Q385 M246 + T444 T334 + T459 N16 + Q385 N54 + H321 Q118 + Y267 L173 + K391 M246 + A445 T334 + P473 N16 + K391 N54 + S323 Q118 + N270 L173 + K393 M246 + Q449 T334 + C474 N16 + K393 N54 + H324 Q118 + G273 L173 + Q395 M246 + T459 T334 + G476 N16 + Q395 N54 + F328 Q118 + S280 L173 + A420 M246 + P473 T334 + G477 N16 + A420 N54 + T334 Q118 + T285 L173 + G423 M246 + C474 T334 + K484 N16 + G423 N54 + D337 Q118 + M286 L173 + T444 M246 + G476 D337 + Q345 N16 + T444 N54 + Q345 Q118 + F289 L173 + A445 M246 + G477 D337 + G346 N16 + A445 N54 + G346 Q118 + V291 L173 + Q449 M246 + K484 D337 + G348 N16 + Q449 N54 + G348 Q118 + Q299 L173 + T459 M248 + L250 D337 + T355 N16 + T459 N54 + T355 Q118 + R320 L173 + P473 M248 + G255 D337 + S376 N16 + P473 N54 + S376 Q118 + H321 L173 + C474 M248 + Q256 D337 + D377 N16 + C474 N54 + D377 Q118 + S323 L173 + G476 M248 + A263 D337 + D379 N16 + G476 N54 + D379 Q118 + H324 L173 + G477 M248 + V264 D337 + Y382 N16 + G477 N54 + Y382 Q118 + F328 L173 + K484 M248 + Y267 D337 + S383 N16 + K484 N54 + S383 Q118 + T334 A174 + G184 M248 + N270 D337 + Q385 V17 + Q32 N54 + Q385 Q118 + D337 A174 + T193 M248 + G273 D337 + K391 V17 + A37 N54 + K391 Q118 + Q345 A174 + N195 M248 + S280 D337 + K393 V17 + T40 N54 + K393 Q118 + G346 A174 + A204 M248 + T285 D337 + Q395 V17 + P45 N54 + Q395 Q118 + G348 A174 + V206 M248 + M286 D337 + A420 V17 + W48 N54 + A420 Q118 + T355 A174 + P211 M248 + F289 D337 + G423 V17 + G50 N54 + G423 Q118 + S376 A174 + I214 M248 + V291 D337 + T444 V17 + T51 N54 + T444 Q118 + D377 A174 + V215 M248 + Q299 D337 + A445 V17 + N54 N54 + A445 Q118 + D379 A174 + L217 M248 + R320 D337 + Q449 V17 + V56 N54 + Q449 Q118 + Y382 A174 + L219 M248 + H321 D337 + T459 V17 + K72 N54 + T459 Q118 + S383 A174 + L235 M248 + S323 D337 + P473 V17 + R87 N54 + P473 Q118 + Q385 A174 + V238 M248 + H324 D337 + C474 V17 + Q98 N54 + C474 Q118 + K391 A174 + M246 M248 + F328 D337 + G476 V17 + M105 N54 + G476 Q118 + K393 A174 + M248 M248 + T334 D337 + G477 V17 + G109 N54 + G477 Q118 + Q395 A174 + L250 M248 + D337 D337 + K484 V17 + F113 N54 + K484 Q118 + A420 A174 + G255 M248 + Q345 Q345 + G346 V17 + R116 V56 + K72 Q118 + G423 A174 + Q256 M248 + G346 Q345 + G348 V17 + Q118 V56 + R87 Q118 + T444 A174 + A263 M248 + G348 Q345 + T355 V17 + Q125 V56 + Q98 Q118 + A445 A174 + V264 M248 + T355 Q345 + S376 V17 + G133 V56 + M105 Q118 + Q449 A174 + Y267 M248 + S376 Q345 + D377 V17 + T134 V56 + G109 Q118 + T459 A174 + N270 M248 + D377 Q345 + D379 V17 + W140 V56 + F113 Q118 + P473 A174 + G273 M248 + D379 Q345 + Y382 V17 + G142 V56 + R116 Q118 + C474 A174 + S280 M248 + Y382 Q345 + S383 V17 + G149 V56 + Q118 Q118 + G476 A174 + T285 M248 + S383 Q345 + Q385 V17 + T165 V56 + Q125 Q118 + G477 A174 + M286 M248 + Q385 Q345 + K391 V17 + W167 V56 + G133 Q118 + K484 A174 + F289 M248 + K391 Q345 + K393 V17 + R171 V56 + T134 Q125 + G133 A174 + V291 M248 + K393 Q345 + Q395 V17 + Q172 V56 + W140 Q125 + T134 A174 + Q299 M248 + Q395 Q345 + A420 V17 + L173 V56 + G142 Q125 + W140 A174 + R320 M248 + A420 Q345 + G423 V17 + A174 V56 + G149 Q125 + G142 A174 + H321 M248 + G423 Q345 + T444 V17 + G184 V56 + T165 Q125 + G149 A174 + S323 M248 + T444 Q345 + A445 V17 + T193 V56 + W167 Q125 + T165 A174 + H324 M248 + A445 Q345 + Q449 V17 + N195 V56 + R171 Q125 + W167 A174 + F328 M248 + Q449 Q345 + T459 V17 + A204 V56 + Q172 Q125 + R171 A174 + T334 M248 + T459 Q345 + P473 V17 + V206 V56 + L173 Q125 + Q172 A174 + D337 M248 + P473 Q345 + C474 V17 + P211 V56 + A174 Q125 + L173 A174 + Q345 M248 + C474 Q345 + G476 V17 + I214 V56 + G184 Q125 + A174 A174 + G346 M248 + G476 Q345 + G477 V17 + V215 V56 + T193 Q125 + G184 A174 + G348 M248 + G477 Q345 + K484 V17 + L217 V56 + N195 Q125 + T193 A174 + T355 M248 + K484 G346 + G348 V17 + L219 V56 + A204 Q125 + N195 A174 + S376 L250 + G255 G346 + T355 V17 + L235 V56 + V206 Q125 + A204 A174 + D377 L250 + Q256 G346 + S376 V17 + V238 V56 + P211 Q125 + V206 A174 + D379 L250 + A263 G346 + D377 V17 + M246 V56 + I214 Q125 + P211 A174 + Y382 L250 + V264 G346 + D379 V17 + M248 V56 + V215 Q125 + I214 A174 + S383 L250 + Y267 G346 + Y382 V17 + L250 V56 + L217 Q125 + V215 A174 + Q385 L250 + N270 G346 + S383 V17 + G255 V56 + L219 Q125 + L217 A174 + K391 L250 + G273 G346 + Q385 V17 + Q256 V56 + L235 Q125 + L219 A174 + K393 L250 + S280 G346 + K391 V17 + A263 V56 + V238 Q125 + L235 A174 + Q395 L250 + T285 G346 + K393 V17 + V264 V56 + M246 Q125 + V238 A174 + A420 L250 + M286 G346 + Q395 V17 + Y267 V56 + M248 Q125 + M246 A174 + G423 L250 + F289 G346 + A420 V17 + N270 V56 + L250 Q125 + M248 A174 + T444 L250 + V291 G346 + G423 V17 + G273 V56 + G255 Q125 + L250 A174 + A445 L250 + Q299 G346 + T444 V17 + S280 V56 + Q256 Q125 + G255 A174 + Q449 L250 + R320 G346 + A445 V17 + T285 V56 + A263 Q125 + Q256 A174 + T459 L250 + H321 G346 + Q449 V17 + 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+ H324 G184 + L235 L250 + Q385 G348 + K393 V17 + S376 V56 + T334 Q125 + F328 G184 + V238 L250 + K391 G348 + Q395 V17 + D377 V56 + D337 Q125 + T334 G184 + M246 L250 + K393 G348 + A420 V17 + D379 V56 + Q345 Q125 + D337 G184 + M248 L250 + Q395 G348 + G423 V17 + Y382 V56 + G346 Q125 + Q345 G184 + L250 L250 + A420 G348 + T444 V17 + S383 V56 + G348 Q125 + G346 G184 + G255 L250 + G423 G348 + A445 V17 + Q385 V56 + T355 Q125 + G348 G184 + Q256 L250 + T444 G348 + Q449 V17 + K391 V56 + S376 Q125 + T355 G184 + A263 L250 + A445 G348 + T459 V17 + K393 V56 + D377 Q125 + S376 G184 + V264 L250 + Q449 G348 + P473 V17 + Q395 V56 + D379 Q125 + D377 G184 + Y267 L250 + T459 G348 + C474 V17 + A420 V56 + Y382 Q125 + D379 G184 + N270 L250 + P473 G348 + G476 V17 + G423 V56 + S383 Q125 + Y382 G184 + G273 L250 + C474 G348 + G477 V17 + T444 V56 + Q385 Q125 + S383 G184 + S280 L250 + G476 G348 + K484 V17 + A445 V56 + K391 Q125 + Q385 G184 + T285 L250 + G477 T355 + S376 V17 + Q449 V56 + K393 Q125 + K391 G184 + M286 L250 + K484 T355 + D377 V17 + T459 V56 + Q395 Q125 + K393 G184 + F289 G255 + Q256 T355 + D379 V17 + P473 V56 + A420 Q125 + Q395 G184 + V291 G255 + A263 T355 + Y382 V17 + C474 V56 + G423 Q125 + A420 G184 + Q299 G255 + V264 T355 + S383 V17 + G476 V56 + T444 Q125 + G423 G184 + R320 G255 + Y267 T355 + Q385 V17 + G477 V56 +

A445 Q125 + T444 G184 + H321 G255 + N270 T355 + K391 V17 + K484 V56 + Q449 Q125 + A445 G184 + S323 G255 + G273 T355 + K393
Q32 + A37 V56 + T459 Q125 + Q449 G184 + H324 G255 + S280 T355 + Q395 Q32 + T40 V56 + P473 Q125 + T459 G184 + F328 G255 + T285
T355 + A420 Q32 + P45 V56 + C474 Q125 + P473 G184 + T334 G255 + M286 T355 + G423 Q32 + W48 V56 + G476 Q125 + C474 G184 + D337
G255 + F289 T355 + T444 Q32 + G50 V56 + G477 Q125 + G476 G184 + Q345 G255 + V291 T355 + A445 Q32 + T51 V56 + K484 Q125 + G477
G184 + G346 G255 + Q299 T355 + Q449 Q32 + N54 K72 + R87 Q125 + K484 G184 + G348 G255 + R320 T355 + T459 Q32 + V56 K72 + Q98
G133 + T134 G184 + T355 G255 + H321 T355 + P473 Q32 + K72 K72 + M105 G133 + W140 G184 + S376 G255 + S323 T355 + C474 Q32 + R87
K72 + G109 G133 + G142 G184 + D377 G255 + H324 T355 + G476 Q32 + Q98 K72 + F113 G133 + G149 G184 + D379 G255 + F328 T355 +
G477 Q32 + M105 K72 + R116 G133 + T165 G184 + Y382 G255 + T334 T355 + K484 Q32 + G109 K72 + Q118 G133 + W167 G184 + S383
G255 + D337 S376 + D377 Q32 + F113 K72 + Q125 G133 + R171 G184 + Q385 G255 + Q345 S376 + D379 Q32 + R116 K72 + G133 G133 +
Q172 G184 + K391 G255 + G346 S376 + Y382 Q32 + Q118 K72 + T134 G133 + L173 G184 + K393 G255 + G348 S376 + S383 Q32 + Q125 K72
+ W140 G133 + A174 G184 + Q395 G255 + T355 S376 + Q385 Q32 + G133 K72 + G142 G133 + G184 G184 + A420 G255 + S376 S376 + K391
Q32 + T134 K72 + G149 G133 + T193 G184 + G423 G255 + D377 S376 + K393 Q32 + W140 K72 + T165 G133 + N195 G184 + T444 G255 +
D379 S376 + Q395 Q32 + G142 K72 + W167 G133 + A204 G184 + A445 G255 + Y382 S376 + A420 Q32 + G149 K72 + R171 G133 + V206 G184
+ Q449 G255 + S383 S376 + G423 Q32 + T165 K72 + Q172 G133 + P211 G184 + T459 G255 + Q385 S376 + T444 Q32 + W167 K72 + L173
G133 + I214 G184 + P473 G255 + K391 S376 + A445 Q32 + R171 K72 + A174 G133 + V215 G184 + C474 G255 + K393 S376 + Q449 Q32 +
Q172 K72 + G184 G133 + L217 G184 + G476 G255 + Q395 S376 + T459 Q32 + L173 K72 + T193 G133 + L219 G184 + G477 G255 + A420 S376
+ P473 Q32 + A174 K72 + N195 G133 + L235 G184 + K484 G255 + G423 S376 + C474 Q32 + G184 K72 + A204 G133 + V238 T193 + N195
G255 + T444 S376 + G476 Q32 + T193 K72 + V206 G133 + M246 T193 + A204 G255 + A445 S376 + G477 Q32 + N195 K72 + P211 G133 +
M248 T193 + V206 G255 + Q449 S376 + K484 Q32 + A204 K72 + I214 G133 + L250 T193 + P211 G255 + T459 D377 + D379 Q32 + V206 K72
+ V215 G133 + G255 T193 + I214 G255 + P473 D377 + Y382 Q32 + P211 K72 + L217 G133 + Q256 T193 + V215 G255 + C474 D377 + S383
Q32 + I214 K72 + L219 G133 + A263 T193 + L217 G255 + G476 D377 + Q385 Q32 + V215 K72 + L235 G133 + V264 T193 + L219 G255 +
G477 D377 + K391 Q32 + L217 K72 + V238 G133 + Y267 T193 + L235 G255 + K484 D377 + K393 Q32 + L219 K72 + M246 G133 + N270 T193
+ V238 Q256 + A263 D377 + Q395 Q32 + L235 K72 + M248 G133 + G273 T193 + M246 Q256 + V264 D377 + A420 Q32 + V238 K72 + L250
G133 + S280 T193 + M248 Q256 + Y267 D377 + G423 Q32 + M246 K72 + G255 G133 + T285 T193 + L250 Q256 + N270 D377 + T444 Q32 +
M248 K72 + Q256 G133 + M286 T193 + G255 Q256 + G273 D377 + A445 Q32 + L250 K72 + A263 G133 + F289 T193 + Q256 Q256 + S280
D377 + Q449 Q32 + G255 K72 + V264 G133 + V291 T193 + A263 Q256 + T285 D377 + T459 Q32 + Q256 K72 + Y267 G133 + Q299 T193 +
V264 Q256 + M286 D377 + P473 Q32 + A263 K72 + N270 G133 + R320 T193 + Y267 Q256 + F289 D377 + C474 Q32 + V264 K72 + G273 G133
+ H321 T193 + N270 Q256 + V291 D377 + G476 Q32 + Y267 K72 + S280 G133 + S323 T193 + G273 Q256 + Q299 D377 + G477 Q32 + N270
K72 + T285 G133 + H324 T193 + S280 Q256 + R320 D377 + K484 Q32 + G273 K72 + M286 G133 + F328 T193 + T285 Q256 + H321 D379 +
Y382 Q32 + S280 K72 + F289 G133 + T334 T193 + M286 Q256 + S323 D379 + S383 Q32 + T285 K72 + V291 G133 + D337 T193 + F289 Q256 +
H324 D379 + Q385 Q32 + M286 K72 + Q299 G133 + Q345 T193 + V291 Q256 + F328 D379 + K391 Q32 + F289 K72 + R320 G133 + G346 T193
+ Q299 Q256 + T334 D379 + K393 Q32 + V291 K72 + H321 G133 + G348 T193 + R320 Q256 + D337 D379 + Q395 Q32 + Q299 K72 + S323
G133 + T355 T193 + H321 Q256 + Q345 D379 + A420 Q32 + R320 K72 + H324 G133 + S376 T193 + S323 Q256 + G346 D379 + G423 Q32 +
H321 K72 + F328 G133 + D377 T193 + H324 Q256 + G348 D379 + T444 Q32 + S323 K72 + T334 G133 + D379 T193 + F328 Q256 + T355 D379
+ A445 Q32 + H324 K72 + D337 G133 + Y382 T193 + T334 Q256 + S376 D379 + Q449 Q32 + F328 K72 + Q345 G133 + S383 T193 + D337
Q256 + D377 D379 + T459 Q32 + T334 K72 + G346 G133 + Q385 T193 + Q345 Q256 + D379 D379 + P473 Q32 + D337 K72 + G348 G133 +
K391 T193 + G346 Q256 + Y382 D379 + C474 Q32 + Q345 K72 + T355 G133 + K393 T193 + G348 Q256 + S383 D379 + G476 Q32 + G346 K72
+ S376 G133 + Q395 T193 + T355 Q256 + Q385 D379 + G477 Q32 + G348 K72 + D377 G133 + A420 T193 + S376 Q256 + K391 D379 + K484
Q32 + T355 K72 + D379 G133 + G423 T193 + D377 Q256 + K393 Y382 + S383 Q32 + S376 K72 + Y382 G133 + T444 T193 + D379 Q256 +
Q395 Y382 + Q385 Q32 + D377 K72 + S383 G133 + A445 T193 + Y382 Q256 + A420 Y382 + K391 Q32 + D379 K72 + Q385 G133 + Q449 T193
+ S383 Q256 + G423 Y382 + K393 Q32 + Y382 K72 + K391 G133 + T459 T193 + Q385 Q256 + T444 Y382 + Q395 Q32 + S383 K72 + K393
G133 + P473 T193 + K391 Q256 + A445 Y382 + A420 Q32 + Q385 K72 + Q395 G133 + C474 T193 + K393 Q256 + Q449 Y382 + G423 Q32 +
K391 K72 + A420 G133 + G476 T193 + Q395 Q256 + T459 Y382 + T444 Q32 + K393 K72 + G423 G133 + G477 T193 + A420 Q256 + P473
Y382 + A445 Q32 + Q395 K72 + T444 G133 + K484 T193 + G423 Q256 + C474 Y382 + Q449 Q32 + A420 K72 + A445 T134 + W140 T193 +
T444 Q256 + G476 Y382 + T459 Q32 + G423 K72 + Q449 T134 + G142 T193 + A445 Q256 + G477 Y382 + P473 Q32 + T444 K72 + T459 T134
+ G149 T193 + Q449 Q256 + K484 Y382 + C474 Q32 + A445 K72 + P473 T134 + T165 T193 + T459 A263 + V264 Y382 + G476 Q32 + Q449
K72 + C474 T134 + W167 T193 + P473 A263 + Y267 Y382 + G477 Q32 + T459 K72 + G476 T134 + R171 T193 + C474 A263 + N270 Y382 +
K484 Q32 + P473 K72 + G477 T134 + Q172 T193 + G476 A263 + G273 S383 + Q385 Q32 + C474 K72 + K484 T134 + L173 T193 + G477 A263
+ S280 S383 + K391 Q32 + G476 R87 + Q98 T134 + A174 T193 + K484 A263 + T285 S383 + K393 Q32 + G477 R87 + M105 T134 + G184 N195
+ A204 A263 + M286 S383 + Q395 Q32 + K484 R87 + G109 T134 + T193 N195 + V206 A263 + F289 S383 + A420 A37 + T40 R87 + F113 T134
+ N195 N195 + P211 A263 + V291 S383 + G423 A37 + P45 R87 + R116 T134 + A204 N195 + I214 A263 + Q299 S383 + T444 A37 + W48 R87 +
Q118 T134 + V206 N195 + V215 A263 + R320 S383 + A445 A37 + G50 R87 + Q125 T134 + P211 N195 + L217 A263 + H321 S383 + Q449 A37 +
T51 R87 + G133 T134 + I214 N195 + L219 A263 + S323 S383 + T459 A37 + N54 R87 + T134 T134 + V215 N195 + L235 A263 + H324 S383 +
P473 A37 + V56 R87 + W140 T134 + L217 N195 + V238 A263 + F328 S383 + C474 A37 + K72 R87 + G142 T134 + L219 N195 + M246 A263 +
T334 S383 + G476 A37 + R87 R87 + G149 T134 + L235 N195 + M248 A263 + D337 S383 + G477 A37 + Q98 R87 + T165 T134 + V238 N195 +
L250 A263 + Q345 S383 + K484 A37 + M105 R87 + W167 T134 + M246 N195 + G255 A263 + G346 Q385 + K391 A37 + G109 R87 + R171 T134
+ M248 N195 + Q256 A263 + G348 Q385 + K393 A37 + F113 R87 + Q172 T134 + L250 N195 + A263 A263 + T355 Q385 + Q395 A37 + R116
R87 + L173 T134 + G255 N195 + V264 A263 + S376 Q385 + A420 A37 + Q118 R87 + A174 T134 + Q256 N195 + Y267 A263 + D377 Q385 +
G423 A37 + Q125 R87 + G184 T134 + A263 N195 + N270 A263 + D379 Q385 + T444 A37 + G133 R87 + T193 T134 + V264 N195 + G273 A263
+ Y382 Q385 + A445 A37 + T134 R87 + N195 T134 + Y267 N195 + S280 A263 + S383 Q385 + Q449 A37 + W140 R87 + A204 T134 + N270
N195 + T285 A263 + Q385 Q385 + T459 A37 + G142 R87 + V206 T134 + G273 N195 + M286 A263 + K391 Q385 + P473 A37 + G149 R87 +
P211 T134 + S280 N195 + F289 A263 + K393 Q385 + C474 A37 + T165 R87 + I214 T134 + T285 N195 + V291 A263 + Q395 Q385 + G476 A37 +
W167 R87 + V215 T134 + M286 N195 + Q299 A263 + A420 Q385 + G477 A37 + R171 R87 + L217 T134 + F289 N195 + R320 A263 + G423
Q385 + K484 A37 + Q172 R87 + L219 T134 + V291 N195 + H321 A263 + T444 K391 + K393 A37 + L173 R87 + L235 T134 + Q299 N195 +
S323 A263 + A445 K391 + Q395 A37 + A174 R87 + V238 T134 + R320 N195 + H324 A263 + Q449 K391 + A420 A37 + G184 R87 + M246 T134
+ H321 N195 + F328 A263 + T459 K391 + G423 A37 + T193 R87 + M248 T134 + S323 N195 + T334 A263 + P473 K391 + T444 A37 + N195 R87
+ L250 T134 + H324 N195 + D337 A263 + C474 K391 + A445 A37 + A204 R87 + G255 T134 + F328 N195 + Q345 A263 + G476 K391 + Q449
A37 + V206 R87 + Q256 T134 + T334 N195 + G346 A263 + G477 K391 + T459 A37 + P211 R87 + A263 T134 + D337 N195 + G348 A263 +
K484 K391 + P473 A37 + I214 R87 + V264 T134 + Q345 N195 + T355 V264 + Y267 K391 + C474 A37 + V215 R87 + Y267 T134 + G346 N195 +
S376 V264 + N270 K391 + G476 A37 + L217 H1 + Q169 T134 + G348 N195 + D377 V264 + G273 K391 + G477 A37 + L219 H1 + A186 T134 +
T355 N195 + D379 V264 + S280 K391 + K484 A37 + L235 H1 + E190 T134 + S376 N195 + Y382 V264 + T285 K393 + Q395 A37 + V238 H1 +
A225 T134 + D377 N195 + S383 V264 + M286 K393 + A420 A37 + M246 H1 + K242 T134 + D379 N195 + Q385 V264 + F289 K393 + G423 A37
+ M248 H1 + S244 T134 + Y382 N195 + K391 V264 + V291 K393 + T444 A37 + L250 H1 + G258 T134 + S383 N195 + K393 V264 + Q299 K393
+ A445 A37 + G255 H1 + N260 T134 + Q385 N195 + Q395 V264 + R320 K393 + Q449 A37 + Q256 H1 + K269 T134 + K391 N195 + A420 V264
+ H321 K393 + T459 A37 + A263 H1 + W284 T134 + K393 N195 + G423 V264 + S323 K393 + P473 A37 + V264 H1 + Y295 T134 + Q395 N195

+ T444 V264 + H324 K393 + C474 A37 + Y267 H1 + S304 T134 + A420 N195 + A445 V264 + F328 K393 + G476 A37 + N270 H1 + V326 T134 + G423 N195 + Q449 V264 + T334 K393 + G477 A37 + G273 T5 + Q169 T134 + T444 N195 + T459 V264 + D337 K393 + K484 A37 + S280 T5 + A186 T134 + A445 N195 + P473 V264 + Q345 Q395 + A420 A37 + T285 T5 + E190 T134 + Q449 N195 + C474 V264 + G346 Q395 + G423 A37 + M286 T5 + A225 T134 + T459 N195 + G476 V264 + G348 Q395 + T444 A37 + F289 T5 + K242 T134 + P473 N195 + G477 V264 + T355 Q395 + A445 A37 + V291 T5 + S244 T134 + C474 N195 + K484 V264 + S376 Q395 + Q449 A37 + Q299 T5 + G258 T134 + G476 A204 + V206 V264 + D377 Q395 + T459 A37 + R320 T5 + N260 T134 + G477 A204 + P211 V264 + D379 Q395 + P473 A37 + H321 T5 + K269 T134 + K484 A204 + I214 V264 + Y382 Q395 + C474 A37 + S323 T5 + W284 W140 + G142 A204 + V215 V264 + S383 Q395 + G476 A37 + H324 T5 + Y295 W140 + G149 A204 + L217 V264 + Q385 Q395 + G477 A37 + F328 T5 + S304 W140 + T165 A204 + L219 V264 + K391 Q395 + K484 A37 + T334 T5 + V326 W140 + W167 A204 + L235 V264 + K393 A420 + G423 A37 + D337 G7 + Q169 W140 + R171 A204 + V238 V264 + Q395 A420 + T444 A37 + Q345 G7 + A186 W140 + Q172 A204 + M246 V264 + A420 A420 + A445 A37 + G346 G7 + E190 W140 + L173 A204 + M248 V264 + G423 A420 + Q449 A37 + G348 G7 + A225 W140 + A174 A204 + L250 V264 + T444 A420 + T459 A37 + T355 G7 + K242 W140 + G184 A204 + G255 V264 + A445 A420 + P473 A37 + S376 G7 + S244 W140 + T193 A204 + Q256 V264 + Q449 A420 + C474 A37 + D377 G7 + G258 W140 + N195 A204 + A263 V264 + T459 A420 + G476 A37 + D379 G7 + N260 W140 + A204 A204 + V264 V264 + P473 A420 + G477 A37 + Y382 G7 + A265 W140 + V206 A204 + Y267 V264 + C474 A420 + K484 A37 + S383 G7 + K269 W140 + P211 A204 + N270 V264 + G476 G423 + T444 A37 + Q385 G7 + W284 W140 + I214 A204 + G273 V264 + G477 G423 + A445 A37 + K391 G7 + Y295 W140 + V215 A204 + S280 V264 + K484 G423 + Q449 A37 + K393 G7 + S304 W140 + L217 A204 + T285 Y267 + N270 G423 + T459 A37 + Q395 G7 + V326 W140 + L219 A204 + M286 Y267 + G273 G423 + P473 A37 + A420 Q11 + Q169 W140 + L235 A204 + F289 Y267 + S280 G423 + C474 A37 + G423 Q11 + A186 W140 + V238 A204 + V291 Y267 + T285 G423 + G476 A37 + T444 Q11 + E190 W140 + M246 A204 + Q299 Y267 + M286 G423 + G477 A37 + A445 Q11 + A225 W140 + M248 A204 + R320 Y267 + F289 G423 + K484 A37 + Q449 Q11 + K242 W140 + L250 A204 + H321 Y267 + V291 T444 + A445 A37 + T459 Q11 + S244 W140 + G255 A204 + S323 Y267 + Q299 T444 + Q449 A37 + P473 Q11 + G258 W140 + Q256 A204 + H324 Y267 + R320 T444 + T459 A37 + C474 Q11 + N260 W140 + A263 A204 + F328 Y267 + H321 T444 + P473 A37 + G476 Q11 + K269 W140 + V264 A204 + T334 Y267 + S323 T444 + C474 A37 + G477 Q11 + W284 W140 + Y267 A204 + D337 Y267 + H324 T444 + G476 A37 + K484 Q11 + Y295 W140 + N270 A204 + Q345 Y267 + F328 T444 + G477 T40 + P45 Q11 + S304 W140 + G273 A204 + G346 Y267 + T334 T444 + K484 T40 + W48 Q11 + V326 W140 + S280 A204 + G348 Y267 + D337 A445 + Q449 T40 + G50 N16 + Q169 W140 + T285 A204 + T355 Y267 + Q345 A445 + T459 T40 + T51 N16 + A186 W140 + M286 A204 + S376 Y267 + G346 A445 + P473 T40 + N54 N16 + E190 W140 + F289 A204 + D377 Y267 + G348 A445 + C474 T40 + V56 N16 + A225 W140 + V291 A204 + D379 Y267 + T355 A445 + G476 T40 + K72 N16 + K242 W140 + Q299 A204 + Y382 Y267 + S376 A445 + G477 T40 + R87 N16 + S244 W140 + R320 A204 + S383 Y267 + D377 A445 + K484 T40 + Q98 N16 + G258 W140 + H321 A204 + Q385 Y267 + D379 Q449 + T459 T40 + M105 N16 + N260 W140 + S323 A204 + K391 Y267 + Y382 Q449 + P473 T40 + G109 N16 + K269 W140 + H324 A204 + K393 Y267 + S383 Q449 + C474 T40 + F113 N16 + W284 W140 + F328 A204 + Q395 Y267 + Q385 Q449 + G476 T40 + R116 N16 + Y295 W140 + T334 A204 + A420 Y267 + K391 Q449 + G477 T40 + Q118 N16 + S304 W140 + D337 A204 + G423 Y267 + K393 Q449 + K484 T40 + Q125 N16 + V326 W140 + Q345 A204 + T444 Y267 + Q395 T459 + P473 T40 + G133 V17 + Q169 W140 + G346 A204 + A445 Y267 + A420 T459 + C474 T40 + T134 V17 + A186 W140 + G348 A204 + Q449 Y267 + G423 T459 + G476 T40 + W140 V17 + E190 W140 + T355 A204 + T459 Y267 + T444 T459 + G477 T40 + G142 V17 + A225 W140 + S376 A204 + P473 Y267 + A445 T459 + K484 T40 + G149 V17 + K242 W140 + D377 A204 + C474 Y267 + Q449 P473 + C474 T40 + T165 V17 + S244 W140 + D379 A204 + G476 Y267 + T459 P473 + G476 T40 + W167 V17 + G258 W140 + Y382 A204 + G477 Y267 + P473 P473 + G477 T40 + R171 V17 + N260 W140 + S383 A204 + K484 G109 + Q169 P473 + K484 T40 + Q172 V17 + K269 W140 + Q385 V206 + P211 G109 + A186 C474 + G476 T40 + L173 V17 + W284 W140 + K391 V206 + I214 G109 + E190 C474 + G477 T40 + Q169 V17 + Y295 N54 + Q169 K72 + Q169 G109 + A225 C474 + K484 T40 + A186 V17 + S304 N54 + A186 K72 + A186 G109 + K242 G476 + G477 T40 + E190 V17 + V326 N54 + E190 K72 + E190 G109 + S244 G476 + K484 T40 + A225 Q32 + Q169 N54 + A225 K72 + A225 G109 + G258 G477 + K484 T40 + K242 Q32 + A186 N54 + K242 K72 + K242 G109 + N260 G133 + Q169 T40 + S244 Q32 + E190 N54 + S244 K72 + S244 G109 + K269 G133 + A186 T40 + G258 Q32 + A225 N54 + G258 K72 + G258 G109 + W284 G133 + E190 T40 + N260 Q32 + K242 N54 + N260 K72 + N260 G109 + Y295 G133 + A225 T40 + K269 Q32 + S244 N54 + K269 K72 + K269 G109 + S304 G133 + K242 T40 + W284 Q32 + G258 N54 + W284 K72 + W284 G109 + V326 G133 + S244 T40 + Y295 Q32 + N260 N54 + Y295 K72 + Y295 F113 + Q169 G133 + G258 T40 + S304 Q32 + K269 N54 + S304 K72 + S304 F113 + A186 G133 + N260 T40 + V326 Q32 + W284 N54 + V326 K72 + V326 F113 + E190 G133 + K269 P45 + Q169 Q32 + Y295 V56 + Q169 R87 + Q169 F113 + A225 G133 + W284 P45 + A186 Q32 + S304 V56 + A186 R87 + A186 F113 + K242 G133 + Y295 P45 + E190 Q32 + V326 V56 + E190 R87 + E190 F113 + S244 G133 + S304 P45 + A225 A37 + Q169 V56 + A225 R87 + A225 F113 + G258 G133 + V326 P45 + K242 A37 + A186 V56 + K242 R87 + K242 F113 + N260 T134 + Q169 P45 + S244 A37 + E190 V56 + S244 R87 + S244 F113 + K269 T134 + A186 P45 + G258 A37 + A225 V56 + G258 R87 + G258 F113 + W284 T134 + E190 P45 + N260 A37 + K242 V56 + N260 R87 + N260 F113 + Y295 T134 + A225 P45 + K269 A37 + S244 V56 + K269 R87 + K269 F113 + S304 T134 + K242 P45 + W284 A37 + G258 V56 + W284 R87 + W284 F113 + V326 T134 + S244 P45 + Y295 A37 + N260 V56 + Y295 R87 + Y295 R116 + Q169 T134 + G258 P45 + S304 A37 + K269 V56 + S304 R87 + S304 R116 + A186 T134 + N260 P45 + V326 A37 + W284 V56 + V326 R87 + V326 R116 + E190 T134 + K269 W48 + Q169 A37 + Y295 Q169 + R171 Q98 + Q169 R116 + A225 T134 + W284 W48 + A186 A37 + S304 Q169 + Q172 Q98 + A186 R116 + K242 T134 + Y295 W48 + E190 A37 + V326 Q169 + L173 Q98 + E190 R116 + S244 T134 + S304 W48 + A225 G142 + Q169 Q169 + A174 Q98 + A225 R116 + G258 T134 + V326 W48 + K242 G142 + A186 Q169 + G184 Q98 + K242 R116 + N260 W140 + Q169 W48 + S244 G142 + E190 Q169 + A186 Q98 + S244 R116 + K269 W140 + A186 W48 + G258 G142 + A225 Q169 + E190 Q98 + G258 R116 + W284 W140 + E190 W48 + N260 G142 + K242 Q169 + T193 Q98 + N260 R116 + Y295 W140 + A225 W48 + K269 G142 + S244 Q169 + N195 Q98 + K269 R116 + S304 W140 + K242 W48 + W284 G142 + G258 Q169 + A204 Q98 + W284 R116 + V326 W140 + S244 W48 + Y295 G142 + N260 Q169 + V206 Q98 + Y295 Q118 + Q169 W140 + G258 W48 + S304 G142 + K269 Q169 + P211 Q98 + S304 Q118 + A186 W140 + N260 W48 + V326 G142 + W284 Q169 + I214 Q98 + V326 Q118 + E190 W140 + K269 G50 + Q169 G142 + Y295 Q169 + V215 M105 + Q169 Q118 + A225 W140 + W284 G50 + A186 G142 + S304 Q169 + L217 M105 + A186 Q118 + K242 W140 + Y295 G50 + E190 G142 + V326 Q169 + L219 M105 + E190 Q118 + S244 W140 + S304 G50 + A225 G149 + Q169 Q169 + A225 M105 + A225 Q118 + G258 W140 + V326 G50 + K242 G149 + A186 Q169 + L235 M105 + K242 Q118 + N260 E190 + L235 G50 + S244 G149 + E190 Q169 + V238 M105 + S244 Q118 + K269 A225 + V238 G50 + G258 G149 + A225 Q169 + K242 M105 + G258 Q118 + W284 A225 + K242 G50 + N260 G149 + K242 Q169 + S244 M105 + N260 Q118 + Y295 A225 + S244 G50 + K269 G149 + S244 Q169 + M246 M105 + K269 Q118 + S304 A225 + M246 G50 + W284 G149 + G258 Q169 + M248 M105 + W284 Q118 + V326 A225 + M248 G50 + Y295 G149 + N260 Q169 + L250 M105 + Y295 Q125 + Q169 A225 + 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A225 + S304 A186 + V206 W167 + A186 Q169 + R320 Q172 + G258 G184 + K269 A225 + R320 A186 + P211 W167 + E190 Q169 + H321 Q172 + N260 G184 + W284 A225 + H321 A186 + I214 W167 + A225 Q169 + S323 Q172 + K269 G184 + Y295 A225 + S323 A186 + V215 W167 + K242 Q169 + H324 Q172 + W284 G184 + S304 A225 + H324 A186 + L217 W167 + S244 Q169 + V326 Q172 + Y295 G184 + V326 A225 + V326 A186 + L219 W167 + G258 Q169 + F328 Q172 + S304 L235 + K242 A225 + F328 A186 + A225 W167 + N260 Q169 + T334 Q172 + V326 L235 + S244 A225 + T334 A186 + L235 W167 + K269 Q169 + D337 L173 + A186 L235 + G258 A225 + D337 A186 + V238 W167 + W284 Q169 + Q345 L173 + E190 L235 + N260 A225 + Q345 A186 + K242 W167 + Y295 Q169 + G346 L173 + A225 L235 + K269 A225 + G346 A186 + S244 W167 + S304 Q169 + G348 L173 + K242 L235 + W284 A225 + G348 A186 + M246 W167 + V326 Q169 + T355 L173 + S244 L235 + Y295 A225 + T355 A186 + M248 E190 + T193 Q169 + S376 L173 + G258 L235 + S304 A225 + S376 A186 + L250 E190 + N195 Q169 + D377 L173 + N260 L235 + V326 A225 + D377 A186 + G255 E190 + A204 Q169 + D379 L173 + K269 V238 + K242 A225 + D379 A186 + Q256 E190 + V206 Q169 + Y382 L173 + W284 V238 + S244 A225 + Y382 A186 + G258 E190 + P211 Q169 + S383 L173 + Y295 V238 + G258 A225 + S383 A186 + N260 E190 + I214 Q169 + Q385 L173 + S304 V238 + N260 A225 + Q385 A186 + A263 E190 + V215 Q169 + K391 L173 + V326 V238 + K269 A225 + K391 A186 + V264 E190 + L217 Q169 + K393 A174 + A186 V238 + W284 A225 + K393 A186 + Y267 E190 + L219 Q169 + Q395 A174 + E190 V238 + Y295 A225 + Q395 A186 + K269 E190 + A225 Q169 + A420 A174 + A225 V238 + S304 A225 + A420 A186 + N270 E190 + L235 Q169 + G423 A174 + K242 V238 + V326 A225 + G423 A186 + G273 E190 + V238 Q169 + T444 A174 + S244 K242 + S244 A225 + T444 A186 + S280 E190 + K242 Q169 + A445 A174 + G258 K242 + M246 A225 + A445 A186 + W284 E190 + S244 Q169 + Q449 A174 + N260 K242 + M248 A225 + Q449 A186 + T285 E190 + M246 Q169 + T459 A174 + K269 K242 + L250 A225 + T459 A186 + M286 E190 + M248 Q169 + P473 A174 + W284 K242 + G255 A225 + P473 A186 + F289 E190 + L250 Q169 + C474 A174 + Y295 K242 + Q256 A225 + C474 A186 + V291 E190 + G255 Q169 + G476 A174 + S304 K242 + G258 A225 + G476 A186 + Y295 E190 + Q256 Q169 + G477 A174 + V326 K242 + N260 A225 + G477 A186 + Q299 E190 + G258 Q169 + K484 L217 + A225 K242 + A263 A225 + K484 A186 + S304 E190 + N260 T193 + A225 L217 + K242 K242 + V264 S244 + M248 A186 + R320 E190 + A263 T193 + K242 L217 + S244 K242 + Y267 S244 + L250 A186 + H321 E190 + V264 T193 + S244 L217 + G258 K242 + K269 S244 + G255 A186 + S323 E190 + Y267 T193 + G258 L217 + N260 K242 + N270 S244 + Q256 A186 + H324 E190 + K269 T193 + N260 L217 + K269 K242 + G273 S244 + G258 A186 + V326 E190 + N270 T193 + K269 L217 + W284 K242 + S280 S244 + N260 A186 + F328 E190 + G273 T193 + W284 L217 + Y295 K242 + W284 S244 + A263 A186 + T334 E190 + S280 T193 + Y295 L217 + S304 K242 + T285 S244 + V264 A186 + D337 E190 + W284 T193 + S304 L217 + V326 K242 + M286 S244 + Y267 A186 + Q345 E190 + T285 T193 + V326 L219 + A225 K242 + F289 S244 + K269 A186 + G346 E190 + M286 N195 + A225 L219 + K242 K242 + V291 S244 + N270 A186 + G348 E190 + F289 N195 + K242 L219 + S244 K242 + Y295 S244 + G273 A186 + T355 E190 + V291 N195 + S244 L219 + G258 K242 + Q299 S244 + S280 A186 + S376 E190 + Y295 N195 + G258 L219 + N260 K242 + S304 S244 + W284 A186 + D377 E190 + Q299 N195 + N260 L219 + K269 K242 + R320 S244 + T285 A186 + D379 E190 + S304 N195 + K269 L219 + W284 K242 + H321 S244 + M286 A186 + Y382 E190 + R320 N195 + W284 L219 + Y295 K242 + S323 S244 + F289 A186 + S383 E190 + H321 N195 + Y295 L219 + S304 K242 + H324 S244 + V291 A186 + Q385 E190 + S323 N195 + S304 L219 + V326 K242 + V326 S244 + Y295 A186 + K391 E190 + H324 N195 + V326 M246 + G258 K242 + F328 S244 + Q299 A186 + K393 E190 + V326 A204 + A225 M246 + N260 K242 + T334 S244 + S304 A186 + Q395 E190 + F328 A204 + K242 M246 + K269 K242 + D337 S244 + R320 A186 + A420 E190 + T334 A204 + S244 M246 + W284 K242 + Q345 S244 + H321 A186 + G423 E190 + D337 A204 + G258 M246 + Y295 K242 + G346 S244 + S323 A186 + T444 E190 + Q345 A204 + N260 M246 + S304 K242 + G348 S244 + H324 A186 + A445 E190 + G346 A204 + K269 M246 + V326 K242 + T355 S244 + V326 A186 + Q449 E190 + G348 A204 + W284 M248 + G258 K242 + S376 S244 + F328 A186 + T459 E190 + T355 A204 + Y295 M248 + N260 K242 + D377 S244 + T334 A186 + P473 E190 + S376 A204 + S304 M248 + K269 K242 + D379 S244 + D337 A186 + C474 E190 + D377 A204 + V326 M248 + W284 K242 + Y382 S244 + Q345 A186 + G476 E190 + D379 V206 + A225 M248 + Y295 K242 + S383 S244 + G346 A186 + G477 E190 + Y382 V206 + K242 M248 + S304 K242 + Q385 S244 + G348 A186 + K484 E190 + S383 V206 + S244 M248 + V326 K242 + K391 S244 + T355 I214 + A225 E190 + Q385 V206 + G258 L250 + G258 K242 + K393 S244 + S376 I214 + K242 E190 + K391 V206 + N260 L250 + N260 K242 + Q395 S244 + D377 I214 + S244 E190 + K393 V206 + K269 L250 + K269 K242 + A420 S244 + D379 I214 + G258 E190 + Q395 V206 + W284 L250 + W284 K242 + G423 S244 + Y382 I214 + N260 E190 + A420 V206 + Y295 L250 + Y295 K242 + T444 S244 + S383 I214 + K269 E190 + G423 V206 + S304 L250 + S304 K242 + A445 S244 + Q385 I214 + W284 E190 + T444 V206 + V326 L250 + V326 K242 + Q449 S244 + K391 I214 + Y295 E190 + A445 P211 + A225 G255 + G258 K242 + T459 S244 + K393 I214 + S304 E190 + Q449 P211 + K242 G255 + N260 K242 + P473 S244 + Q395 I214 + V326 E190 + T459 P211 + S244 G255 + K269 K242 + C474 S244 + A420 V215 + A225 E190 + P473 P211 + G258 G255 + W284 K242 + G476 S244 + G423 V215 + K242 E190 + C474 P211 + N260 G255 + Y295 K242 + G477 S244 + T444 V215 + S244 E190 + G476 P211 + K269 G255 + S304 K242 + K484 S244 + A445 V215 + G258 E190 + G477 P211 + W284 G255 + V326 N260 + A263 S244 + Q449 V215 + N260 E190 + K484 P211 + Y295 Q256 + G258 N260 + V264 S244 + T459 V215 + K269 G258 + N260 P211 + S304 Q256 + N260 N260 + Y267 S244 + P473 V215 + W284 G258 + A263 P211 + V326 Q256 + K269 N260 + K269 S244 + C474 V215 + Y295 G258 + V264 A236 + K269 Q256 + W284 N260 + N270 S244 + G476 V215 + S304 G258 + Y267 A236 + W284 Q256 + Y295 N260 + G273 S244 + G477 V215 + V326 G258 + K269 A236 + Y295 Q256 + S304 N260 + S280 S244 + K484 N270 + W284 G258 + N270 A236 + S304 Q256 + V326 N260 + W284 K269 + N270 N270 + Y295 G258 + G273 A236 + V326 W284 + T285 N260 + T285 K269 + G273 N270 + S304 G258 + S280 V264 + K269 W284 + M286 N260 + M286 K269 + S280 N270 + V326 G258 + W284 V264 + W284 W284 + F289 N260 + F289 K269 + W284 G273 + W284 G258 + T285 V264 + Y295 W284 + V291 N260 + V291 K269 + T285 G273 + Y295 G258 + M286 V264 + S304 W284 + Y295 N260 + Y295 K269 + M286 G273 + S304 G258 + F289 V267 + V326 W284 + Q299 N260 + Q299 K269 + F289 G273 + V326 G258 + V291 Y267 + K269 W284 + S304 N260 + S304 K269 + V291 S280 + W284 G258 + Y295 Y267 + W284 W284 + R320 N260 + R320 K269 + Y295 S280 + Y295 G258 + Q299 Y267 + Y295 W284 + H321 N260 + H321 K269 + Q299 S280 + S304 G258 + S304 Y267 + S304 W284 + S323 N260 + S323 K269 + S304 S280 + V326 G258 + R320 Y267 + V326 W284 + H324 N260 + H324 K269 + R320 T285 + Y295 G258 + H321 A288 + F289 W284 + V326 N260 + V326 K269 + H321 T285 + S304 G258 + S323 A288 + V291 W284 + F328 N260 + F328 K269 + S323 T285 + V326 G258 + H324 A288 + Y295 W284 + T334 N260 + T334 K269 + H324 M286 + Y295 G258 + V326 A288 + Q299 W284 + D337 N260 + D337 K269 + V326 M286 + S304 G258 + F328 A288 + S304 W284 + Q345 N260 + Q345 K269 + F328 M286 + V326 G258 + T334 A288 + R320 W284 + G346 N260 + G346 K269 + T334 F289 + Y295 G258 + D337 A288 + H321 W284 + G348 N260 + G348 K269 + D337 F289 + S304 G258 + Q345 A288 + S323 W284 + T355 N260 + T355 K269 + Q345 F289 + V326 G258 + G346 A288 + H324 W284 + S376 N260 + S376 K269 + G346 V291 + Y295 G258 + G348 A288 + V326 W284 + D377 N260 + D377 K269 + G348 V291 + S304 G258 + T355 A288 + F328 W284 + D379 N260 + D379 K269 + T355 V291 + V326 G258 + S376 A288 + T334 W284 + Y382 N260 + Y382 K269 + S376 Y295 + Q299 G258 + D377 A288 + D337 W284 + S383 N260 + S383 K269 + D377 Y295 + S304 G258 + D379 A288 + Q345 W284 + Q385 N260 + Q385 K269 + D379 Y295 + R320 G258 + Y382 A288 + G346 W284 + K391 N260 + K391 K269 + Y382 Y295 + H321 G258 + S383 A288 + G348 W284 + K393 N260 + K393 K269 + S383 Y295 + S323 G258 + Q385 A288 + T355 W284 + Q395 N260 + Q395 K269 + Q385 Y295 + H324 G258 + K391 A288 + S376 W284 + A420 N260 + A420 K269 + K391 Y295 + V326 G258 + K393 A288 + D377 W284 + G423 N260 + G423 K269 + K393 Y295 + F328 G258 + Q395 A288 + D379 W284 + T444 N260 + T444 K269 + Q395 Y295 + T334 G258 + A420 A288 + Y382 W284 + A445 N260 + A445 K269 + A420 Y295 + D337 G258 + G423 A288 + S383 W284 + Q449 N260 + Q449 K269 + G423 Y295 + Q345 G258 + T444 A288 + Q385 W284 + T459 N260 + T459 K269 + T444 Y295 + G346 G258 + A445 A288 + K391 W284 + P473 N260 + P473 K269 + A445 Y295 + G348 G258 + Q449 A288 + K393 W284 + C474 N260 + C474 K269 + Q449 Y295 + T355 G258 + T459 A288 + Q395 W284 + G476 N260 + G476 K269 + T459 Y295 + S376 G258 + P473 A288 + A420 W284 + G477 N260 + G477 K269 + P473 Y295 + D377 G258 + C474 A288 + G423 W284 + K484 N260 + K484 K269 + C474 Y295 + D379 G258 + G476 A288 + T444 S304 + V326 S304 + K393 K269 + G476 Y295 + Y382 G258 + G477 A288 + A445 S304 + Y295 S304 + Q395 K269 + G477 Y295 + S383 G258 + K484 A288 + Q449 S304 + S304 S304 + A420 K269 + K484 Y295 + Q385 Q299 + S304 A288 + T459 S304 + V326 S304 + G423 S304 + T355 Y295 + K391 Q299 + V326 A288 + P473 S304 + Q299 S304 + T444 S304 + S376 Y295 +

K393 R320 + V326 A288 + C474 S304 + S304 S304 + A445 S304 + D377 Y295 + Q395 H321 + V326 A288 + G476 S304 + R320 S304 + Q449 S304 + D379 Y295 + A420 S323 + V326 A288 + G477 S304 + H321 S304 + T459 S304 + Y382 Y295 + G423 H324 + V326 A288 + K484 S304 + S323 S304 + P473 S304 + S383 Y295 + T444 Y295 + P473 S304 + Q345 S304 + H324 S304 + C474 S304 + Q385 Y295 + A445 Y295 + C474 S304 + G346 S304 + V326 S304 + G476 S304 + K391 Y295 + Q449 Y295 + G476 S304 + G348 S304 + F328 S304 + G477 S304 + D337 Y295 + T459 Y295 + G477 Y295 + K484 S304 + T334 S304 + K484

wherein numbering is according to SEQ ID NO: 13; or

(64) TABLE-US-00003 H1 + A37 N54 + F113 K72 + G109 R116 + A174 T165 + A174 N195 + V206 H1 + T40 N54 + R116 K72 + F113 R116 + G184 T165 + G184 N195 + L228 H1 + N54 N54 + N125 K72 + R116 R116 + N195 T165 + N195 N195 + G255 H1 + V56 N54 + F133 K72 + N125 R116 + G196 T165 + G196 N195 + A265 H1 + A60 N54 + T134 K72 + F133 R116 + A204 T165 + A204 N195 + S280 H1 + K72 N54 + T165 K72 + T134 R116 + V206 T165 + V206 N195 + S304 H1 + G109 N54 + Q172 K72 + T165 R116 + L228 T165 + L228 N195 + R320 H1 + F113 N54 + L173 K72 + Q172 R116 + G255 T165 + G255 N195 + H321 H1 + R116 N54 + A174 K72 + L173 R116 + A265 T165 + A265 N195 + S323 H1 + N125 N54 + G184 K72 + A174 R116 + S280 T165 + S280 N195 + K391 H1 + F133 N54 + N195 K72 + G184 R116 + S304 T165 + S304 G196 + A204 H1 + T134 N54 + G196 K72 + N195 R116 + R320 T165 + R320 G196 + V206 H1 + T165 N54 + A204 K72 + G196 R116 + H321 T165 + H321 G196 + L228 H1 + Q172 N54 + V206 K72 + A204 R116 + S323 T165 + S323 G196 + G255 H1 + L173 N54 + L228 K72 + V206 R116 + K391 T165 + K391 G196 + A265 H1 + A174 N54 + G255 K72 + L228 N125 + F133 Q172 + L173 G196 + S280 H1 + G184 N54 + A265 K72 + G255 N125 + T134 Q172 + A174 G196 + S304 H1 + N195 N54 + S280 K72 + A265 N125 + T165 Q172 + G184 G196 + R320 H1 + G196 N54 + S304 K72 + S280 N125 + Q172 Q172 + N195 G196 + H321 H1 + A204 N54 + R320 K72 + S304 N125 + L173 Q172 + G196 G196 + S323 H1 + V206 N54 + H321 K72 + R320 N125 + A174 Q172 + A204 G196 + K391 H1 + L228 N54 + S323 K72 + H321 N125 + G184 Q172 + V206 A204 + V206 H1 + G255 N54 + K391 K72 + S323 N125 + N195 Q172 + L228 A204 + L228 H1 + A265 V56 + A60 K72 + K391 N125 + G196 Q172 + G255 A204 + G255 H1 + S280 V56 + K72 G109 + F113 N125 + A204 Q172 + A265 A204 + A265 H1 + S304 V56 + G109 G109 + R116 N125 + V206 Q172 + S280 A204 + S280 H1 + R320 V56 + F113 G109 + N125 N125 + L228 Q172 + S304 A204 + S304 H1 + H321 V56 + R116 G109 + F133 N125 + G255 Q172 + R320 A204 + R320 H1 + S323 V56 + N125 G109 + T134 N125 + A265 Q172 + H321 A204 + H321 H1 + K391 V56 + F133 G109 + T165 N125 + S280 Q172 + S323 A204 + S323 A37 + T40 V56 + T134 G109 + Q172 N125 + S304 Q172 + K391 A204 + K391 A37 + N54 V56 + T165 G109 + L173 N125 + R320 L173 + A174 V206 + L228 A37 + V56 V56 + Q172 G109 + A174 N125 + H321 L173 + G184 V206 + G255 A37 + A60 V56 + L173 G109 + G184 N125 + S323 L173 + N195 V206 + A265 A37 + K72 V56 + A174 G109 + N195 N125 + K391 L173 + G196 V206 + S280 A37 + G109 V56 + G184 G109 + G196 F133 + T134 L173 + A204 V206 + S304 A37 + F113 V56 + N195 G109 + A204 F133 + T165 L173 + V206 V206 + R320 A37 + R116 V56 + G196 G109 + V206 F133 + Q172 L173 + L228 V206 + H321 A37 + N125 V56 + A204 G109 + L228 F133 + L173 L173 + G255 V206 + S323 A37 + F133 V56 + V206 G109 + G255 F133 + A174 L173 + A265 V206 + K391 A37 + T134 V56 + L228 G109 + A265 F133 + G184 L173 + S280 L228 + G255 A37 + T165 V56 + G255 G109 + S280 F133 + N195 L173 + S304 L228 + A265 A37 + Q172 V56 + A265 G109 + S304 F133 + G196 L173 + R320 L228 + S280 A37 + L173 V56 + S280 G109 + R320 F133 + A204 L173 + H321 L228 + S304 A37 + A174 V56 + S304 G109 + H321 F133 + V206 L173 + S323 L228 + R320 A37 + G184 V56 + R320 G109 + S323 F133 + L228 L173 + K391 L228 + H321 A37 + N195 V56 + H321 G109 + K391 F133 + G255 A174 + G184 L228 + S323 A37 + G196 V56 + S323 F113 + R116 F133 + A265 A174 + N195 L228 + K391 A37 + A204 V56 + K391 F113 + N125 F133 + S280 A174 + G196 G255 + A265 A37 + V206 A60 + K72 F113 + F133 F133 + S304 A174 + A204 G255 + S280 A37 + L228 A60 + G109 F113 + T134 F133 + R320 A174 + V206 G255 + S304 A37 + G255 A60 + F113 F113 + T165 F133 + H321 A174 + L228 G255 + R320 A37 + A265 A60 + R116 F113 + Q172 F133 + S323 A174 + G255 G255 + H321 A37 + S280 A60 + N125 F113 + L173 F133 + K391 A174 + A265 G255 + S323 A37 + S304 A60 + F133 F113 + A174 T134 + T165 A174 + S280 G255 + K391 A37 + R320 A60 + T134 F113 + G184 T134 + Q172 A174 + S304 A265 + S280 A37 + H321 A60 + T165 F113 + N195 T134 + L173 A174 + R320 A265 + S304 A37 + S323 A60 + Q172 F113 + G196 T134 + A174 A174 + H321 A265 + R320 A37 + K391 A60 + L173 F113 + A204 T134 + G184 A174 + S323 A265 + H321 T40 + N54 A60 + A174 F113 + V206 T134 + N195 A174 + K391 A265 + S323 T40 + V56 A60 + G184 F113 + L228 T134 + G196 G184 + N195 A265 + K391 T40 + A60 A60 + N195 F113 + G255 T134 + A204 G184 + G196 S280 + S304 T40 + K72 A60 + G196 F113 + A265 T134 + V206 G184 + A204 S280 + R320 T40 + G109 A60 + A204 F113 + S280 T134 + L228 G184 + V206 S280 + H321 T40 + F113 A60 + V206 F113 + S304 T134 + G255 G184 + L228 S280 + S323 T40 + R116 A60 + L228 F113 + R320 T134 + A265 G184 + G255 S280 + K391 T40 + N125 A60 + G255 F113 + H321 T134 + S280 G184 + A265 S304 + R320 T40 + F133 A60 + A265 F113 + S323 T134 + S304 G184 + S280 S304 + H321 T40 + T134 A60 + S280 F113 + K391 T134 + R320 G184 + S304 S304 + S323 T40 + T165 A60 + S304 R116 + N125 T134 + H321 G184 + R320 S304 + K391 T40 + Q172 A60 + R320 R116 + F133 T134 + S323 G184 + H321 R320 + H321 T40 + L173 A60 + H321 R116 + T134 T134 + K391 G184 + S323 R320 + S323 T40 + A174 A60 + S323 R116 + T165 T165 + Q172 G184 + K391 R320 + K391 T40 + G184 A60 + K391 R116 + Q172 T165 + L173 N195 + G196 H321 + S323 T40 + N195 T40 + S323 R116 + L173 H1 + Q169 N195 + A204 H321 + K391 T40 + G196 T40 + K391 T40 + A265 H1 + A186 S244 + G255 S323 + K391 T40 + A204 N54 + V56 T40 + S280 H1 + E190 S244 + G258 A265 + K269 T40 + V206 N54 + A60 T40 + S304 H1 + A225 S244 + N260 A265 + W284 T40 + L228 N54 + K72 T40 + R320 H1 + K242 S244 + A265 A265 + Y295 T40 + G255 N54 + G109 T40 + H321 H1 + S244 S244 + K269 A265 + S304 T40 + Q169 R116 + Q169 A204 + A225 H1 + G258 S244 + S280 A265 + V326 T40 + A186 R116 + A186 A204 + K242 H1 + N260 S244 + W284 K269 + S280 T40 + E190 R116 + E190 A204 + S244 H1 + K269 S244 + Y295 K269 + W284 T40 + A225 R116 + A225 A204 + G258 H1 + W284 S244 + S304 K269 + Y295 T40 + K242 R116 + K242 A204 + N260 H1 + Y295 S244 + R320 K269 + S304 T40 + S244 R116 + S244 A204 + K269 H1 + S304 S244 + H321 K269 + R320 T40 + G258 R116 + G258 A204 + W284 H1 + V326 S244 + S323 K269 + H321 T40 + N260 R116 + N260 A204 + Y295 A37 + Q169 S244 + V326 K269 + S323 T40 + K269 R116 + K269 A204 + S304 A37 + A186 S244 + K391 K269 + V326 T40 + W284 R116 + W284 V206 + A225 A37 + E190 S244 + I405 K269 + K391 T40 + Y295 R116 + Y295 V206 + K242 A37 + A225 S244 + A421 K269 + I405 T40 + S304 R116 + S304 V206 + S244 A37 + K242 S244 + A422 K269 + A421 T40 + V326 N125 + Q169 V206 + G258 A37 + S244 S244 + A428 K269 + A422 N54 + Q169 N125 + A186 V206 + N260 A37 + G258 S244 + G448 K269 + A428 N54 + A186 N125 + E190 V206 + K269 A37 + N260 S244 + D476 K269 + G448 N54 + E190 N125 + A225 V206 + W284 A37 + K269 S244 + G477 K269 + D476 N54 + A225 N125 + K242 V206 + Y295 A37 + W284 G255 + G258 K269 + G477 N54 + K242 N125 + S244 V206 + S304 A37 + Y295 G255 + N260 S280 + W284 N54 + S244 N125 + G258 V206 + V326 A37 + S304 G255 + K269 S280 + Y295 N54 + G258 N125 + N260 A225 + L228 A37 + V326 G255 + W284 S280 + S304 N54 + N260 N125 + K269 A225 + K242 Q172 + A186 G255 + Y295 S280 + V326 N54 + K269 N125 + W284 A225 + S244 Q172 + E190 G255 + S304 W284 + Y295 N54 + W284 N125 + Y295 A225 + G255 Q172 + A225 G255 + V326 W284 + S304 N54 + Y295 N125 + S304 A225 + G258 Q172 + K242 G258 + N260 W284 + R320 N54 + S304 F133 + Q169 A225 + N260 Q172 + S244 G258 + A265 W284 + H321 N54 + V326 F133 + A186 A225 + A265 Q172 + G258 G258 + K269 W284 + S323 V56 + Q169 F133 + E190 A225 + K269 Q172 + N260 G258 + S280 W284 + V326 V56 + A186 F133 + A225 A225 + S280 Q172 + K269 G258 + W284 W284 + K391 V56 + E190 F133 + K242 A225 + W284 Q172 + W284 G258 + Y295 W284 + I405 V56 + A225 F133 + S244 A225 + Y295 Q172 + Y295 G258 + S304 W284 + A421 V56 + K242 F133 + G258 A225 + S304 Q172 + S304 G258 + R320 W284 + A422 V56 + S244 F133 + N260 A225 + R320 L173 + A186 G258 + H321 W284 + A428 V56 + G258 F133 + K269 A225 + H321 L173 + E190 G258 + S323 W284 + G448 V56 + N260 F133 + W284 A225 + S323 L173 + A225 G258 + V326 W284 + D476 V56 + K269 F133 + Y295 A225 + V326 L173 + K242 G258 + K391 W284 + G477 V56 + W284 F133 + S304 A225 + K391 L173 + S244 G258 + I405 Y295 + S304 V56 + Y295 T134 + Q169 A225 + I405 L173 + G258 G258 + A421 Y295 + R320 V56 + S304 T134 + A186 A225 + A421 L173 + N260 G258 + A422 Y295 + H321 V56 + V326 T134 + E190 A225 + A422 L173 + K269 G258 + A428 Y295 + S323 A60 + Q169 T134 + A225 A225 + A428 L173 + W284 G258 + G448 Y295 + V326 A60 + A186 T134 + K242 A225 + G448 L173 + Y295 G258 + D476 Y295 + K391 A60 + E190 T134 + S244 A225 + D476 L173 + S304 G258 + G477 Y295 + I405 A60 + A225 T134 + G258 A225 + G477 A174 + A186 N260 + A265 Y295 + A421 A60 + K242 T134 + N260 L228 + K242 A174 + E190 N260 + K269 Y295 + A422 A60 + S244 T134 + K269 L228 + S244 A174 + A225 N260 + S280 Y295 + A428 A60 + G258

T134 + W284 L228 + G258 A174 + K242 N260 + W284 Y295 + G448 A60 + N260 T134 + Y295 L228 + N260 A174 + S244 N260 + Y295 Y295 + D476 A60 + K269 T134 + S304 L228 + K269 A174 + G258 N260 + S304 Y295 + G477 A60 + W284 T134 + V326 L228 + W284 A174 + N260 N260 + R320 S304 + R320 A60 + Y295 T165 + Q169 L228 + Y295 A174 + K269 N260 + H321 S304 + H321 A60 + S304 T165 + A186 L228 + S304 A174 + W284 N260 + S323 S304 + S323 A60 + V326 T165 + E190 L228 + V326 A174 + Y295 N260 + V326 S304 + V326 K72 + Q169 T165 + N195 K242 + S244 A174 + S304 N260 + K391 S304 + K391 K72 + A186 T165 + A225 K242 + G255 G184 + A186 N260 + I405 S304 + I405 K72 + E190 T165 + K242 K242 + G258 G184 + E190 N260 + A421 S304 + A421 K72 + A225 T165 + S244 K242 + N260 G184 + A225 N260 + A422 S304 + A422 K72 + K242 T165 + G258 K242 + A265 G184 + K242 N260 + A428 S304 + A428 K72 + S244 T165 + N260 K242 + K269 G184 + S244 N260 + G448 S304 + G448 K72 + G258 T165 + K269 K242 + S280 G184 + G258 N260 + D476 S304 + D476 K72 + N260 T165 + W284 K242 + W284 G184 + N260 N260 + G477 S304 + G477 K72 + K269 T165 + Y295 K242 + Y295 G184 + K269 N195 + A225 R320 + H321 K72 + W284 T165 + S304 K242 + S304 G184 + W284 N195 + K242 R320 + V326 K72 + Y295 Q169 + Q172 K242 + R320 G184 + Y295 N195 + S244 H321 + V326 K72 + S304 Q169 + L173 K242 + H321 G184 + S304 N195 + G258 S323 + V326 G109 + Q169 Q169 + A174 K242 + S323 A186 + E190 N195 + N260 V326 + K391 G109 + A186 Q169 + G184 K242 + V326 A186 + N195 N195 + K269 V326 + I405 G109 + E190 Q169 + A186 K242 + K391 A186 + G196 N195 + W284 V326 + A421 G109 + A225 Q169 + E190 K242 + I405 A186 + A204 N195 + Y295 V326 + A422 G109 + K242 Q169 + N195 K242 + A421 A186 + V206 N195 + S304 V326 + A428 G109 + S244 Q169 + G196 K242 + A422 A186 + A225 N195 + V326 V326 + G448 G109 + G258 Q169 + A204 K242 + A428 A186 + L228 G196 + A225 V326 + D476 G109 + N260 Q169 + V206 K242 + G448 A186 + K242 G196 + K242 V326 + G477 G109 + K269 Q169 + A225 K242 + D476 A186 + S244 G196 + S244 E190 + Y295 G109 + W284 Q169 + L228 K242 + G477 A186 + G255 G196 + G258 E190 + S304 G109 + Y295 Q169 + K242 E190 + A204 A186 + G258 G196 + N260 E190 + R320 G109 + S304 Q169 + S244 E190 + V206 A186 + N260 G196 + K269 E190 + H321 G109 + V326 Q169 + G255 E190 + A225 A186 + A265 G196 + W284 E190 + S323 F113 + Q169 Q169 + G258 E190 + L228 A186 + K269 G196 + Y295 E190 + V326 F113 + A186 Q169 + N260 E190 + K242 A186 + S280 G196 + S304 E190 + K391 F113 + E190 Q169 + A265 E190 + S244 A186 + W284 G196 + V326 E190 + I405 F113 + A225 Q169 + K269 E190 + G255 A186 + Y295 A186 + G448 E190 + A421 F113 + K242 Q169 + S280 E190 + G258 A186 + S304 A186 + D476 E190 + A422 F113 + S244 Q169 + W284 E190 + N260 A186 + R320 A186 + G477 E190 + A428 F113 + G258 Q169 + Y295 E190 + A265 A186 + H321 E190 + N195 E190 + G448 F113 + N260 Q169 + S304 E190 + K269 A186 + S323 E190 + G196 E190 + D476 F113 + K269 Q169 + R320 E190 + S280 A186 + V326 Q169 + A422 E190 + G477 F113 + W284 Q169 + H321 E190 + W284 A186 + K391 Q169 + A428 A186 + A422 F113 + Y295 Q169 + S323 Q169 + I405 A186 + I405 Q169 + G448 A186 + A428 F113 + S304 Q169 + V326 Q169 + A421 A186 + A421 Q169 + D476 F113 + V326 Q169 + K391 Q169 + G477

wherein numbering is according to SEQ ID NO: 14.

(65) Stability of a parent polypeptide may be improved by introducing a pairwise deletion within the B domain of parent polypeptide. Thus, in one embodiment, the said variant comprises a) a deletion and/or a substitution at two or more positions corresponding to positions R181, G182, D183, and G184 of the amino acid sequence as set forth in SEQ ID NOs: 7 or 10, and b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q172, L173, A174, G184, N195, G196, A204, V206, L228, G255, A265, S280, S304, R320, H321, S323, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14.

(66) The term “pairwise deletion” as used herein, refers to a deletion of an amino acid in two positions. Said two positions may be adjacent to one another but may also be separated by one, two, three, four, or five amino acids.

(67) In a particular embodiment, the deletion a) is selected from the group consisting of R181+G182, R181+D183, R181+G184, G182+D183, G182+G184, or D183+G184.

(68) In an embodiment, variant comprises one or more of the following alterations; H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, G133S, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, A174S, A174D, A174P, A174M, A174T, A174H, A174K, A174G, A174Q, A174N, A174V, A174L, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, N195F, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, 1214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, Q256A, N260G, A263G, V264I, V264T, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295F, Y295N, Q299V, S304R, S304N, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, Q395P, T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S of SEQ ID NO: 13; or H1*, G7A, A37H, T40D, W48Y, W48F, N54S, V56T, A60V, K72R, Q98A, G109A, F113Q, R116H, R116V, R116Q, N125D, F133Q, T134E, T165G, Q169E, Q172D, Q172G, Q172N, L173V, A174S, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, N195F, G196R, A204G, V206L, V206Y, A225V, L228I, K242Q, S244Q, G255A, G255S, N260G, A265G, K269Q, N270T, S280Q, W284H, A288L, A288V, F289L, Y295F, Y295N, S304N, S304R, S304N, R320A, H321Y, S323N, V326L, K391A, I405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q of SEQ ID NO: 14.

(69) In a particular embodiment, the variant has an IF of >1 when compared to said parent polypeptide, and wherein said variant is selected from G182*+D183*+N195F H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K

H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K
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(70) V56T+D183*+G182*, wherein numbering is according to SEQ ID NO: 13; and N54S+G182*+D183*+N195F, G109A+G182*+D183*+N195F,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476G,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477S,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+G182*+D183*+N195F+V206L+I405L+A421+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S304N+I405L+A421H+A422P+A428T+D476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476G+G477T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477Q,

H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476N,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476Y,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206I+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+G196R+V206L+G255A+I405L+A421H+A422P+A428T+N475S,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+T40D+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T+G448D,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A42
H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+F116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+M105F+G182*+D183*+N195F+V206L+L228I+R320A+S323N+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+ N195F+V206L+I405L+A421H+A422P+A428T,
wherein numbering is according to SEQ ID NO:14.

(71) As can be seen from the data obtained in Examples 3 and 4, all tested variants have an Improvement Factor (IF) of at least 1.0, i.e. at least on par with a parent polypeptide having SEQ ID NO: 15, when tested in either (i) a detergent Model A at 15° C. and an enzyme concentration of 0.05 mg/L, (ii) a detergent Model A at 15° C. and an enzyme concentration of 0.2 mg/L, (iii) a detergent Model J at 15° C. and an enzyme concentration of 0.05 mg/L, or (iv) a detergent Model J at 15° C. and an enzyme concentration of 0.2 mg/L. Thus, the variants of the present invention have even in different model detergent compositions, at a low temperature as well as in a low concentrations, a wash performance at least that of the parent polypeptide having SEQ ID NO: 15. Specifications of the model detergents and specific test conditions for the AMSA, can be seen in Examples 3 and 4.

(72) In another embodiment, the variant is selected from the group consisting of;
G182*+D183*+N195F+V206F+N260P+L312I+L313I+R320A+T355L+L3891+T400P,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G50A+T51A+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+T40K+N54S+V56T+G109A+G149H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+Q22N+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+H28N+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N29S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+H28N+N29S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+R176K+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+R176K+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+N175G+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+Y295N+Q299T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K269S+A274K+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L235I+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L235V+V238A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q365S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+H402R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320S+H321N+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L235V+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A265G+K391A+G476K,

H1*+T51Q+N54S+V56T+G109A+A174S+G182*+D183*+N195F+Y203F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+N175Q+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+H28N+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+G50A+T51A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G7K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+M202V+V206L+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+T51G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+Y203G+V206L+K391A+G476K,
H1*+T51G+N54S+V56T+G109A+G110N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125S+A174S+N175G+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A265G+V291A+F328L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246F+A265G+V291A+F328L+Q365S+K391A+G476K,
H1*+N16Y+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+G337E+D377H+K391A+G476K,
H1*+19L+N54S+V56T+G109A+Q125S+V131T+A174*+G182*+D183*+V206L+Y267F+F289H+Q361E+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+P473R+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206F+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295N+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+A186N+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A288V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A225V+K391A+G476K,
H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K, G182*+D183*+N195F,
H1*+N54S+V56T+G109A+Q169E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+W284H+K391A+G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+S304R+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N270G+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295F+K391 A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174T+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W140Y+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+F328L+K391 A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G476K, F113Q+R116H+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A288L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270G+K391A+G476K, Q169E+Q172K+A174*+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, F113Q+R116Q+D183*+G182*+N195F,
N54S+V56T+D183*+G182*+N195F, H1*+N54S+V56T+G109A+A174T+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116W+A174S+G182*+D183*+N195F+V206L+K391A+G476K, Q169E+Q172K+G182*+D183*+N195F,

G182*+D183*+A186D, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238A+K391A+G476K, G182*+D183*+A186N, H1*+N54S+V56T+G109A+A174S+V206L+G182*+D183*+K391A+G476K, H1*+N54S+V56T+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206F+K391A+G476K, H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+A186N+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+F289L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y295F+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S244Q+K391A+G476K, G182*+D183*+F328L, G182*+D183*+N195F+A288V, G182*+D183*+A186N+N195F, G182*+D183*+E190P+N195F, W140Y+G182*+D183*+N195F, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A225V+K391A+G476K, G182*+D183*+A186D+N195F, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K, H1*+W48Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, G182*+D183*+N195F+V238A, G182*+D183*+Y295N, Q172G+G182*+D183*, F113Q+G182*+D183*+N195F, H1*+N54S+V56T+Q98A+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, G182*+D183*+N195F+Y295F, G182*+D183*+A186E, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206Y+K391A+G476K, G182*+D183*+N195F+W284H, G182*+D183*+A186Q+N195F, K72R+G182*+D183*, D183*+G182*+N195F+V326L, D183*+G182*+A186E+N195F, Q169E+G182*+D183*+N195F, D183*+G182*+N195F+K242Q, G7A+G182*+D183*+N195F, D183*+G182*+N195F+N260G, D183*+G182*+R321A+S323N, D183*+G182*+W284H, H1*+D183*+G182*, D183*+G182*+A186H+N195F, D183*+G182*+F289V, D183*+G182*+K391A, D183*+G182*+S304R, W167F+G182*+D183*+N195F, F113Q+R116H+D183*+G182*, D183*+G182*+N270G, D183*+G182*+S244Q, H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K, D183*+G182*+N195F+Y295N, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K, D183*+G182*+A186Q, D183*+G182*+N195F+V206F, D183*+G182*+N195F+R320A, G182*+D183*+G184T+N195F, N54S+V56T+D183*+G182*, W48Y+D183*+G182*, A174S+D183*+G182*, G7A+D183*+G182*, N54S+D183*+G182*, D183*+G182*+N195F+M246V, W48Y+D183*+G182*+N195F, D183*+G182*+N195F+S304R, wherein numbering is according to SEQ ID NO: 13, and wherein the IF is determined by use of Model A detergent.

(73) In one embodiment, the IF is determined by use of Model A detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1 as compared to the parent polypeptide, wherein the IF is determined by use of Model A detergent at a temperature of 15° C. and a concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.

(74) In another embodiment, the variant has an IF of at least 1.0 when compared to the parent polypeptide when tested in Model J detergent, and wherein the variant is selected from the group consisting of: G182*+D183*+N195F, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+F113W+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+F113S+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+F113N+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+F113Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+F113R+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q118N+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+R116K+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+R116W+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+R116L+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+G142T+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q125P+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q125K+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381G+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246T+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V213T+K391A+G476K, H1*+A37L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q118K+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q125A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377S+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377G+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377R+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K, D183*+G182*+A186Q, W140Y+D183*+G182*,
D183*+G182*+K242Q, D183*+G182*+V206L, D183*+G182*+N195F+V206F, Q169E+D183*+G182*, D183*+G182*+N195F+R320A,
G182*+D183*+G184T+N195F, D183*+G182*+G184T, D183*+G182*+P473R, W48Y+D183*+G182*, D183*+G182*+R323A,
G7A+D183*+G182*, H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
A174S+D183*+G182*, D183*+G182*+N195F+S323N, D183*+G182*+N195F+M246V, W48Y+D183*+G182*+N195F, V56T+D183*+G182*,
wherein numbering is according to SEQ ID NO: 13, G182*+D183*+N195F+A428T, G182*+D183*+N195F, H1*+G182*+D183*+N195F,
N54S+G182*+D183*+N195F, V56T+G182*+D183*+N195F, G182*+D183*+N195F+I405L,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T+D476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476G,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477S,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280Q+H321Y+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S304N+I405L+A421H+A422P+A428T+D476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476G+G477T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477Q,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476Y,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206I+I405L+A421H+A422P+A428T
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+G196R+V206L+G255A+I405L+A421H+A422P+A428T+N475S,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+T40D+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
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H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T+G448D,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
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H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
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H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
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H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,

H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+M105F+G182*+D183*+N195F+V206L+L228I+R320A+S323N+I405L+A421H+A422P+A428T,
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H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
wherein numbering is according to SEQ ID NO: 14.

(75) In one embodiment, the IF is determined by use of Model J detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1 as compared to the parent polypeptide, wherein the IF is determined by use of Model J detergent at a temperature of 15° C. and a concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.

(76) In another embodiment, the variant has an IF of at least >1.5 when compared to said parent polypeptide, and wherein said variant is selected from H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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G182*+D183*+A186D, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238A+K391A+G476K, G182*+D183*+A186N, H1*+N54S+V56T+G109A+A174S+V206L+G182*+D183*+K391A+G476K, H1*+N54S+V56T+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206F+K391A+G476K, H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+A186N+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+F289L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y295F+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+K391A+G476K, G182*+D183*+A186N+N195F, G182*+D183*+A186D+N195F, G182*+D183*+N195F+W284H, D183*+G182*+N195F+V326L, wherein numbering is according to SEQ ID NO: 13, and H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476G+G477T, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477Q, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477A, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476N, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476Y, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206I+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+T40D+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A42 H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+F113Q+F116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T, H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T, H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14, and wherein the IF is determined by use of Model A detergent.

(77) In one embodiment, the IF is determined by use of Model A detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1.5 as compared to the parent polypeptide, wherein the IF is determined by use of Model A detergent at a temperature of 15° C. and a concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.

(78) In another embodiment, the variant has an IF of at least 1.5 when compared to the parent polypeptide and when determined by use of Model J detergent, wherein the variant is selected from the group consisting of: G182*+D183*+N195F, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+G142T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125P+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V213T+K391A+G476K,
H1*+A37L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381A+K391A+G476K,
H1*+N54S+V56T+G109A+W167I+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346P+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214T+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473G+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473T+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+Q395P+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444S+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444D+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444Y+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280W+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255N+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T285L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T285Q+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+Q449T+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473K+G476K,
H1*+N54S+V56T+K72S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T40K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K,
H1*+G50A+N54S+V56T+G109A++Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484A,

H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484G,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484P,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484E,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484S,
H1*K+G50A+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+L173V+A174S+G182*+D183*+N195F+A204T+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W140Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+Q98S+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+H324L+K391A+G476K,
H1*+N54S+V56T+G109A+G149Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K+G477A,
H1*+N54S+V56T+G109AG255A++A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+F113L+R116L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q345D+K391A+G476K+G477Q,
H1*+G50A+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V264I+K391A+G423H+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320V+K391A+G476Y,
H1*+T5K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, G182*+D183*+N195F+V238A,
G182*+D183*+Y295N, Q172G+G182*+D183*, F113Q+G182*+D183*+N195F,
H1*+N54S+V56T+Q98A+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, G182*+D183*+N195F+Y295F,
G182*+D183*+A186E, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206Y+K391A+G476K, G182*+D183*+N195F+W284H,
G182*+D183*+A186Q+N195F, K72R+G182*+D183*, D183*+G182*+N195F+V326L, D183*+G182*+A186E+N195F,
Q169E+Gi182*+D183*+N195F, D183*+G182*+N195F+K242Q, G7A+G182*+D183*+N195F, D183*+G182*+N195F+N260G,
D183*+G182*+R321A+S323N, D183*+G182*+W284H, D183*+G182*+A186H+N195F, D183*+G182*+K391A, wherein numbering is according to SEQ ID NO: 13, V56T+G182*+D183*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477S,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476G+G477T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+G477A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476K+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476N,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+I405L+A421H+A422P+A428T+D476Y,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206I+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+G196R+V206L+G255A+I405L+A421H+A422P+A428T+N475S,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+T40D+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T+G448D,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,

H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A42
H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+M105F+G182*+D183*+N195F+V206L+L228I+R320A+S323N+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
wherein numbering is according to SEQ ID NO: 14.

(79) In one embodiment, the IF is determined by use of Model J detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1.5 as compared to the parent polypeptide, wherein the IF is determined by use of Model J detergent at a temperature of 15° C. and a concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.

(80) Some variants according to the invention have an even further improved wash performance indicated by an IF of at least 2.0 when compared the a parent polypeptide having an amino acid sequence set forth in SEQ ID NO: 15, and wherein the IF is determined by use of Model A detergent at a temperature of 15° C.

(81) Thus, in one embodiment, the variant has an IF of at least >2.0 when compared to said parent polypeptide, and wherein said variant is selected from H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125P+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246T+K391A+G476K,
H1*+A37L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346P+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+A420S+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K+G477A,

H1*+N54S+V56T+G109AG255A++A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+F113L+R116L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+K72Q+G109A+W167H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109S+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109S+F113Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+Q395P+T444Q+P4
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
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H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+K391A+G476K,
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H1*+A37H+N54S+V56T+G109A+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+T459N+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116H+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
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H1*+N54S+V56T+G109A+G182*+D183*+N195F+K391A+G476K, H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L,
N54S+V56T+G109A+G182*+D183*+N195F, N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+A174S+G182*+D183*+N195F+V206L+G273R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L, N54S+V56T+G109A+A174S+G182*+D183*+N195F+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+R116Q+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+R116A+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
H1*+V56A+G109A+A174S+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+A174S+G182*+D183*+G184T+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109S+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172R+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+R116H+Q172R+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+R116H+T165G+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
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H1*+A37V+N54S+V56T+G109A+R116H+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+P45A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+P364A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+G50A+T51A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V264F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382L+K391A+G476K,
H1*+G7K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+Q256A+K391A+G476K,
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H1*+N54S+V56T+G109A+W167H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T165V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174Q+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382F+K391A+G476K,
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q385L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+R171H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+P473R+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206F+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295N+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+A186N+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A288V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G476K,

H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A225V+K391A+G476K,
H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K, wherein numbering is according to SEQ ID NO: 13, and H1*+T40D+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T
H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14, and wherein the IF is determined by use of Model A detergent.

(82) In one embodiment, the variant has an IF of at least 2.0 when compared to the parent polypeptide, wherein the IF is determined by use of Model A detergent at a temperature of 15° C. and a concentration of 0.05 mg enzyme/L.

(83) In one embodiment, the variant has an IF of at least 2.0 when compared to the parent polypeptide and when determined by use of Model J detergent, and wherein the variant is selected from the group consisting of:
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+Q125P+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
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H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T+G448D,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,

H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
wherein numbering is according to SEQ ID NO: 14.

(84) In one embodiment, the variant has an IF of at least 2.0 when compared to the parent polypeptide, wherein the IF is determined by use of Model J detergent at a temperature of 15° C. and an enzymes concentration of 0.05 mg/L.

(85) Other variants according to the invention have an IF of at least 3.0, indicating that at least these variants have a significantly improved wash performance when compared the a parent polypeptide having an amino acid sequence set forth in SEQ ID NO: 15.

(86) In a particular embodiment, the variant is selected from a group consisting of:
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116Q+W167F+A174S+G182*+D183*+G184T+N195F+V206L+
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+G133Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
and
H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+Q395P+T444Q+P4
wherein numbering is according to SEQ ID NO: 13.

(87) In another embodiment, the variant has an IF of at least 3.0 when compared to the parent polypeptide and determined by use of Model J detergent, and wherein the variant is selected from the group consisting of:
H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444D+G476K,
H1*+N54S+V56T+K72S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+L173V+A174S+G182*+D183*+N195F+A204V+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K+G477A,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174Q+G182*+D183*+N195F+V206L+K391A+G476K+G477A,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+G184T+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238G+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+Q172E+L173P+A174K+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250M+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+G182*+D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+C474V+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+C474V+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+R116Q+Q172D+A174S+G182*+D183*+N195F+V206L+G346T+K391A+T444Q+G477A+G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q172N+A174S+G182*+D183*+N195F+V206L+V264I+K391A+C474V+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+T165G+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,

H1*+N54S+V56T+G109A+G133Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+Q395P+K391A+G476K,
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H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+G184T+N195F+V206L+K391A+T444Q+P473R+G47
H1*+N54S+V56T+G109S+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109S+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109S+F113Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+Q395P+T444Q+P47
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H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295N+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+A186N+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A288V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K, wherein numbering is according to SEQ ID

NO: 13, H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14.

(88) The variants according to the present invention may further comprise one or more additional alterations than those described above at one or more (e.g., several) other positions.

(89) The amino acid changes may be of a minor nature, that is conservative amino acid substitutions or insertions that do not significantly affect the folding and/or activity of the protein; small deletions, typically of 1-30 amino acids; small amino- or carboxyl-terminal extensions, such as an amino-terminal methionine residue; a small linker peptide of up to 20-25 residues; or a small extension that facilitates purification by changing net charge or another function, such as a poly-histidine tract, an antigenic epitope or a binding domain.

(90) Examples of conservative substitutions are within the groups of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine and asparagine), hydrophobic amino acids (leucine, isoleucine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine, serine, threonine and methionine). Amino acid substitutions that do not generally alter specific activity are known in the art and are described, for example, by H. Neurath and R. L. Hill, 1979, In, The Proteins, Academic Press, New York. Common substitutions are Ala/Ser, Val/Ile, Asp/Glu, Thr/Ser, Ala/Gly, Ala/Thr, Ser/Asn, Ala/Val, Ser/Gly, Tyr/Phe, Ala/Pro, Lys/Arg, Asp/Asn, Leu/Ile, Leu/Val, Ala/Glu, and Asp/Gly.

(91) Alternatively, the amino acid changes are of such a nature that the physico-chemical properties of the polypeptides are altered. For example, amino acid changes may improve the thermal stability of the polypeptide, alter the substrate specificity, change the pH optimum, and the like.

(92) Essential amino acids in a polypeptide may be identified according to procedures known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (Cunningham and Wells, 1989, *Science* 244: 1081-1085). In the latter technique, single alanine mutations are introduced at every residue in the molecule, and the resultant mutant molecules are tested for alpha-amylase activity to identify amino acid residues that are critical to the activity of the molecule. See also, Hilton et al., 1996, *J. Biol. Chem.* 271: 4699-4708. The active site of the enzyme or other biological interaction can also be determined by physical analysis of structure, as determined by such techniques as nuclear magnetic resonance, crystallography, electron diffraction, or photoaffinity labeling, in conjunction with mutation of putative contact site amino acids. See, for example, de Vos et al., 1992, *Science* 255: 306-312; Smith et al., 1992, *J. Mol. Biol.* 224: 899-904; Wlodaver et al., 1992, *FEBS Lett.* 309: 59-64. The identity of essential amino acids can also be inferred from an alignment with a related polypeptide.

(93) The variants according to the present invention may consist of 400 to 485 amino acids, e.g., 410 to 485, 425 to 485, and 440 to 485 amino acids.

(94) Polynucleotides

(95) The present invention also relates to polynucleotides encoding a variant of the present invention. Thus, in particular, the present invention relates to a polynucleotide endocing a variant comprising an alteration in one or more positions corresponding to the H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, G448, D476, and G477 of SEQ ID NO: 14.

(96) The term “polynucleotides encoding” as used herein, refers to a polynucleotide that encodes a mature polypeptide having alpha-amylase having alpha-amylase activity. In one aspect, the polypeptide coding sequence is the nucleotide sequence set forth in SEQ ID NO: 1, 2, or 3.

(97) In one embodiment, the polynucleotide encoding a variant according to the present invention as at least 60%, such as at least 65%, such as at least 70%, such as at least 75%, such as at least 80%, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99% but less than 100% sequence identity to the polynucleotide of SEQ ID NOs: 1, 2 and 3.

(98) Nucleic Acid Constructs

(99) The present invention also relates to nucleic acid constructs comprising a polynucleotide encoding a variant of the present invention operably linked to one or more control sequences that direct the expression of the coding sequence in a suitable host cell under conditions compatible with the control sequences. Thus, in particular, the present invention relates to a nucleic acid construct comprising a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, wherein the polynucleotide is operately linked to one or more control sequences.

(100) The term “nucleic acid construct” as used herein, refers to a nucleic acid molecule, either single- or double-stranded, which is isolated from a naturally occurring gene or is modified to contain segments of nucleic acids in a manner that would not otherwise exist in nature or which is synthetic, which comprises one or more control sequences.

(101) The term “operably linked” as used herein, refers to a configuration in which a control sequence is placed at an appropriate position relative to the coding sequence of a polynucleotide such that the control sequence directs expression of the coding sequence.

(102) The polynucleotide may be manipulated in a variety of ways to provide for expression of a variant. Manipulation of the polynucleotide prior to its insertion into a vector may be desirable or necessary depending on the expression vector. The techniques for modifying polynucleotides utilizing recombinant DNA methods are well known in the art.

(103) The control sequence may be a promoter, a polynucleotide which is recognized by a host cell for expression of the polynucleotide. The promoter comprises transcriptional control sequences that mediate the expression of the variant. The promoter may be any polynucleotide that shows transcriptional activity in the host cell including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the host cell.

(104) Expression Vectors

(105) The present invention also relates to recombinant expression vectors comprising a polynucleotide encoding a variant of the present invention, a promoter, and transcriptional and translational stop signals. Thus, the present invention relates to an expression vector, optionally recombinant, comprising a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113,

R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, a promotor, and transcriptional and translational stop signals.

(106) The term "expression vector" as used herein, refers to a linear or circular DNA molecule that comprises a polynucleotide encoding a variant and is operably linked to control sequences that provide for its expression.

(107) The various nucleotide and control sequences may be joined together to produce a recombinant expression vector that may include one or more convenient restriction sites to allow for insertion or substitution of the polynucleotide encoding the variant at such sites. Alternatively, the polynucleotide may be expressed by inserting the polynucleotide or a nucleic acid construct comprising the polynucleotide into an appropriate vector for expression. In creating the expression vector, the coding sequence is located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

(108) The recombinant expression vector may be any vector (e.g., a plasmid or virus) that can be conveniently subjected to recombinant DNA procedures and can bring about expression of the polynucleotide. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vector may be a linear or closed circular plasmid.

(109) The vector may be an autonomously replicating vector, i.e., a vector that exists as an extrachromosomal entity, the replication of which is independent of chromosomal replication, e.g., a plasmid, an extrachromosomal element, a minichromosome, or an artificial chromosome. The vector may contain any means for assuring self-replication. Alternatively, the vector may be one that, when introduced into the host cell, is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, a single vector or plasmid or two or more vectors or plasmids that together contain the total DNA to be introduced into the genome of the host cell, or a transposon, may be used.

(110) The vector preferably contains one or more selectable markers that permit easy selection of transformed, transfected, transduced, or the like cells. A selectable marker is a gene the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like.

(111) The skilled person would know which expression vector is the most suitable for specific expression systems. Thus, the present invention is not limited to any specific expression vector, but any expression vector comprising the polynucleotide encoding a variant according to the invention is considered part of the present invention.

(112) Host Cells

(113) The present invention also relates to recombinant host cells, comprising a polynucleotide encoding a variant of the present invention operably linked to one or more control sequences that direct the production of a variant of the present invention. Thus, the present invention relates to a host cell, optionally a recombinant host cell, comprising a polynecleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, operably linked to one or more control sequences that direct the production of the variant.

(114) The term "host cell" as used herein, refers to any cell type that is susceptible to transformation, transfection, transduction, or the like with a nucleic acid construct or expression vector comprising a polynucleotide of the present invention. The term "host cell" encompasses any progeny of a parent cell that is not identical to the parent cell due to mutations that occur during replication.

(115) A construct or vector comprising a polynucleotide is introduced into a host cell so that the construct or vector is maintained as a chromosomal integrant or as a self-replicating extra-chromosomal vector as described earlier. The choice of a host cell will to a large extent depend upon the gene encoding the variant and its source.

(116) The host cell may be any cell useful in the recombinant production of a variant, e.g., a prokaryote or a eukaryote.

(117) The prokaryotic host cell may be any Gram-positive or Gram-negative bacterium. Gram-positive bacteria include, but are not limited to, *Bacillus, Clostridium, Enterococcus, Geobacillus, Lactobacillus, Lactococcus, Oceanobacillus, Staphylococcus, Streptococcus*, and *Streptomyces*. Gram-negative bacteria include, but are not limited to, *Campylobacter, E. coli, Flavobacterium, Fusobacterium, Helicobacter, Ilyobacter, Neisseria, Pseudomonas, Salmonella*, and *Ureaplasma.*

(118) Methods According to the Invention

(119) The present invention also relates to methods of producing a variant, comprising: (a) cultivating a host cell of the present invention under conditions suitable for expression of the variant; and (b) recovering the variant. Thus, the present invention relates to a method of producing a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, wherein the method comprises the steps of a) cultivating a host cell according to the invention under conditions suitable for expression of the variant, and b) recovering the variant.

(120) The host cells are cultivated in a nutrient medium suitable for production of the variant using methods known in the art. For example, the cell may be cultivated by shake flask cultivation, or small-scale or large-scale fermentation (including continuous, batch, fed-batch, or solid state fermentations) in laboratory or industrial fermentors performed in a suitable medium and under conditions allowing the variant to be expressed and/or isolated. The cultivation takes place in a suitable nutrient medium comprising carbon and nitrogen sources and inorganic salts, using procedures known in the art. Suitable media are available from commercial suppliers or may be prepared according to published compositions (e.g., in catalogues of the American Type Culture Collection). If the variant is secreted into the nutrient medium, the variant can be recovered directly from the medium. If the variant is not secreted, it can be recovered from cell lysates.

(121) The variant may be detected using methods known in the art that are specific for the polypeptides having alpha amylase activity. These detection methods include, but are not limited to, use of specific antibodies, formation of an enzyme product, or disappearance of an enzyme substrate. For example, an enzyme assay may be used to determine the activity of the variant.

(122) The variant may be recovered using methods known in the art. For example, the variant may be recovered from the nutrient medium by conventional procedures including, but not limited to, collection, centrifugation, filtration, extraction, spray-drying, evaporation, or precipitation.

(123) The variant may be purified by a variety of procedures known in the art including, but not limited to, chromatography (e.g., ion exchange, affinity, hydrophobic, chromatofocusing, and size exclusion), electrophoretic procedures (e.g., preparative isoelectric focusing), differential solubility (e.g., ammonium sulfate precipitation), SDS-PAGE, or extraction (see, e.g., *Protein Purification*, Janson and Ryden, editors, VCH Publishers, New York, 1989) to obtain substantially pure variants.

(124) In an alternative aspect, the variant is not recovered, but rather a host cell of the present invention expressing the variant is used as a source of the variant.
(125) Furthermore, the present invention relates to methods for obtaining a variant, comprising introducing into a parent alpha-amylase having at least 80% sequence identity to the polypeptide of SEQ ID NO: 4 (a) a substitution and/or deletion of two or more positions in the parent alpha-amylase said positions corresponding to positions 181, 182, 183, and 184 of the mature polypeptide of SEQ ID NO: 4; and (b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, wherein the resulting variant has at least 80%, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 97%, but less than 100% sequence identity with the mature polypeptide of SEQ ID NO: 4, wherein said variant has alpha-amylase activity; (c) recovering said variant.
(126) The variants may be prepared using any mutagenesis procedure known in the art, such as site-directed mutagenesis, synthetic gene construction, semi-synthetic gene construction, random mutagenesis, shuffling, etc.
(127) Site-directed mutagenesis is a technique in which one or more (e.g., several) mutations are introduced at one or more defined sites in a polynucleotide encoding the parent.
(128) Site-directed mutagenesis can be accomplished in vitro by PCR involving the use of oligonucleotide primers containing the desired mutation. Site-directed mutagenesis can also be performed in vitro by cassette mutagenesis involving the cleavage by a restriction enzyme at a site in the plasmid comprising a polynucleotide encoding the parent polypeptide and subsequent ligation of an oligonucleotide containing the mutation in the polynucleotide. Usually the restriction enzyme that digests the plasmid and the oligonucleotide is the same, permitting sticky ends of the plasmid and the insert to ligate to one another. See, e.g., Scherer and Davis, 1979, *Proc. Natl. Acad. Sci. USA* 76: 4949-4955; and Barton et al., 1990, *Nucleic Acids Res.* 18: 7349-4966.
(129) Site-directed mutagenesis can also be accomplished in vivo by methods known in the art. See, e.g., U.S. Patent Application Publication No. 2004/0171154; Storici et al., 2001, *Nature Biotechnol.* 19: 773-776; Kren et al., 1998, *Nat. Med.* 4: 285-290; and Calissano and Macino, 1996, *Fungal Genet. Newslett.* 43: 15-16.
(130) Any site-directed mutagenesis procedure may be used in the present invention. There are many commercial kits available that can be used to prepare variants. The skilled person in the art is well-aware of such commercial kits and how to use them.
(131) Synthetic gene construction entails in vitro synthesis of a designed polynucleotide molecule to encode a polypeptide of interest. Gene synthesis may be performed utilizing a number of techniques, such as the multiplex microchip-based technology described by Tian et al. (2004, *Nature* 432: 1050-1054) and similar technologies wherein oligonucleotides are synthesized and assembled upon photo-programmable microfluidic chips.
(132) Single or multiple amino acid substitutions, deletions, and/or insertions may be made and tested using known methods of mutagenesis, recombination, and/or shuffling, followed by a relevant screening procedure, such as those disclosed by Reidhaar-Olson and Sauer, 1988, *Science* 241: 53-57; Bowie and Sauer, 1989, *Proc. Natl. Acad. Sci. USA* 86: 2152-2156; WO 95/17413; or WO 95/22625. Other methods that may be used include error-prone PCR, phage display (e.g., Lowman et al., 1991, *Biochemistry* 30: 10832-10837; U.S. Pat. No. 5,223,409; WO 92/06204) and region-directed mutagenesis (Derbyshire et al., 1986, *Gene* 46: 145; Ner et al., 1988, *DNA* 7: 127).
(133) Mutagenesis/shuffling methods may be combined with high-throughput, automated screening methods to detect activity of cloned, mutagenized polypeptides expressed by host cells (Ness et al., 1999, *Nature Biotechnology* 17: 893-896). Mutagenized DNA molecules that encode active polypeptides may be recovered from the host cells and rapidly sequenced using standard methods in the art. These methods allow the rapid determination of the importance of individual amino acid residues in a polypeptide.
(134) Semi-synthetic gene construction is accomplished by combining aspects of synthetic gene construction, and/or site-directed mutagenesis, and/or random mutagenesis, and/or shuffling. Semi-synthetic construction is typified by a process utilizing polynucleotide fragments that are synthesized, in combination with PCR techniques. Defined regions of genes may thus be synthesized de novo, while other regions may be amplified using site-specific mutagenic primers, while yet other regions may be subjected to error-prone PCR or non-error prone PCR amplification. Polynucleotide subsequences may then be shuffled.
(135) The present invention also relates to a method of improving wash performance of a parent polypeptide having the amino acid sequence of SEQ ID NO: 13 or 14, or having at least 80% sequence identity thereto, said method comprising the steps of: a) a substitution and/or deletion of two, three or four positions in the parent alpha-amylase said positions corresponding to positions 181, G182, D183, and G184 of the mature polypeptide of SEQ ID NO: 4; and b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, wherein the variant has at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 98%, at least 99%, but less than 100% sequence identity with the polypeptide of SEQ ID NO: 4 or 7, and wherein the variant has alpha-amylase activity and improved wash performance compared to the parent polypeptide.
(136) In one embodiment, the variant has at least 50%, such as at least 60%, or at least 70%, or at least 80%, or at least 90%, or at least 100% of the activity of the parent polypeptide having the amino acid sequence of SEQ ID NO: 7.
(137) In one embodiment, the activity is determined according to a Phadebas assay.
(138) The alpha-amylase activity may be determined by a method using the Phadebas substrate (from for example Magle Life Sciences, Lund, Sweden). A Phadebas tablet includes interlinked starch polymers that are in the form of globular microspheres that are insoluble in water. A blue dye is covalently bound to these microspheres. The interlinked starch polymers in the microsphere are degraded at a speed that is proportional to the alpha-amylase activity. When the alpha-amylase degrades the starch polymers, the released blue dye is water soluble and concentration of dye can be determined by measuring absorbance at 620 nm. The concentration of blue is proportional to the alpha-amylase activity in the sample.
(139) The variant sample to be analyzed is diluted in activity buffer with the desired pH. Two substrate tablets are suspended in 5 mL activity buffer and mixed on magnetic stirrer. During mixing of substrate transfer 150 μl to microtiter plate (MTP) or PCR-MTP. Add 30 μl diluted amylase sample to 150 μl substrate and mix. Incubate for 15 minutes at 37° C. The reaction is stopped by adding 30 μl 1M NaOH and mix. Centrifuge MTP for 5 minutes at 4000×g. Transfer 100 μl to new MTP and measure absorbance at 620 nm.
(140) The alpha-amylase sample should be diluted so that the absorbance at 620 nm is between 0 and 2.2, and is within the linear range of the activity assay.
(141) Thus, in one embodiment, the activity is determined by a method comprising the steps of; a) incubating an alpha-amylase variant according to

the invention with a dyed amylose substrate for 15 minute at 37° C.; and b) measuring the absorption at OD 620 nm.

(142) In a further embodiment, the activity is determined by a method comprising the steps of; a) incubating an alpha-amylase variant according to the invention with a dyed amylose substrate for 15 minute at 37° C.; and b) centrifuging the sample; c) transferring the supernatant to reader plate, and measuring the absorption at OD 620 nm.

Fermentation Broth Formulations or Cell Compositions

(143) The present invention also relates to a fermentation broth formulation or a cell composition comprising a polypeptide of the present invention. Thus, in one embodiment, the fermentation broth formulation or the cell composition comprises a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, a nucleic acid construct encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, or an expression vector encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q269, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14. The fermentation broth product may further comprise additional ingredients used in the fermentation process, such as, for example, cells (including, the host cells containing the gene encoding the polypeptide of the present invention which are used to produce the polypeptide of interest), cell debris, biomass, fermentation media and/or fermentation products. In some embodiments, the composition is a cell-killed whole broth containing organic acid(s), killed cells and/or cell debris, and culture medium.

(144) The term "fermentation broth" as used herein refers to a preparation produced by cellular fermentation that undergoes no or minimal recovery and/or purification. For example, fermentation broths are produced when microbial cultures are grown to saturation, incubated under carbon-limiting conditions to allow protein synthesis (e.g., expression of enzymes by host cells) and secretion into cell culture medium. The fermentation broth can contain unfractionated or fractionated contents of the fermentation materials derived at the end of the fermentation. Typically, the fermentation broth is unfractionated and comprises the spent culture medium and cell debris present after the microbial cells (e.g., filamentous fungal cells) are removed, e.g., by centrifugation. In some embodiments, the fermentation broth contains spent cell culture medium, extracellular enzymes, and viable and/or nonviable microbial cells.

(145) In one embodiment, the fermentation broth formulation and cell compositions comprise a first organic acid component comprising at least one 1-5 carbon organic acid and/or a salt thereof and a second organic acid component comprising at least one 6 or more carbon organic acid and/or a salt thereof. In a particular embodiment, the first organic acid component is acetic acid, formic acid, propionic acid, a salt thereof, or a mixture of two or more of the foregoing and the second organic acid component is benzoic acid, cyclohexanecarboxylic acid, 4-methylvaleric acid, phenylacetic acid, a salt thereof, or a mixture of two or more of the foregoing.

(146) In one embodiment, the composition contains an organic acid(s), and optionally further contains killed cells and/or cell debris. In one embodiment, the killed cells and/or cell debris are removed from a cell-killed whole broth to provide a composition that is free of these components.

(147) The fermentation broth formulations or cell compositions may further comprise a preservative and/or anti-microbial (e.g., bacteriostatic) agent, including, but not limited to, sorbitol, sodium chloride, potassium sorbate, and others known in the art.

(148) The cell-killed whole broth or composition may comprise the unfractionated contents of the fermentation materials derived at the end of the fermentation. Typically, the cell-killed whole broth or composition comprises the spent culture medium and cell debris present after the microbial cells (e.g., filamentous fungal cells) are grown to saturation, incubated under carbon-limiting conditions to allow protein synthesis. In some embodiments, the cell-killed whole broth or composition comprises the spent cell culture medium, extracellular enzymes, and killed filamentous fungal cells. In some embodiments, the microbial cells present in the cell-killed whole broth or composition may be permeabilized and/or lysed using methods known in the art.

(149) A whole broth or cell composition as described herein is typically a liquid, but may comprise insoluble components, such as killed cells, cell debris, culture media components, and/or insoluble enzyme(s). In some embodiments, insoluble components may be removed to provide a clarified liquid composition.

(150) The whole broth formulations and cell compositions of the present invention may be produced by a method described in WO 90/15861 or WO 2010/096673.

(151) Enzyme Compositions

(152) The present invention also relates to compositions comprising a variant according to the invention. Thus, the invention relates to a composition comprising a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14.

(153) In one embodiment, the composition comprises a variant comprising a) a deletion and/or a substitution at two or more positions corresponding to positions R181, G182, D183, and G184 of the amino acid sequence as set forth in SEQ ID NOs: 7 or 10, and b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109,

F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, T173, A174, A186, E190, T193, N195, A204, V206, P211, V213, I214, L217, A225, L235, V238, K242, S244, M246, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, G337, Q345, G346, T355, S376, D377, S381, Y382, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14. Preferably, the compositions are enriched in such a variant. The term “enriched” as used herein, refers to that the alpha-amylase activity of the composition has been increased, e.g., with an enrichment factor of at least 1.1.

(154) The compositions may comprise a variant of the present invention as the major enzymatic component, e.g., a mono-component composition. In another embodiment, the composition further comprises at least one further active component.

(155) The term “active component” as used herein, refers to any biological or non-biological molecule which in itself is active. For example, an active component is an enzyme.

(156) Thus, in one embodiment, the further active component is an enzyme, such as a protease, lipase, cellulose, pectate lyase and mannanase. Thus, the compositions may comprise multiple enzymatic activities, such as one or more (e.g., several) enzymes selected from the group consisting of hydrolase, isomerase, ligase, lyase, oxidoreductase, or transferase, e.g., an alpha-galactosidase, alpha-glucosidase, aminopeptidase, amylase, beta-galactosidase, beta-glucosidase, beta-xylosidase, carbohydrase, carboxypeptidase, catalase, cellobiohydrolase, cellulase, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, endoglucanase, esterase, glucoamylase, invertase, laccase, lipase, mannosidase, mutanase, oxidase, pectinolytic enzyme, peroxidase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, or xylanase.

(157) In one embodiment, the composition is a liquid laundry or liquid dish wash composition, such as an Automatic Dish Wash (ADW) liquid detergent composition, or a powder laundry, such as a soap bar, or powder dish wash composition, such as an ADW unit dose detergent composition and such as a Hand Dish Wash (HDW) detergent composition.

(158) The choice of additional components is within the skills of the skilled person in the art and includes conventional ingredients, including the exemplary non-limiting components set forth below. The choice of components may include, for fabric care, the consideration of the type of fabric to be cleaned, the type and/or degree of soiling, the temperature at which cleaning is to take place, and the formulation of the detergent product. Although components mentioned below are categorized by general header according to a particular functionality, this is not to be construed as a limitation, as a component may comprise additional functionalities as will be appreciated by the skilled artisan.

(159) In one embodiment of the present invention, the variant of the present invention may be added to a detergent composition in an amount corresponding to 0.001-100 mg of protein, such as 0.01-100 mg of protein, preferably 0.005-50 mg of protein, more preferably 0.01-25 mg of protein, even more preferably 0.05-10 mg of protein, most preferably 0.05-5 mg of protein, and even most preferably 0.01-1 mg of protein per liter of wash liquor. The term “protein” in this context is contemplated to be understood to include a variant according to the present invention.

(160) A composition for use in automatic dish wash (ADW), for example, may include 0.0001%-50%, such as 0.001%-20%, such as 0.01%-10%, such as 0.05-5% of enzyme protein by weight of the composition.

(161) A composition for use in laundry granulation, for example, may include 0.0001%-50%, such as 0.001%-20%, such as 0.01%-10%, such as 0.05%-5% of enzyme protein by weight of the composition.

(162) A composition for use in laundry liquid, for example, may include 0.0001%-10%, such as 0.001-7%, such as 0.1%-5% of enzyme protein by weight of the composition.

(163) The variants of the invention as well as the further active components, such as additional enzymes, may be stabilized using conventional stabilizing agents, e.g., a polyol such as propylene glycol or glycerol, a sugar or sugar alcohol, lactic acid, boric acid, or a boric acid derivative, e.g., an aromatic borate ester, or a phenyl boronic acid derivative such as 4-formylphenyl boronic acid, and the composition may be formulated as described in, for example, WO92/19709 and WO92/19708.

(164) In certain markets different wash conditions and, as such, different types of detergents are used. This is disclosed in e.g. EP 1 025 240. For example, In Asia (Japan) a low detergent concentration system is used, while the United States uses a medium detergent concentration system, and Europe uses a high detergent concentration system.

(165) A low detergent concentration system includes detergents where less than about 800 ppm of detergent components are present in the wash water. Japanese detergents are typically considered low detergent concentration system as they have approximately 667 ppm of detergent components present in the wash water.

(166) A medium detergent concentration includes detergents where between about 800 ppm and about 2000 ppm of detergent components are present in the wash water. North American detergents are generally considered to be medium detergent concentration systems as they have approximately 975 ppm of detergent components present in the wash water.

(167) A high detergent concentration system includes detergents where greater than about 2000 ppm of detergent components are present in the wash water. European detergents are generally considered to be high detergent concentration systems as they have approximately 4500-5000 ppm of detergent components in the wash water.

(168) Latin American detergents are generally high suds phosphate builder detergents and the range of detergents used in Latin America can fall in both the medium and high detergent concentrations as they range from 1500 ppm to 6000 ppm of detergent components in the wash water. Such detergent compositions are all embodiments of the invention.

(169) A variant of the present invention may also be incorporated in the detergent formulations disclosed in WO97/07202, which is hereby incorporated by reference.

(170) Examples are given herein of preferred uses of the compositions of the present invention. The dosage of the composition and other conditions under which the composition is used may be determined on the basis of methods known in the art.

(171) In particular, a composition according to the present invention further comprises a chelator.

(172) The term “chelator” as used herein, refers to chemicals which form molecules with certain metal ions, inactivating the ions so that they cannot react with other elements. Thus, a chelator may be defined as a binding agent that suppresses chemical activity by forming chelates.

(173) Chelation is the formation or presence of two or more separate bindings between a ligand and a single central atom. The ligand may be any organic compound, a silicate or a phosphate. In the present context the term “chelating agents” comprises chelants, chelating agent, chelating agents, complexing agents, or sequestering agents that forms water-soluble complexes with metal ions such as calcium and magnesium. The chelate effect describes the enhanced affinity of chelating ligands for a metal ion compared to the affinity of a collection of similar nonchelating ligands for the same metal. Chelating agents having binding capacity with metal ions, in particular calcium (Ca2+) ions, and has been used widely in detergents and compositions in general for wash, such as laundry or dish wash. Chelating agents have however shown themselves to inhibit enzymatic activity. The term chelating agent is used in the present application interchangeably with “complexing agent” or “chelating agent” or “chelant”.

(174) Since most alpha-amylases are calcium sensitive the presence of chelating agents these may impair the enzyme activity. The calcium sensitivity of alpha-amylases can be determined by incubating a given alpha-amylase in the presence of a strong chelating agent and analyze the impact of this incubation on the activity of the alpha-amylase in question. A calcium sensitive alpha-amylase will lose a major part or all of its activity during the incubation. Chelating agent may be present in the composition in an amount from 0.0001 wt % to 20 wt %, preferably from 0.01 to 10 wt %, more preferably from 0.1 to 5 wt %.

(175) Non-limiting examples of chelating agents are; EDTA, DTMPA, HEDP, and citrate. Thus, in one embodiment, the composition comprises a

variant according to the invention and a chelating agent, such as EDTA, DTMPA, HEDP or citrate.
(176) The term "EDTA" as used herein, refers to ethylene-diamine-tetra-acetic acid which falls under the definition of "strong chelating agents".
(177) The term "DTMPA" as used herein, refers to diethylenetriamine penta(methylene phosphonic acid). DTMPA can inhibit the scale formation of carbonate, sulfate and phosphate.
(178) The term "HEDP" as used herein, refers to hydroxy-ethane diphosphonic acid, which falls under the definition of "strong chelating agents".
(179) The chelate effect or the chelating effect describes the enhanced affinity of chelating ligands for a metal ion compared to the affinity of a collection of similar nonchelating ligands for the same metal. However, the strength of this chelate effect can be determined by various types of assays or measure methods thereby differentiating or ranking the chelating agents according to their chelating effect (or strength).
(180) In an assay the chelating agents may be characterized by their ability to reduce the concentration of free calcium ions (Ca2+) from 2.0 mM to 0.10 mM or less at pH 8.0, e.g. by using a test based on the method described by M. K. Nagarajan et al., JAOCS, Vol. 61, no. 9 (September 1984), pp. 1475-1478.
(181) For reference, a chelator having the same ability to reduce the concentration of free calcium ions (Ca2+) from 2.0 mM to 0.10 mM at pH as EDTA at equal concentrations of the chelator are said to be strong chelators.
(182) The composition of the present invention may be in any convenient form, e.g., a bar, a homogenous tablet, a tablet having two or more layers, a pouch having one or more compartments, a regular or compact powder, a granule, a paste, a gel, or a regular, compact or concentrated liquid. There are a number of detergent formulation forms such as layers (same or different phases), pouches, as well as forms for machine dosing unit.
(183) Pouches can be configured as single or multicompartments. It can be of any form, shape and material which is suitable for hold the composition, e.g. without allowing the release of the composition from the pouch prior to water contact. The pouch is made from water soluble film which encloses an inner volume. Said inner volume can be divided into compartments of the pouch. Preferred films are polymeric materials preferably polymers which are formed into a film or sheet. Preferred polymers, copolymers or derivatives thereof are selected polyacrylates, and water soluble acrylate copolymers, methyl cellulose, carboxy methyl cellulose, sodium dextrin, ethyl cellulose, hydroxyethyl cellulose, hydroxypropyl methyl cellulose, malto dextrin, poly methacrylates, most preferably polyvinyl alcohol copolymers and, hydroxypropyl methyl cellulose (HPMC). Preferably the level of polymer in the film for example PVA is at least about 60%. Preferred average molecular weight will typically be about 20,000 to about 150,000. Films can also be of blend compositions comprising hydrolytically degradable and water soluble polymer blends such as polyactide and polyvinyl alcohol (known under the Trade reference M8630 as sold by Chris Craft In. Prod. Of Gary, Ind., US) plus plasticisers like glycerol, ethylene glycerol, Propylene glycol, sorbitol and mixtures thereof. The pouches can comprise a solid laundry cleaning composition or part components and/or a liquid cleaning composition or part components separated by the water soluble film. The compartment for liquid components can be different in composition than compartments containing solids. Ref: (US2009/0011970 A1).
(184) Detergent ingredients may be separated physically from each other by compartments in water dissolvable pouches or in different layers of tablets. Thereby negative storage interaction between components may be avoided. Different dissolution profiles of each of the compartments can also give rise to delayed dissolution of selected components in the wash solution.
(185) A liquid or gel detergent, which is not unit dosed, may be aqueous, typically containing at least 20% by weight and up to 95% water, such as up to about 70% water, up to about 65% water, up to about 55% water, up to about 45% water, up to about 35% water. Other types of liquids, including without limitation, alkanols, amines, diols, ethers and polyols may be included in an aqueous liquid or gel. An aqueous liquid or gel detergent may contain from 0-30% organic solvent. A liquid or gel detergent may be non-aqueous.
(186) Another form of composition is in the form of a soap bar, such as a laundry soap bar, and may be used for hand washing laundry, fabrics and/or textiles. The term "soap bar" as used herein, refers to includes laundry bars, soap bars, combo bars, syndet bars and detergent bars. The types of bar usually differ in the type of surfactant they contain, and the term laundry soap bar includes those containing soaps from fatty acids and/or synthetic soaps. The laundry soap bar has a physical form which is solid and not a liquid, gel or a powder at room temperature. The term "solid" as used herein, refers to a physical form which does not significantly change over time, i.e. if a solid object (e.g. laundry soap bar) is placed inside a container, the solid object does not change to fill the container it is placed in. The bar is a solid typically in bar form but can be in other solid shapes such as round or oval.
(187) The soap bar may also comprise complexing agents like EDTA and HEDP, perfumes and/or different type of fillers, surfactants e.g. anionic synthetic surfactants, builders, polymeric soil release agents, detergent chelators, stabilizing agents, fillers, dyes, colorants, dye transfer inhibitors, alkoxylated polycarbonates, suds suppressers, structurants, binders, leaching agents, bleaching activators, clay soil removal agents, anti-redeposition agents, polymeric dispersing agents, brighteners, fabric softeners, perfumes and/or other compounds known in the art.
(188) The soap bar may be processed in conventional laundry soap bar making equipment such as but not limited to: mixers, plodders, e.g. a two stage vacuum plodder, extruders, cutters, logo-stampers, cooling tunnels and wrappers. The invention is not limited to preparing the soap bars by any single method. The premix of the invention may be added to the soap at different stages of the process. For example, the premix comprising a soap, an enzyme, optionally one or more additional enzymes, a protease inhibitor, and a salt of a monovalent cation and an organic anion may be prepared and the mixture may then plodded. The enzyme and optional additional enzymes may be added at the same time as an enzyme inhibitor, e.g. a protease inhibitor, for example in liquid form. Besides the mixing step and the plodding step, the process may further comprise the steps of milling, extruding, cutting, stamping, cooling and/or wrapping.
(189) Uses
(190) The present invention further relates to the use of a variant according to the present invention in a cleaning process such as laundry or hard surface cleaning including automated dish wash and industrial cleaning. The soils and stains that are important for cleaning are composed of many different substances, and a range of different enzymes, all with different substrate specificities, have been developed for use in detergents both in relation to laundry and hard surface cleaning, such as dishwashing. These enzymes are considered to provide an enzyme detergency benefit, since they specifically improve stain removal in the cleaning process that they are used in, compared to the same process without enzymes. Stain removing enzymes that are known in the art include enzymes such as proteases, amylases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xanthanases, peroxidaes, haloperoxygenases, catalases and mannanases.
(191) In one embodiment, the invention relates the use of variants of the present invention in detergent compositions, for use in cleaning hard-surfaces, such as dish wash, or in laundering or for stain removal. In another embodiment, the invention relates to the use of an alpha-amylase variant according to the invention in a cleaning process such as laundry or hard surface cleaning including, but not limited to, dish wash and industrial cleaning. Thus, in one embodiment, the invention relates to the use of a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14 in a cleaning process such as laundry or hard surface cleaning including dish wash and industrial cleaning.
(192) In a particular embodiment, the invention relates to the use of a variant comprising a) a substitution and/or deletion of two, three or four positions in the parent alpha-amylase said positions corresponding to positions 181, G182, D183, and G184 of the mature polypeptide of SEQ ID

NO: 4; and b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, in a cleaning process such as laundry or hard surface cleaning including dish wash and industrial cleaning.

(193) In one embodiment of the invention relates the use of a composition according to the invention comprising a variant of the present invention together with one or more surfactants and optionally one or more detergent components, selected from the list comprising of hydrotropes, builders and co-builders, bleaching systems, polymers, fabric hueing agents and adjunct materials, or any mixture thereof in detergent compositions and in detergent applications.

(194) A further embodiment is the use of the composition according to the invention comprising a variant of the present invention together with one or more surfactants, and one or more additional enzymes selected from the group comprising of proteases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xanthanases, peroxidaes, haloperoxygenases, catalases and mannanases, or any mixture thereof in detergent compositions and in detergent applications.

(195) In another aspect, the invention relates to a laundering process which may be for household laundering as well as industrial laundering. Furthermore, the invention relates to a process for the laundering of textiles (e.g. fabrics, garments, cloths etc.) where the process comprises treating the textile with a washing solution containing a detergent composition and an alpha-amylase of the present invention. The laundering can for example be carried out using a household or an industrial washing machine or be carried out by hand using a detergent composition containing a glucoamylase of the invention.

(196) In another aspect, the invention relates to a dish wash process which may be for household dish wash as well as industrial dish wash. The term “dish wash” as used herein, refers to both manual dish wash and automated dish wash. Furthermore, the invention relates to a process for the washing of hard surfaces (e.g. cutlery such as knives, forks, spoons; crockery such as plates, glasses, bowls; and pans) where the process comprises treating the hard surface with a washing solution containing a detergent composition and an alpha-amylase variant of the present invention. The hard surface washing can for example be carried out using a household or an industrial dishwasher or be carried out by hand using a detergent composition containing an alpha-amylase of the invention, optionally together with one or more further enzymes selected from the group comprising of proteases, amylases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xanthanases, peroxidaes, haloperoxygenases, catalases, mannanases, or any mixture thereof.

(197) In a further aspect, the invention relates to a method for removing a stain from a surface comprising contacting the surface with a composition comprising an alpha-amylase of the present invention together with one or more surfactants and optionally one or more detergent components selected from the list comprising of hydrotropes, builders and co-builders, bleaching systems, polymers, fabric hueing agents and adjunct materials, or any mixture thereof in detergent compositions and in detergent applications. A further aspect is a method for removing a stain from a surface comprising contacting the surface with a composition comprising an alpha-amylase variant of the present invention together with one or more surfactants, one or more additional enzymes selected from the group comprising of proteases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xanthanases, peroxidaes, haloperoxygenases, catalases and mannanases, or any mixture thereof in detergent compositions and in detergent applications.

(198) The present invention is further described by the following examples that should not be construed as limiting the scope of the invention.

EXAMPLES

Example 1a: Construction of the Parent Polypeptide as Set Forth in SEQ ID NO: 13

(199) Construction of a hybrid between the A and B domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 10 (the first parent polypeptide) and the C domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 11 (the second parent polypeptide).

(200) Based on 3D structural alignment of the two amylases, amino acid no 1 to amino acid no 397 of the first parent polypeptide are defined as domain A and B, and amino acid 398 to amino acid no. 485 of the second parent polypeptide are defined as the C domain. Synthetic DNA fragments coding for part of the A domain from the first parent polypeptide and the C domain from the second parent polypeptide were designed and purchased from an external vendor. In addition to combining fragments of the two different amylases, the following stabilizing substitutions were designed into the synthetic amylase gene: G182* and D183*.

(201) The new gene consisting of a gene fragment encoding the A and B domain of the first parent polypeptide and the C domain of the second parent polypeptide coded by the synthetic gene fragments was constructed by triple SOE (splicing by overlap extension) PCR method. The two fragments were assembled by splicing by overlap extension polymerase chain reaction and transformed into a competent *B. subtilis* host for overexpression. The resulting amylase consisting of the A and B domain from the first parent polypeptide and the C domain from the second parent polypeptide, and having the deletions of G182* and D183* is the polypeptide as set forth in SEQ ID NO: 13.

Example 1b: Construction of the Parent Polypeptide as Set Forth in SEQ ID NO: 14

(202) Construction of a hybrid between the A and B domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 10 (the first parent polypeptide) and the C domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 12 (the third parent polypeptide).

(203) Based on 3D structural alignment of the two amylases, amino acid no 1 to amino acid no 397 of the first parent polypeptide are defined as domain A and B, and amino acid 398 to amino acid no. 485 of the third parent polypeptide are defined as the C domain. Synthetic DNA fragments coding for part of the A domain from the first parent polypeptide and the C domain from the second parent polypeptide were designed and purchased from an external vendor. In addition to combining fragments of the two different amylases, the following stabilizing substitutions were designed into the synthetic amylase gene: G182* and D183*.

(204) The new gene consisting of a gene fragment encoding the A and B domain of the first parent polypeptide and the C domain of the second parent polypeptide coded by the synthetic gene fragments was constructed by triple SOE (splicing by overlap extension) PCR method. A DNA fragment was amplified from an expression clone of the first amylase. The fragments were finally assembled by triple SOE and the derived PCR fragment was transformed into a suitable *B. subtilis* host and the gene integrated into the *B. subtilis* chromosome by homologous recombination into the pectate lyase (pel) locus. The gene coding for chloramphenicol acetyltransferase was used as maker (as described in (Diderichsen et al., 1993, Plasmid 30: 312-315). Chloramphenicol resistant clones were analyzed by DNA sequencing to verify the correct DNA sequence of the construct. The resulting amylase consisting of the A and B domain from the first parent polypeptide and the C domain from the second parent polypeptide, and having the deletions of G182* and D183* is the polypeptide as set forth in SEQ ID NO: 14.

Example 2: Generation of Variants According to the Invention

(205) The variants of the present invention have been generated by site-directed mutagenesis. Genomic DNA prepared from the organism containing amylase gene at the Pel locus was used as template for generating the site-directed mutants.

(206) Mutagenic forward primer and PnMi4490 (CAATCCAAGAGAACCCTGATACGGATG—SEQ ID NO: 16) reverse primer was used to generate a˜3.8 kb fragment. This fragment was used as a megaprimer along with PnMi4491 (CGGAACGCCTGGCTGACAACACG—SEQ ID NO: 17) forward primer to get 6 kb insertion cassette. To enable integration in the Pel locus by double cross-over upon transformation, along with the

amylase and cat genes, the cassette contained upstream and downstream Pel sequences at the ends. Selection was done on LB Agar containing chloramphenicol and the mutation was confirmed by DNA sequencing of amylase gene.
(207) A library of variants were generated where different combinations of alterations in the following positions were made; H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, A60, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, G196, A204, V206, P211, V213, I214, L217, A225, L228, L235, V238, K242, S244, M246, L2502, G255, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, G337, Q345, G346, T355, S376, D377, D381, Y382, Q385, K391, and Q395 of SEQ ID NOs: 7 or 10; A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; and 1405, A421, A422, A428, G448, D476, and G477 of SEQ ID NO: 14.
(208) The generated variants are listed in the following examples.
Example 3: Wash Performance of Generated Variants Using Automatic Mechanical Stress Assay (AMSA)
(209) In order to assess the wash performance of the variants of the present invention in a detergent base composition, washing experiments may be performed using Automatic Mechanical Stress Assay (AMSA). With the AMSA test the wash performance of a large quantity of small volume enzyme-detergent solutions can be examined. The AMSA plate has a number of slots for test solutions and a lid firmly squeezing the textile swatch to be washed against all the slot openings. During the washing time, the plate, test solutions, textile and lid were vigorously shaken to bring the test solution in contact with the textile and apply mechanical stress in a regular, periodic oscillating manner. For further description see WO 02/42740, especially the paragraph Special method embodiments at page 23-24.
(210) General Wash Performance Description
(211) A test solution comprising water (6°dH), 0.79 g/L detergent, e.g. Model detergent J as described below, and the enzyme of the invention at concentration of 0, 0.05 or 0.2 mg enzyme protein/L, is prepared. Fabrics stained with starch (CS-28 from Center For Test materials BV, P.O. Box 120, 3133 KT, Vlaardingen, The Netherlands) was added and washed for 20 minutes at 15° C. and 30° C., or alternatively 20 minutes at 15° C. and 40° C. as specified below. After thorough rinse underrunning tap water and drying in the dark, the light intensity values of the stained fabrics were subsequently measured as a measure for wash performance. The test with 0 mg enzyme protein/L was used as a blank and corresponded to the contribution from the detergent. Preferably mechanical action is applied during the wash step, e.g. in the form of shaking, rotating or stirring the wash solution with the fabrics. The AMSA wash performance experiments was conducted under the experimental conditions specified below:
(212) TABLE-US-00004 TABLE A Experimental condition Detergent Liquid Model detergent J (see Table B) Detergent dosage 0.79 g/L Test solution volume 160 micro L pH As is Wash time 20 minutes Temperature 15° C. or 30° C. Water hardness 6° dH Enzyme concentration 0.2 mg enzyme protein/L and in test 0.05 mg enzyme protein/L Test material CS-28 (Rice starch cotton)
(213) TABLE-US-00005 TABLE B Model detergent J Content of % active compound component Compound (% w/w) (% w/w) LAS 5.15 5.00 linear alkylbenzene sulfonates AS 5.00 4.50 alkylbenzene sulfonates AEOS 14.18 10.00 alkyl ethoxy sulfate Coco fatty acid 1.00 1.00 AEO 5.00 5.00 alkyl ethoxylate MEA 0.30 0.30 monoethanolamine MPG 3.00 3.00 monopropylene glycol Ethanol 1.50 1.35 DTPA (as Na5 salt) 0.25 0.10 pentasodium diethylenetriaminepenta- acetic acid Sodium citrate 4.00 4.00 Sodium formate 1.00 1.00 Sodium hydroxide 0.66 0.66 H.sub.2O, ion exchanged 58.95 58.95
Water hardness was adjusted to 6°dH by addition of CaCl.sub.2), MgCl.sub.2, and NaHCO.sub.3(Ca.sup.2+:Mg.sup.2+:HCO.sub.3−=2:1:4.5) to the test system. After washing the textiles were flushed in tap water and dried.
(214) TABLE-US-00006 TABLE C Experimental condition Detergent Liquid Model detergent A (see Table D) Detergent dosage 3.33 g/L Test solution volume 160 micro L pH As is Wash time 20 minutes Temperature 15° C. or 40° C. Water hardness 15° dH Enzyme concentration 0.2 mg enzyme protein/L, in test 0.05 mg enzyme protein/L Test material CS-28 (Rice starch cotton)
(215) TABLE-US-00007 TABLE D Model detergent A Content of % active compound component Compound (% w/w) (% w/w) LAS 12.00 11.60 (linear alkylbenzene sulfonates) AEOS 17.63 4.90 (alkyl ethoxy sulfate), SLES (sodium lauryl ether sulfate) Soy fatty acid 2.75 2.48 Coco fatty acid 2.75 2.80 AEO (alkyl ethoxylate) 11.00 11.00 Sodium hydroxide 1.75 1.80 Ethanol/Propan-2-ol 3.00 2.70/0.30 MPG 6.00 6.00 monopropylene glycol Glycerol 1.71 1.70 TEA (triethanolamine) 3.33 3.30 Sodium formate 1.00 1.00 Sodium citrate 2.00 2.00 DTMPA 0.48 0.20 (diethylenetriaminepenta- acetic acid) PCA 0.46 0.18 polycarboxylic acid type polymer Phenoxy ethanol 0.50 0.50 H.sub.2O, ion exchanged 33.64 33.64
Water hardness was adjusted to 15°dH by addition of CaCl.sub.2, MgCl.sub.2, and NaHCO.sub.3 (Ca.sup.2+:Mg.sup.2+:HCO.sub.3−=4:1:7.5) to the test system. After washing the textiles were flushed in tap water and dried.
(216) The wash performance was measured as the brightness expressed as the intensity of the light reflected from the sample when illuminated with white light. When the sample was stained the intensity of the reflected light was lower, than that of a clean sample. Therefore the intensity of the reflected light can be used to measure wash performance.
(217) Color measurements were made with a professional flatbed scanner (EPSON Expression 10000XL, EPSON) used to capture an image of the washed textile.
(218) To extract a value for the light intensity from the scanned images, 48.fwdarw.24 Bit Color pixel values from the image were converted into values for red, green and blue (RGB). The intensity value (Int) is calculated by adding the RGB values together as vectors and then taking the length of the resulting vector:
(219) Int = $\sqrt{r^2 + g^2 + b^2}$
(220) The term “improved wash performance” of the present experiment is defined as displaying an alteration of the wash performance of a variant of the present invention relative to the wash performance of the polypeptide having an amino acid sequence as set forth in SEQ ID NO: 15, The alteration may e.g. be seen as increased stain removal. Improved wash performance was determined as described above. The wash performance was considered to be improved if the Improvement Factor (IF) is at least 1.0, preferably at least 1.05 in one or more of the conditions listed above; i.e. either in Model detergent A at 15° C. or 40° C., where the variant concentration was 0.05 or 0.2 mg/L or in Model detergent J at 15° C. or 30° C., where the variant concentration was 0.05 or 0.2 mg/L. The wash conditions were as described above in Tables A and C.
(221) The terms “Delta intensity” or “Delta intensity value” as used herein refers to the result of an intensity measurement of a test material, e.g. a swatch CS-28 (Center For Testmaterials BV, P.O. Box 120, 3133 KT Vlaardingen, the Netherlands). The swatch was measured with a portion of the swatch, washed under identical conditions, as background. The delta intensity is the intensity value of the test material washed with amylase subtracting the intensity value of the test material washed without amylase.
(222) The wash performance of the variants according to the invention obtained by AMSA are the following;
(223) TABLE-US-00008 A-15- A-15- A-40- A-40- J-15- J-15- J-30- J-30- 0.05 0.2 0.05 0.2 0.05 0.2 0.05 0.2 Modifications in SEQ ID NO: 13 mg/L mg/L mg/L mg/L mg/L mg/L mg/L mg/L Reference 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 G182* + D183* + N195F 1.0 1.3 1.1 1.0 1.5 1.3 1.0 1.0 H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.1 1.9 1.2 1.2 2.6 2.0 1.5 1.3 N195F + V206L + K391A + G476K H1* + N54S + V56T + R87S + G109A + A174S + G182* + 2.8 2.2 1.3 1.1 2.6 2.4 1.9 1.5 D183* + N195F + V206L + K391A + G476K H1* + T40G + N54S + V56T + G109A + A174S + G182* + 2.2 1.7 1.3 1.1 2.5 2.2 1.9 1.6 D183* + N195F + V206L + K391A + G476K H1* + T51K + N54S + V56T + G109A + A174S + G182* + 2.7 2.1 1.3 1.1 3.2 2.3 2.0 1.5 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + A174S + G182* + 2.6 2.2 1.2 1.0 2.6 2.5 1.7 1.5 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72H + G109A + A174S + G182* + 2.5 2.4 1.4 1.1 2.6 2.3 1.7 1.6 D183* + N195F + V206L + K391A + G476K H1* + A37H + N54S + V56T + G109A + A174S + G182* + 2.8 2.7 1.4 1.2 2.8 2.8 1.9 1.6 D183* + N195F + V206L + K391A + G476K H1* + A37M + N54S + V56T + G109A + A174S + G182* + 2.0 2.3 1.4 1.2 2.7 2.5

2.1 1.6 D183* + N195F + V206L + K391A + G476K H1* + A37V + N54S + V56T + G109A + A174S + G182* + 3.0 2.6 1.4 1.2 3.1 2.5 2.2 1.7 D183* + N195F + V206L + K391A + G476K H1* + A37S + N54S + V56T + G109A + A174S + G182* + 2.9 2.5 1.2 1.2 2.9 2.3 1.5 1.3 D183* + N195F + V206L + K391A + G476K H1* + A37Y + N54S + V56T + G109A + A174S + G182* + 2.6 2.0 1.3 1.2 2.5 2.1 1.5 1.3 D183* + N195F + V206L + K391A + G476K H1* + A37R + N54S + V56T + G109A + A174S + G182* + 2.2 1.9 1.3 1.2 2.5 2.1 1.4 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + F113W + A174S + G182* + 2.8 2.2 1.3 1.2 2.7 2.1 1.5 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + F113S + A174S + G182* + 2.3 1.9 1.3 1.1 2.4 1.7 1.3 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + F113N + A174S + G182* + 2.0 1.9 1.2 1.2 2.3 2.1 1.4 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + F113Y + A174S + G182* + 1.9 2.1 1.2 1.2 3.1 2.3 1.5 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + F113R + A174S + G182* + 1.0 1.4 1.2 1.2 2.0 1.8 1.4 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q118N + A174S + G182* + 1.9 1.8 1.3 1.2 2.9 2.3 1.5 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + R116Q + A174S + G182* + 2.6 2.0 1.2 1.2 3.5 2.6 1.7 1.4 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + R116V + A174S + G182* + 2.1 2.0 1.1 1.1 2.7 2.3 1.6 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + R116K + A174S + G182* + 2.2 2.1 1.2 1.1 2.4 2.2 1.4 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + R116W + A174S + G182* + 1.8 2.0 0.7 0.9 1.1 1.9 1.0 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + R116L + A174S + G182* + 1.8 1.8 1.2 1.1 1.9 2.0 1.3 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + G142T + A174S + G182* + 1.6 1.7 1.2 1.1 1.9 2.0 1.3 1.1 D183* + N195F + V206L + K391A + G476K 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1.2 1.2 N195F + V206L + K391A + Q449T + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 1.7 1.3 1.1 2.2 2.0 1.3 1.2 N195F + V206L + K391A + P473K + G476K H1* + N54S + V56T + K72S + G109A + A174S + G182* + 2.5 2.3 1.2 1.1 3.7 2.2 1.5 1.4 D183* + N195F + V206L + K391A + G476K H1* + T40K + N54S + V56T + G109A + A174S + G182* + 2.3 2.1 1.2 1.1 2.0 1.7 1.5 1.3 D183* + N195F + V206L + G346T + K391A + G476K H1* + G50A + N54S + V56T + G109A ++ Q172N + 2.4 2.1 1.2 1.2 2.0 1.9 1.4 1.3 A174S + G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.9 1.7 1.3 1.0 2.4 1.6 1.5 1.3 N195F + V206L + K391A + G476K + K484A H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 1.8 1.2 1.0 2.2 1.8 1.4 1.2 N195F + V206L + K391A + G476K + K484G H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 1.7 1.1 1.0 2.4 2.0 1.4 1.3 N195F + V206L + K391A + G476K + K484P H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.9 1.4 1.2 1.1 2.5 1.7 1.3 1.2 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D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.1 2.0 1.2 1.2 2.3 1.7 1.2 1.2 N195F + V206L + G346T + K391A + G476K + G477A H1* + N54S + V56T + G109AG255A ++ A174S + G182* + 2.1 2.0 1.2 1.2 2.3 1.8 1.2 1.2 D183* + N195F + V206L + K391A + G476K H1* + G50A + N54S + V56T + G109A + F113L + R116L + 2.3 2.2 1.2 1.1 2.6 2.2 1.3 1.2 A174S + G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + W167F + A174S + G182* + 2.0 1.9 1.1 1.0 2.3 2.0 1.4 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + L173V + A174S + G182* + 1.8 1.9 1.0 1.0 2.2 1.8 1.4 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + W167F + A174S + 2.1 1.7 1.1 1.1 2.9 2.0 1.4 1.3 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 1.4 1.0 1.1 2.4 1.9 1.4 1.3 N195F + V206L + K391A + G476K + G477Q H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.6 1.5 1.0 1.0 2.2 1.7 1.3 1.2 N195F + V206L + Q345D + K391A + G476K + G477Q H1* + G50A + N54S + V56T + G109A + Q172D + A174S + 1.9 1.9 0.9 1.1 2.1 1.9 1.6 1.3 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.9 1.9 0.7 1.0 2.2 2.0 1.4 1.3 N195F + V206L + V264I + K391A + G423H + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.1 2.0 0.8 1.0 2.1 2.1 1.5 1.3 N195F + V206L + R320V + K391A + G476Y H1* + T5K + N54S + V56T + G109A + A174S + G182* + 1.7 1.8 0.9 1.0 1.8 1.7 1.4 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + R116M + A174S + G182* + 2.9 2.2 1.2 1.1 2.8 2.1 1.2 1.1 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q172Y + A174S + G182* + 2.1 1.8 1.1 1.1 1.9 1.8 1.4 1.1 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.7 1.1 1.1 1.9 1.7 1.3 1.2 N195F + V206L + I214L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.2 1.9 1.2 1.2 2.2 2.2 1.3 1.2 N195F + V206L + L217V + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.2 1.7 1.2 1.2 2.3 2.1 1.4 1.2 N195F + V206L + L217H + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.1 2.0 1.1 1.1 2.3 2.1 1.2 1.2 N195F + V206L + P211D + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.1 1.9 1.1 1.2 2.3 1.9 1.3 1.3 N195F + V206L + K391A + K391Q + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.9 1.1 1.1 2.3 1.9 1.4 1.2 N195F + V206L + K391A + T444H + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.2 1.7 1.1 1.0 2.3 1.8 1.2 1.2 N195F + V206L + K391A + T444V + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.6 1.2 1.1 1.9 1.8 1.3 1.2 N195F + V206L + K391A + T444R + G476K H1* + G50A + N54S + V56T + G109A + L173V + A174S + 2.4 2.1 1.2 1.2 3.2 2.2 1.6 1.4 G182* + D183* + N195F + A204V + V206L + K391A + G476K H1* + G50A + N54S + V56T + G109A + F113L + W167F + 1.9 1.7 1.0 1.1 2.1 1.8 1.2 1.2 A174S + G182* + D183* + N195F + V206L + K391A + G476K H1* + G7K + N54S + V56T + G109A + A174S + G182* + 2.0 1.8 1.1 1.1 2.7 2.1 1.6 1.2 D183* + N195F + V206L + R320A + K391A + G476K H1* + N54S + V56T + G109A + Q118G + A174S + G182* + 1.7 1.8 1.1 1.1 2.4 1.6 1.5 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.6 1.6 1.0 1.1 1.6 1.6 1.3 1.2 N195F + V206L + S280K + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.4 1.6 1.1 1.1 1.8 1.6 1.4 1.2 N195F + V206L + N270G + K391A + G476K H1* + N54S + V56T + G109A + Q172D + A174S + G182* + 2.1 1.9 1.2 1.1 3.1 2.3 1.6 1.4 D183* + N195F + V206L + G346T + K391A + G476K + G477A H1* + N54S + V56T + G109A + Q118S + A174S + G182* + 2.1 1.9 1.1 1.1 1.8 1.5 1.4 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.9 1.2 1.2 2.3 1.7 1.5 1.3 N195F + V206L + R320A + K391A + G476K H1* + N54S + V56T + G109A + T134E + A174S + G182* + 2.4 2.1 1.2 1.1 2.8 2.4 1.4 1.4 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.8 1.0 1.0 1.8 1.8 1.2 1.2 N195F + V206L + G337H + Q385L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.8 1.1 1.1 2.2 1.9 1.3 1.2 N195F + V206L + Y382M + K391A + G476K H1* + N54S + V56T + K72R + G109A + A174S + G182* + 1.9 1.9 1.1 1.1 2.7 2.2 1.4 1.3 D183* + N195F + V206L + G255A + D377H + K391A + G476K H1* + N54S + V56T + K72R + G109A + T134E + A174S + 2.4 2.0 1.2 1.1 3.0 2.3 1.4 1.3 G182* + D183* + N195F + V206L + G255A + K391A + G476K H1* + N54S + V56T + K72R + M105F + G109A + A174S + 2.0 1.7 1.2 1.1 2.5 2.0 1.4 1.2 G182* + D183* + N195F + V206L + G255A + K391A + G476K H1* + N54S + V56T + M105F + G109A + T134E + A174S + 2.0 1.9 1.0 1.0 2.3 2.1 1.2 1.1 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + M105F + G109A + Q118F + T134E + 1.9 1.7 1.1 1.1 1.7 1.9 1.3 1.2 A174S + G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.9 1.1 1.0 2.3 2.0 1.4 1.3 N195F + V206L + R320A + S323N + D377H + K391A + G476K H1* + T51E + N54S + V56T + G109A + A174H + G182* + 2.3 1.9 1.2 1.1 2.6 2.2 1.6 1.4 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + L173A + A174S + G182* + 2.4 2.0 1.3 1.2 2.4 2.0 1.6 1.4 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.5 2.0 1.3 1.2 2.4 1.6 1.6 1.4 N195F + V206L + Q299V + K391A + G476K H1* + N54S + V56T + G109A + L173G + A174S + G182* + 2.2 2.0 1.3 1.2 2.3 1.9 1.7 1.4 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.1 2.0 1.3 1.2 2.4 2.0 1.5 1.3 N195F + V206L + T355L + K391A + G476K H1* + N54S +

V56T + G109A + A174S + G182* + D183* + 1.3 1.7 1.0 1.0 1.9 1.6 1.2 1.1 N195F + V206L + T355F + K391A + G476K H1* + G50A + N54S + V56T + G109A + A174S + G182* + 2.3 2.0 1.2 1.1 2.1 1.8 1.4 1.3 D183* + N195F + A204T + V206L + K391A + G476K H1* + G50A + N54S + V56T + G109A + A174E + G182* + 1.8 1.6 1.1 1.1 2.1 1.8 1.2 1.2 D183* + N195F + A204T + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + A174S + G182* + 2.2 1.8 1.2 1.1 2.7 2.1 1.4 1.2 D183* + N195F + V206L + R320A + S323N + K391A + G476K H1* + N54S + V56T + M105I + G109A + A174S + G182* + 1.8 1.8 1.1 1.1 1.5 1.5 1.3 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.8 1.1 1.0 1.8 1.6 1.3 1.2 N195F + V206L + M246L + K391A + G476K H1* + T51G + N54S + V56T + G109A + A174N + G182* + 1.6 1.5 1.0 1.0 2.1 1.7 1.3 1.1 D183* + N195F + V206L + K391A + G476K H1* + T51A + N54S + V56T + G109A + A174G + G182* + 1.4 1.5 0.9 0.9 1.6 1.5 1.3 1.0 D183* + N195F + V206L + K391A + G476K H1* + N54S 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H1* + Q11H + N54S + V56T + G109A + Q118R + T134E + 1.9 1.6 1.1 1.0 2.2 1.6 1.3 1.3 A174S + G184T + G182* + D183* + N195F + V206L + D377H + K391A + G476K H1* + N54S + V56T + K72R + G109A + T134E + A174S + 2.3 2.1 1.2 1.1 3.2 1.9 1.3 1.3 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + T134E + A174S + 2.2 2.0 1.1 1.0 3.8 1.9 1.3 1.2 G182* + D183* + N195F + V206L + D377H + K391A + G476K H1* + N54S + V56T + K72R + G109A + T134E + A174S + 2.2 2.0 1.2 1.1 3.9 2.4 1.4 1.3 G182* + D183* + G184T + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q118R + T134E + A174S + 2.0 1.9 1.0 1.2 2.5 2.0 1.2 1.1 G182* + D183* + N195F + V206L + G255A + K391A + G476K H1* + N54S + V56T + G109A + Q118R + A174S + G182* + 2.2 1.9 1.1 1.1 2.8 1.8 1.2 1.2 D183* + N195F + V206L + D377H + K391A + G476K H1* + N54S + V56T + G109A + T134E + A174S + G182* + 2.1 2.1 1.0 1.1 3.4 2.3 1.3 1.3 D183* + G184T + N195F + V206L + D377H + K391A + G476K H1* + N54S + V56T + G109A + Q172D + A174Q + G182* + 2.0 1.9 1.2 1.1 3.0 2.2 1.1 1.2 D183* + N195F + V206L + K391A + G476K + G477A H1* + N54S + V56T + K72A + G109A + W167F + A174S + 1.1 1.4 0.6 0.8 1.5 1.3 1.0 0.8 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + W167S + G182* + D183* + 1.4 1.3 0.8 1.0 1.5 1.6 1.3 1.1 N195F + A174S + V206L + K391A + G476K H1* + N54S + V56T + K72H + G109A + W167S + A174S + 1.4 1.4 0.5 1.0 1.0 1.4 0.8 1.1 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + W167T + A174S + 1.3 1.2 0.8 0.8 1.2 1.0 1.1 1.0 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72E + G109A + A174S + G182* + 1.3 1.4 0.8 0.9 1.4 1.3 0.9 0.9 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + W167L + A174S + 1.6 1.6 1.1 1.1 2.0 1.5 1.4 1.2 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + W167M + A174S + 1.5 1.6 0.8 1.0 1.5 1.3 1.2 1.2 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + W167H + A174S + 1.3 1.4 0.6 1.0 1.6 1.3 0.9 1.1 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + A174S + G182* + 2.0 1.9 1.2 1.1 2.5 2.0 1.4 1.3 D183* + G184T + N195F + V206L + G255A + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.2 1.9 1.1 1.1 2.3 1.9 1.5 1.4 G184T + N195F + V206L + D377H + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.1 2.0 1.2 1.0 2.1 1.7 1.3 1.3 N195F + V206L + G255A + D377H + K391A + G476K H1* + N54S + V56T + G109A + Q118R + A174S + G182* + 1.2 1.5 1.0 1.0 1.8 1.2 1.0 1.0 D183* + N195F + V206L + G255A + D377H + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.2 1.3 0.9 1.0 1.3 1.3 1.0 1.0 G184A + T193K + N195F + V206L + K391A + G476K H1* + N54S + V56T + M105F + G109A + A174S + G182* + 1.9 2.0 1.1 1.2 2.1 1.9 1.4 1.3 D183* + G184T + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + T134E + A174S + G182* + 1.7 2.1 1.1 1.1 2.1 2.0 1.3 1.3 D183* + N195F + V206L + R320A + S323N + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.8 1.2 1.1 2.1 1.9 1.4 1.3 N195F + V206L + V291G + D377H + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.9 1.1 1.1 2.1 1.9 1.3 1.3 N195F + V206L + V291A + F328L + D377H + K391A + G476K H1* + V56T + G109A + A174S + G182* + D183* + N195F + 1.3 1.8 1.1 1.1 1.5 1.5 1.3 1.2 V206L + D377H + K391A + G476K H1* + N54S + V56T + G109A + T134E + A174S + G182* + 2.4 2.2 1.1 1.1 3.0 2.4 1.5 1.3 D183* + G184T + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.8 1.2 1.1 2.1 1.9 1.4 1.2 N195F + V206L + V238A + D377H + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.8 1.1 1.1 2.5 1.9 1.3 1.2 N195F + V206L + V238T + D377H + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.9 1.6 1.1 1.1 2.3 1.8 1.2 1.2 N195F + V206L + V238G + D377H + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 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G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.3 1.5 1.0 1.0 1.7 1.6 1.2 1.2 N195F + V206L + G346D + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.9 1.7 1.1 1.2 2.5 1.7 1.4 1.3 N195F + V206L + D377A + K391A + G476K H1* + N54S + V56T + G109A + T134E + A174S + G182* + 2.2 2.1 1.2 1.1 2.3 2.1 1.1 1.2 D183* + N195F + V206L + D377H + K391A + G476K H1* + N54S + V56T + G109A + Q118R + T134E + A174S + 1.9 1.7 1.3 1.2 2.0 1.7 1.2 1.2 G182* + D183* + G184T + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + Q118R + T134E + 1.9 1.7 1.1 1.1 2.1 1.9 1.0 1.1 A174S + G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.6 1.2 1.1 2.3 1.9 1.2 1.1 N195F + V206L + G346P + K391A + G476K H1* + N54S + V56T + G109A + R116S + A174S + G182* + 2.0 1.8 1.2 1.1 2.1 1.9 1.1 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.8 1.2 1.1 2.0 2.0 1.2 1.2 N195F + V206L + 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A174S + G182* + D183* + N195F + V206L + G255A + K391A + Q395P + G476K H1* + N54S + V56T + K72R + G109A + T134E + W167F + 2.6 2.4 1.3 1.2 3.2 2.5 1.7 1.4 Q172N + A174S + G182* + D183* + N195F + V206L + G255A + A265G + K391A + T444Q + P473R + G476K H1* + N54S + V56T + K72R + G109A + G133Q + W167F + 2.4 2.2 1.3 1.2 3.2 2.4 1.6 1.4 Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + 2.9 2.3 1.3 1.2 3.7 2.8 1.8 1.4 W167F + Q172G + A174S + G182* + D183* + N195F + A204G + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + 2.0 2.0 1.3 1.2 2.7 2.1 1.5 1.3 W167F + Q172N + L173V + A174S + G182* + D183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K H1* + A37H + N54S + V56T + G109A + F113Q + W167F + 2.5 1.9 1.2 1.1 2.6 1.8 1.6 1.3 Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K H1* + A37H + N54S + V56T + G109A + F113Q + A174S + 1.9 1.7 1.3 1.2 2.8 2.0 1.3 1.3 G182* + D183* + N195F + V206L + 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N54S + V56T + K72R + G109A + A174S + G182* + 2.6 1.5 1.2 1.0 3.1 1.8 1.6 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 2.5 1.8 1.2 1.0 2.2 1.7 1.6 1.2 G182* + D183* + G184T + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 2.4 1.6 1.1 1.0 2.6 1.9 1.5 1.2 G182* + D183* + N195F + V206L + Y295N + K391A + G476K H1* + N54S + V56T + G109A + W167F + A174S + G182* + 2.4 1.7 1.2 1.0 2.5 1.9 1.7 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 2.4 1.6 1.4 1.0 4.9 1.8 1.8 1.2 G182* + D183* + A186N + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 2.4 1.6 1.1 1.0 2.4 1.6 1.3 1.1 G182* + D183* + N195F + V206L + A288V + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 2.4 1.9 1.2 1.0 3.2 2.2 2.0 1.3 G182* + D183* + N195F + V206L + G255A + K391A + G476K H1* + N54S + G109A + Q169E + Q172K + A174* + G182* + 2.3 1.7 1.1 1.1 3.2 2.2 1.5 1.2 D183* + N195F + V206L + K391A + G476K H1* + W48F + N54S + V56T + G109A + Q169E + Q172K + 2.3 1.6 1.1 1.0 2.3 1.6 1.4 1.2 A174* + G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 2.3 1.8 1.2 1.1 1.8 1.8 1.6 1.2 G182* + D183* + N195F + V206L + V291A + K391A + G476K H1* + N54S + V56T + G109A + F113Q + A174S + G182* + 2.3 1.7 1.3 1.0 2.3 2.0 1.7 1.3 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 2.3 1.6 1.2 1.0 3.3 2.0 1.6 1.2 G182* + D183* + N195F + V206L + R320A + S323N + K391A + G476K H1* + V56T + G109A + Q169E + Q172K + A174* + G182* + 2.3 1.6 1.2 0.9 2.0 1.7 1.6 1.1 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + F113Q + R116H + 2.3 1.7 1.1 1.0 2.0 1.7 1.7 1.2 W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + K391A + P473R + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.2 1.7 1.1 1.0 2.6 2.1 1.8 1.4 N195F + V206L + N270T + K391A + 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K72R + G109A + F113Q + R116Q + 2.0 1.6 1.2 1.1 2.0 1.6 1.3 1.1 W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + K391A + P473R H1* + N54S + V56T + G109A + A174S + G182* + D183* + 2.0 1.7 1.2 1.1 1.9 1.9 1.5 1.3 N195F + V206L + M246V + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 1.9 1.4 1.2 1.1 2.7 2.0 1.4 1.2 G182* + D183* + N195F + V206L + K391A H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 1.9 1.5 1.1 1.0 2.2 1.7 1.4 1.1 G182* + D183* + N195F + V206L + N270G + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 1.9 1.6 1.2 1.0 2.7 1.8 1.4 1.2 G182* + D183* + N195F + V206L + Y295F + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.9 1.5 1.2 1.1 2.9 1.8 1.3 1.3 N195F + V206L + R320A + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174T + 1.9 1.9 1.2 1.0 2.4 2.1 1.6 1.4 G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + Q169E + Q172K + G182* + 1.9 1.7 1.1 1.0 2.6 1.7 1.8 1.2 D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + W140Y + Q169E + Q172K + 1.9 1.4 1.1 1.0 2.1 1.8 1.5 1.4 A174* + G182* + D183* + N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.9 1.6 1.1 1.1 2.2 1.9 1.7 1.4 N195F + V206L + K391A H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + 1.8 1.6 1.1 1.0 1.7 1.4 1.5 1.1 G182* + D183* + N195F + V206L + F328L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 2.0 1.5 1.1 5.2 2.0 1.8 1.5 N195F + V206L + G476K F113Q + R116H + D183* + G182* + N195F 1.8 1.7 1.1 1.0 1.9 1.9 1.7 1.3 H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 1.3 1.1 1.0 2.6 1.8 1.5 1.2 N195F + V206L + A288L + K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 1.6 1.2 1.1 1.9 1.8 1.6 1.2 N195F + V206L + N270G + K391A + G476K Q169E + Q172K + A174* + D183* + G182* + N195F 1.8 1.2 1.2 1.0 3.3 1.4 1.5 1.3 H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.8 1.0 1.1 0.9 1.8 1.4 1.5 1.1 N195F + V206L + K269Q + K391A + G476K H1* + W48F + N54S + V56T + G109A + A174S + G182* + 1.8 1.4 1.1 1.0 2.2 1.7 1.5 1.2 D183* + N195F + V206L + K391A + G476K F113Q + R116Q + D183* + G182* + N195F 1.8 1.7 1.1 1.0 2.1 1.7 1.7 1.4 N54S + V56T + D183* + G182* + N195F 1.8 1.2 0.8 0.9 1.3 1.3 1.0 1.0 H1* + N54S + V56T + G109A + A174T + G182* + D183* + 1.7 1.7 1.2 1.1 2.4 1.9 1.8 1.3 N195F + V206L + K391A + G476K H1* + N54S + V56T + G109A + R116W + A174S + G182* + 1.7 1.6 1.2 1.1 1.8 1.8 1.6 1.2 D183* + N195F + V206L + K391A + G476K Q169E + Q172K + G182* + D183* + N195F 1.7 1.4 1.3 1.0 4.9 1.6 1.8 1.3 G182* + D183* + A186D 1.7 1.6 1.0 1.0 1.6 1.7 1.3 1.1 H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.5 1.1 1.0 1.8 1.6 1.4 1.1 N195F + V206L + V238A + K391A + G476K G182*

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0.9 0.7 1.1 0.9 D183* + G182* + P473R 0.8 0.9 1.0 1.0 1.1 0.8 1.0 1.0 W48Y + D183* + G182* 0.8 0.9 1.1 1.1 1.0 0.6 1.0 0.9 D183* + G182* + R323A 0.8 0.8 0.9 1.0 1.1 0.8 1.1 0.8 G7A + D183* + G182* 0.8 0.8 1.1 0.9 1.3 1.1 1.2 1.0 G109A + D183* + G182* 0.8 0.8 0.8 0.8 0.0 0.6 1.0 1.1 W167F + D183* + G182* 0.7 0.7 1.0 0.9 1.4 0.9 1.1 1.0 H1* + N54S + V56T + G109A + Q169E + Q172K + A174S + 0.7 0.7 0.8 0.9 1.1 0.5 0.8 0.8 G182* + D183* + N195F + V206L + K391A + G476K A174S + D183* + G182* 0.7 1.2 0.8 0.9 2.2 0.9 1.1 1.3 R116Q + D183* + G182* 0.7 0.7 0.9 0.9 0.7 0.6 1.1 0.9 D183* + G182* + N195F + V291A 0.7 0.9 1.0 1.1 0.6 0.9 0.9 1.0 Q172K + D183* + G182* 0.7 0.9 0.9 1.0 0.4 0.5 1.1 0.9 N54S + D183* + G182* 0.7 1.2 1.0 1.1 1.7 1.1 1.0 1.3 D183* + G182* + N195F + S323N 0.7 0.9 0.8 0.9 1.8 1.0 1.0 1.0 D183* + G182* + N195F + M246V 0.7 0.8 1.1 1.0 0.6 0.5 0.9 0.9 W48Y + D183* + G182* + N195F 0.6 1.2 1.1 1.1 2.0 1.5 1.2 1.4 D183* + G182* + N195F + S304R 0.6 1.1 0.7 1.0 1.6 0.7 1.0 1.2 V56T + D183* + G182* 0.3 0.8 0.2 0.6 1.9 0.6 0.3 0.7 A-15- A-15- A-40- A-40- J-15- J-15- J-30- J-30- 0.05 0.2 0.05 0.2 0.05 0.2 0.05 0.2 Modifications in SEQ ID NO: 14 mg/L mg/L mg/L mg/L mg/L mg/L mg/L mg/L Reference 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 G182* + D183* + N195F 1.4 1.2 1.0 1.0 1.2 1.3 1.1 1.1 H1* + G182* + D183* + N195F 1.1 1.0 1.0 1.1 1.1 1.0 1.1 1.1 N54S + G182* + D183* + N195F 1.2 1.1 0.9 1.1 1.3 1.1 1.1 1.1 V56T + G182* + D183* + N195F 1.4 1.1 1.0 1.0 1.5 1.2 1.1 1.1 G182* + D183* + N195F + I405L 1.1 1.1 1.0 1.0 1.1 1.1 1.0 1.0 H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.0 1.1 0.9 0.9 1.4 1.0 1.1 0.9 N195F + V206L + I405L + A421H + A422P + A428T + D476K A174S + G182* + D183* + N195F 1.2 1.0 1.0 0.9 0.9 0.9 1.0 1.0 H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.4 1.3 1.1 1.0 1.6 1.3 1.3 1.1 N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + A174S + D183* + N195F + 1.2 1.3 1.0 1.0 1.6 1.3 1.2 1.1 V206L + K391A + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.3 1.3 1.1 1.0 1.4 1.3 1.1 1.0 N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.2 1.1 1.1 1.0 1.4 1.2 1.2 1.1 N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476G H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.3 1.3 1.0 1.0 1.5 1.2 1.1 1.0 N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K + G477S H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.3 1.2 1.0 1.0 1.5 1.2 1.2 1.0 N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K H1* + N54S + V56T + K72R + G109A + W167F + A174S + 1.1 1.2 1.0 1.1 1.4 1.4 1.3 1.1 G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + A174S + G182* + D183* + 0.9 1.2 1.1 0.9 1.3 1.2 1.3 1.0 N195F + V206L + S280Q + H321Y + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.1 1.2 1.0 1.0 1.2 1.1 1.1 1.0 N195F + V206L + S304N + I405L + A421H + A422P + A428T + D476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.2 1.2 1.0 1.0 1.2 1.0 1.0 0.9 N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K + G477A H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.6 1.4 1.1 1.1 1.9 1.4 1.3 1.2 N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476G + G477T H1* + N54S + V56T + G109A + F113Q + R116Q + Q172N + 0.9 1.2 0.8 1.1 0.8 1.7 0.9 1.1 A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.7 1.3 1.1 1.0 1.9 1.4 1.3 1.1 N195F + V206L + K391A + I405L + A421H + A422P + A428T + G477Q H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.5 1.4 1.1 1.0 1.9 1.4 1.3 1.1 N195F + V206L + K391A + I405L + A421H + A422P + A428T + G477A H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.3 1.3 1.0 1.0 1.7 1.2 1.1 1.0 N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K + G477Q H1* + N54S + V56T + G109A + A174S + G182* + D183* + 1.5 1.3 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K72R + G109A + T134E + A174S + 1.4 1.3 1.1 1.1

2.5 2.3 1.5 1.3 G182* + D183* + N195F + G196R + V206L + G255A + I405L + A421H + A422P + A428T + N475S H1* + N54S + V56T + G109A + T134E + A174S + G182* + 1.8 1.5 1.2 1.0 2.3 1.8 1.5 1.2 D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + T40D + N54S + V56T + G109A + A174S + G182* + 2.0 1.5 1.2 1.1 1.8 1.6 1.5 1.3 D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + N125D + A174S + 2.1 1.6 1.3 1.0 2.2 1.7 1.5 1.2 G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + A60V + G109A + R116Q + A174S + 1.6 1.6 1.1 1.0 1.9 1.6 1.4 1.2 G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + R116V + A174S + G182* + 1.4 1.4 1.0 1.0 2.1 1.4 1.4 1.2 D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + R116Q + G133Q + T165G + 1.1 1.4 1.0 1.1 2.0 2.3 1.5 1.3 Q172G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + R116H + Q172G + 1.7 1.6 1.2 1.1 3.0 1.9 1.5 1.2 A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + R116Q + Q172G + 1.4 1.5 1.1 1.1 2.1 1.8 1.3 1.2 A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T + G448D H1* + N54S + V56T + G109A + F113Q + R116H + Q172N + 2.2 1.6 1.3 1.2 2.5 1.6 1.6 1.3 A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + T134E + W167F + 2.2 1.6 1.3 1.2 2.4 1.6 1.7 1.2 Q172N + A174S + G182* + D183* + N195F + V206L + G255A + A265G + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + R116Q + W167F + 2.0 1.6 1.2 1.2 2.3 1.6 1.7 1.2 Q172R + L173V + A174S + G182* + D183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + R116Q + W167F + 2.1 1.4 1.3 1.2 2.3 1.6 1.7 1.2 Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + F113Q + T134E + 2.0 1.5 1.1 1.1 2.6 1.9 1.4 1.2 W167F + Q172G + A174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + F113Q + R116H + 1.7 1.6 1.0 0.9 1.8 1.5 1.1 1.1 W167F + Q172G + L173V + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + W167F + Q172G + 1.2 1.2 1.0 0.9 2.2 1.3 1.2 1.1 A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + A174S + G182* + 1.5 1.2 1.2 1.0 2.2 1.1 1.5 1.1 D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + R116Q + A174S + 1.6 1.3 1.1 1.0 1.4 1.2 1.3 1.1 G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + F113Q + F116H + 1.8 1.6 1.0 1.0 0.9 1.4 0.9 1.1 W167F + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + 1.8 1.5 1.1 1.0 1.9 1.5 1.3 1.1 W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + R116Q + W167F + 2.0 1.5 1.0 1.0 1.8 1.5 1.2 1.0 Q172G + A174S + G182* + D183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + R116H + W167F + 1.5 1.3 0.9 0.9 1.4 1.3 1.2 1.1 Q172G + A174S + G182* + D183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + R116Q + A174S + G182* + 1.8 1.6 1.0 0.9 1.6 1.3 1.4 1.1 D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + A60V + G109A + R116Q + W167F + 1.4 1.4 0.7 0.9 1.1 1.3 0.8 1.0 Q172N + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + A37H + N54S + V56T + A60V + G109A + R116Q + 1.9 1.6 1.1 1.1 1.7 1.6 1.4 1.2 T165G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + A174S + M105F + G182* + 1.4 1.2 1.0 1.0 1.5 1.2 1.2 1.0 D183* + N195F + V206L + L228I + R320A + S323N + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + F113Q + T134E + 1.8 1.5 1.1 1.0 1.7 2.1 1.4 1.2 A174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T H1* + N54S + V56T + K72R + G109A + R116Q + V120L + 2.1 1.8 1.2 1.1 2.8 2.5 1.6 1.3 A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + R116H + W167F + 2.6 1.9 1.2 1.1 2.8 2.6 1.6 1.3 Q172G + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T H1* + N54S + V56T + G109A + F113Q + R116Q + Q172N + 2.3 1.6 1.1 1.1 3.0 2.1 1.5 1.3 A174S + G182* + D183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T H1* + A37H + N54S + V56T + A60V + G109A + R116Q + 2.1 1.7 1.2 1.1 2.5 2.1 1.5 1.2 W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T

Example 4: Wash Performance of a Subset of Variants According to the Invention (AMSA)

(224) The following variants have been tested as described above. The color measurements were done by use of a professional flatbed scanner (Kodak iQsmart, Kodak) used to capture an image of the washed textile and with a controlled digital imaging system (DigiEye) for capture an image of the washed melamine plates.

(225) To extract a value for the light intensity from the scanned images, 24-bit pixel values from the image are converted into values for red, green and blue (RGB). The intensity value (Int) is calculated by adding the RGB values together as vectors and then taking the length of the resulting vector:

(226) Int = $\sqrt{r^2 + g^2 + b^2}$

(227) The results obtained were the following;

(228) TABLE-US-00009 A-15- A-15- A-40- A-40- J-15- J-15- J-30- J-30- 0.05 0.2 0.05 0.02 0.05 0.2 0.05 0.2 Modifications in SEQ ID NO: 13 mg/L mg/L mg/L mg/L mg/L mg/L mg/L mg/L Reference 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 G182* + D183* + N195F + V206F + N260P + L312I + L313I + R320A + T355L + 0.7 1.0 0.6 0.7 1.3 0.9 0.8 1.0 L389I + T400P G132* + D183* + N195F 0.1 0.7 1.0 1.0 1.0 0.8 1.0 1.0 H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + 1.7 1.9 1.7 1.3 2.5 1.9 1.2 1.2 G476K H1* + G50A + T51A + V56T + G109A + A174S + G182* + D183* + N195F + V206L + 1.5 1.6 1.5 1.4 2.5 1.6 1.3 1.2 G476K H1* + N54S + V56T + M105F + G109A + A174S + G182* + D183* + N195F + V206L + 2.5 2.2 1.6 1.4 3.8 1.9 1.2 1.1 R320A + S323N + K391A + G476K H1* + T40K + N54S + V56T + G109A + G149H + A174S + G182* + D183* + N195F + 0.9 1.2 1.0 1.0 0.8 1.2 0.9 1.1 V206L + K391A + G476K H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + 1.9 1.7 1.4 1.4 2.8 1.5 1.2 1.1 G255A + K391A + G476K H1* + N16H + V17L + N54S + V56T + G109A + A174S + G182* + D183* + N195F + 2.6 1.8 1.7 1.4 3.6 1.8 1.3 1.1 V206L + K391A + G476K H1* + Q22N + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + 2.5 1.9 1.8 1.4 3.8 2.1 1.3 1.1 K391A + G476K H1* + Q32S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + 2.3 1.9 1.9 1.5 3.3 2.0 1.2 1.2 K391A + G476K H1* + H28N + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + 2.7 1.7 1.6 1.5 4.2 2.0 1.3 1.1 K391A + G476K H1* + N29S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + 1.6 1.3 1.3 1.4 2.6 1.6 1.2 1.0 K391A + G476K H1* + H28N + N29S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + 1.8 1.9 1.8 1.4 3.1 1.8 1.2 1.1 V206L + K391A + G476K H1* + N54S + V56T + G109A + A174S + R176K + N195F + V206L + K391A + G476K 2.3 2.0 1.7 1.5 5.1 1.9 1.3 1.2 H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + R176K + N195F + V206L + 3.8 2.4 2.1 1.7 5.6 2.8 1.4 1.2 K391A + G476K H1* + N54S + V56T + G109A + A174S + N175G + N195F + V206L + K391A + G476K 2.6 1.8 1.8 1.4 4.9 2.0 1.3 1.2 H1* + N54S + V56T + G109A + A174S + N195F + V206L + V291A + Y295N + Q299T + 1.7 1.4 1.4 1.3 2.7 1.6 1.0 1.1 K391A + G476K H1* + N54S + V56T + G109A + A174S + N195F + V206L + K269S + A274K + K391A + 1.7 1.4 1.6 1.2 1.4 1.8 1.0 1.0 G476K H1* + N54S + V56T + G109A + A174S + N195F + V206L + L235I + K391A + G476K 2.6 1.5 2.0 1.3 1.3 2.1 1.0 1.1 H1* + N54S + V56T + G109A + A174S + N195F + V206L + L235V + V238A + K391A + 2.8 1.4 1.7 1.2 0.6 2.2 1.1 1.0 G476K H1* + N54S + V56T + G109A + A174S + N195F + V206L + Q365S + K391A + G476K 1.9 1.5 1.9 1.3 2.0 1.7 1.2 1.0 H1* + N54S + V56T + G109A + A174S + N195F + V206L + K391A + H402R + G476K 2.2 1.4 2.0 1.2 0.6 1.2 1.1 1.1 H1* + N54S + V56T + G109A + A174S + N19SF + V206L + R320S + H321N + K391A + 5.6 2.7 2.5 1.6 4.2 2.9 1.2 1.2 G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + N195F + V206L + K391A + 3.8 2.6 3.0 1.7 3.6 2.8 1.3 1.2 G476K H1* + N54S + V56T + G109A + Q125S + A174S + N195F + V206L + K391A + G476K 3.1 2.2 2.4 1.4 2.0 2.3 1.3 1.1 H1* + N54S + V56T + G109A + A174S + N19SF + V206L + L235V + K391A + G476K 4.2 1.9 1.9 1.5 1.8 2.4 1.0 1.0 H1* + N54S + V56T + G109A + A174S + N195F + V206L + A265G + K391A + G476K 4.2 1.5 2.2 1.4 1.2 2.6 1.1 1.1 H1* + T51Q + N54S + V56T + G109A + A174S + N195F + Y203F + V206L + K391A + 3.7 2.1 2.6 1.6 3.6 2.7 1.3 1.1 G476K H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + N175Q + N195F + V206L + 4.0 2.0 2.7 1.5 3.3 2.8 1.2 1.2

K391A + G476K H1* + H28N + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + 4.5 2.3 2.5 1.6 3.1 2.7 1.2 1.1 K391A + G476K H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + N195F + V206L + 3.9 2.1 2.6 1.6 3.3 3.0 1.4 1.1 K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377H + 5.3 2.9 3.0 1.7 3.8 2.9 1.3 1.2 K391A + G476K H1* + G50A + T51A + N54S + V56T + G109A + A174S + G182* + D183* + N195F + 4.4 2.1 2.3 1.4 2.6 2.3 1.3 1.2 V206L + K391A + G476K H1* + G7K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + 4.1 2.3 2.7 1.6 2.4 2.5 1.3 1.2 K391A + G476K H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + M202V + V206L + 2.7 1.9 1.4 1.4 1.9 2.2 0.9 1.0 K391A + G476K H1* + N54S + V56T + M105F + G109A + A174S + N195F + V206L + M246F + K391A + 3.7 2.1 2.4 1.5 2.7 2.5 1.2 1.1 G476K H1* + N16H + V17L + N54S + V56T + G109A + A174S + G182* + D183* + N195F + 4.5 2.7 1.8 1.4 4.3 2.2 1.5 1.3 V206L + D377H + K391A + G476K H1* + N54S + V56T + G109A + A174S + N195F + V206L 1.1 1.9 1.3 1.1 3.0 1.9 1.3 1.1 H1* + T51G + N54S + V56T + G109A + A174S + N195F + Y203G + V206L + K391A + 2.2 1.8 1.1 1.0 2.7 2.0 0.9 1.1 G476K H1* + T51G + N54S + V56T + G109A + G110N + A174S + N195F + V206L + K391A + 1.6 2.1 1.0 1.1 1.9 1.9 1.1 1.1 G476K H1* + N54S + V56T + G109A + Q125S + A174S + N175G + N195F + V206L + K391A + 2.0 2.3 1.5 1.2 2.6 1.8 1.3 1.2 G476K H1* + N54S + V56T + G109A + A174S + N195F + V206L + A265G + V291A + F328L + 0.7 2.1 1.1 1.0 1.8 1.7 1.0 1.1 K391A + G476K H1* + N54S + V56T + G109A + A174S + N195F + V206L + M246F + A265G + V291A + 3.9 2.6 1.9 1.4 4.0 2.1 1.5 1.2 F328L + Q365S + K391A + G476K H1* + N16Y + N54S + V56T + G109A + R116H + A174S + N195F + V206L + G337E + 4.8 3.5 2.0 1.5 4.9 2.7 1.5 1.3 D377H + K391A + G476K H1* + I9L + N54S + V56T + G109A + Q125S + V131T + A174* + V206L + Y267F + 3.1 2.9 1.4 1.2 2.5 2.0 1.2 1.1 F289H + Q361E + K391A + G476K

(229) The invention described and claimed herein is not to be limited in scope by the specific aspects herein disclosed, since these aspects are intended as illustrations of several aspects of the invention. Any equivalent aspects are intended to be within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims. In the case of conflict, the present disclosure including definitions will control.

Claims

1. A variant of a parent alpha-amylase having alpha-amylase activity, which: a) is a polypeptide having at least 85% sequence identity to SEQ ID NO: 13 or SEQ ID NO: 14, and b) comprises a substitution or a deletion at a position corresponding to N195, N54, V56, G109, R116, or Q172 of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14, and comprises a substitution and/or deletion in two, three or four positions corresponding to positions R181, G182, D183, and G184 of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14, c) has alpha-amylase activity, and d) has an improved wash performance at a low temperature of 5° C. to 40° C., wherein the improved wash performance is an Improvement Factor (IF) of >1, when compared to the parent alpha-amylase.
2. The variant of claim 1, which has a substitution at a position corresponding to N195 of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14.
3. The variant of claim 1, wherein the parent alpha-amylase comprises an A and a B domain having at least 85%, but less than 100% sequence identity to SEQ ID NO: 10.
4. The variant of claim 1, wherein the parent alpha-amylase comprises a C domain having at least 85% sequence identity to SEQ ID NO: 11 or 12.
5. The variant of claim 1, which has a substitution at a position corresponding to N195F of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14.
6. The variant of claim 1 wherein the parent alpha-amylase has at least 90% sequence identity to SEQ ID NO: 13 or 14.
7. The variant of claim 1 wherein the parent alpha-amylase has at least 95% sequence identity to SEQ ID NO: 13 or 14.
8. The variant of claim 3, wherein the A and B domains further comprise a modification in one or more positions corresponding to positions: H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, A60, K72, R87, Q98, M105, F113, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, L173, A186, E190, T193, G196, A204, V206, P211, 1214, V215, L217, L219, A225, L228, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, and Q395 of SEQ ID NO: 13 or 14.
9. The variant of claim 8, wherein the modification is selected from the group consisting of H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40D, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, A60V, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, N125D, G133S, F133Q, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, G196R, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L228I, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, Q256A, N260G, A263G, V264I, V264T, A265G, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S304N, S304R, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, and Q395P.
10. The variant of claim 4, wherein the C domain further comprises a modification in one or more positions corresponding to positions: T400, H402, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13 or I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14.
11. The variant of claim 10, wherein the modification is selected from the group consisting of: T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S, wherein the positions correspond to SEQ ID NO: 13; or I405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q, wherein the positions correspond to SEQ ID NO: 14.
12. The variant of claim 1, which comprises: a) a deletion and/or a substitution at two or more positions corresponding to positions R181, G182, D183, and G184 of SEQ ID NO: 13 or 14, and b) a further modification at one or more positions: i) H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, K72, R87, Q98, M105, F113, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, L173, A186, E190, T193, A204, V206, P211, 1214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484,

wherein the positions correspond to SEQ ID NO: 13; or ii) A37, T40, A60, K72, F113, N125, F133, T134, T165, Q169, L173, A186, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, 1405, A421, A422, A428, R439, G448, W467, D476, and G477, wherein the positions correspond to SEQ ID NO: 14.

13. The variant of claim 12, wherein the deletion a) is selected from the group consisting of R181+G182, R181+D183, R181+G184, G182+D183, G182+G184, or D183+G184.

14. The variant of claim 12, which comprises one or more of the following modifications: a) H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, 0125P, Q125K, Q125A, Q125T, G133S, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, 0169E, R171H, 0172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, Q256A, N260G, A263G, V264I, V264T, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S340R, S304N, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, Q395P, T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S, wherein the positions correspond to SEQ ID NO: 13; or b) G7A, A37H, T40D, W48Y, W48F, N54S, V56T, A60V, K72R, Q98A, G109A, F113Q, R116H, R116V, R116Q, N125D, F133Q, T134E, T165G, Q169E, Q172D, Q172G, Q172N, L173V, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, G196R, A204G, V206L, V206Y, A225V, L228I, K242Q, S244O, G255A, G255S, N260G, A265G, K269Q, N270T, S280Q, W284H, A288L, A288V, F289L, Y295N, Y295F, S304N, S340R, S304N, R320A, H321Y, S323N, V326L, K391 A, 1405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q, wherein the positions correspond to SEQ ID NO: 14.

15. The variant of claim 1, which has an IF of >1 when compared to the parent alpha-amylase, and which comprises: TABLE-US-00010 N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + R87S + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, T40G + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, T51K + N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72H + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, A37M + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, A37S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, A37Y + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, A37R + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113W + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113S + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113N + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Y + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113R + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118N + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116V + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116K + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116W + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116L + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + G142T + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125P + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + S381G + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246T + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V213T + K391A + G476K, A37L + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377S + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377G + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377R + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377A + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + I214G + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + S381A + K391A + G476K, N54S + V56T + G109A + W167G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G346S + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G346P + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + I214H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + I214S + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + I214T + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + A420Q + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + A420S + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + A420K + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + A420L + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + K391Y + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + K391R + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + P473A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + P473G + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473T + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + Q395P + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444S + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444D + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444Y + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280W + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280T + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280A + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* +

N195F + V206L + G255N + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T285L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T285Q + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + Q449T + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473K + G476K, N54S + V56T + K72S + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, T40K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346T + K391A + G476K, G50A + N54S + V56T + G109A + + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484A, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484G, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484P, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484E, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484Q, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484S, H1*K + G50A + N54S + V56T + G109A + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173V + A174S + G182* + D183* + N195F + A204T + V206L + K391A + G476K, N54S + V56T + G109A + W140Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + Q98S + G109A + A174S + G182* + D183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + H324L + K391A + G476K, N54S + V56T + G109A + G149Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G346T + K391A + G476K + G477A, N54S + V56T + G109AG255A ++ A174S + G182* + 0183* + N195F + V206L + K391A + G476K, G50A + N54S + V56T + G109A + F113L + R116L + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167F + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173V + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K + G477Q, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Q345O + K391A + G476K + G477Q, G50A + N54S + V56T + G109A + Q172O + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V264I + K391A + G423H + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + R320V + K391A + G476Y, T5K + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172Y + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + I214L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L217V + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L217H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + P2110 + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + K391Q + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + T444H + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + T444V + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + T444R + G476K, G50A + N54S + V56T + G109A + L173V + A174S + G182* + O183* + N195F + A204V + V206L + K391A + G476K, G50A + N54S + V56T + G109A + F113L + W167F + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, G7K + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + Q118G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + S280K + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + N270G + K391A + G476K, N54S + V56T + G109A + Q172O + A174S + G182* + 0183* + N195F + V206L + G346T + K391A + G476K + G477A, N54S + V56T + G109A + Q118S + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + T134E + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G337H + Q385L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y382M + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + N195F + V206L + G255A + O377H + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + M105F + G109A + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + M105F + G109A + T134E + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + M105F + G109A + Q118F + T134E + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + R320A + S323N + O377H + K391A + G476K, T51E + N54S + V56T + G109A + A174H + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Q299V + K391A + G476K, N54S + V56T + G109A + L173G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + T355L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + T355F + K391A + G476K, G50A + N54S + V56T + G109A + A174S + G182* + 0183* + N195F + A204T + V206L + K391A + G476K, G50A + N54S + V56T + G109A + A174E + G182* + 0183* + N195F + A204T + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + 0183* + N195F + V206L + R320A + S323N + K391A + G476K, N54S + V56T + M105F + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246L + K391A + G476K, T51G + N54S + V56T + G109A + A174N + G182* + 0183* + N195F + V206L + K391A + G476K, T51A + N54S + V56T + G109A + A174G + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A1740 + G182* + O183* + N195F + V206L + K391A + G476K, T51S + N54S + V56T + G109A + A174P + G182* + O183* + N195F + V206L + K391A + G476K, T51G + N54S + V56T + G109A + A174M + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72A + G109A + W167S + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + W167H + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118R + T134E + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, Q11H + N54S + V56T + G109A + Q118R + T134E + A174S + G184T + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118R + T134E + A174S + G182* + 0183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + Q118R + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + T134E + A174S + G182* + O183* + G184T + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + Q1720 + A174Q + G182* + 0183* + N195F + V206L + K391A + G476K + G477A, N54S + V56T + K72A + G109A + W167F + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167S + G182* + O183* + N195F + A174S + V206L + K391A + G476K, N54S + V56T + K72H + G109A + W167S + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167T + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72E + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167L + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + G184T + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G255A + O377H + K391A + G476K, N54S + V56T + G109A + Q118R + A174S + G182* + 0183* + N195F + V206L + G255A + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + G184A + T193K + N195F + V206L + K391A + G476K, N54S + V56T + M105F + G109A + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + T134E + A174S + G182* + O183* + N195F + V206L + R320A + S323N + K391A + G476K, N54S

+ V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V291G + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V291A + F328L + O377H + K391A + G476K, V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + T134E + A174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V238A + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V238T + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V238G + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + F328M + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + F328L + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + F328V + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + F328I + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V291T + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V291A + O377H + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + 0183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + L173H + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125T + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N16S + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72N + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72N + G109A + W167L + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72A + G109A + W167Y + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118R + T134E + A174S + G182* + 0183* + N195F + V206L + G255A + O377H + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + N195F + G184T + V206L + K391A + G476K, N54S + V56T + K72R + G109A + Q118R + A174S + G182* + O183* + N195F + V206L + G255A + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + G184T + N195F + V206L + R320A + S323N + K391A + G476K, N54S + G109A + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377H + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377H + K391A, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377R + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L217T + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377S + K391A + G476K, N54S + V56T + G109A + R116E + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L217Q + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G346O + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377A + K391A + G476K, N54S + V56T + G109A + T134E + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + Q118R + T134E + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + Q118R + T134E + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G346P + K391A + G476K, N54S + V56T + G109A + R116S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + O377Q + K391A + G476K, N54S + V56T + G109A + R116I + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174F + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174T + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118R + T134E + A174S + G182* + 0183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + G109A + Q118R + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, T51G + N54S + V56T + G109A + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51A + N54S + V56T + G109A + Q172O + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51S + N54S + V56T + G109A + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51A + N54S + V56T + G109A + Q172S + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51E + N54S + V56T + G109A + Q172T + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51A + N54S + V56T + G109A + Q172A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51S + N54S + V56T + G109A + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51G + N54S + V56T + G109A + Q172K + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51A + N54S + V56T + G109A + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51E + N54S + V56T + G109A + Q172L + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51E + N54S + V56T + G109A + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172K + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51O + N54S + V56T + G109A + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51A + N54S + V56T + G109A + Q172K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + Q172S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72H + G109A + Q172T + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + Q172R + A174S + K179E + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + Q172M + A174S + G182* + 0183* + N195F + V206L + Y295N + K391A + G476K, N54S + V56T + K72R + G109A + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + Q172T + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72M + G109A + Q172S + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + Q172M + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172M + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72S + G109A + A174H + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A1740 + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A1740 + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A174K + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174G + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A174Q + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A174G + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72S + G109A + A174N + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A174H + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + K391S + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + T444A + G476K, N54S + V56T + G109A + F113L + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, G50A + N54S + V56T + G109A + A174E + G182* + 0183* + N195F + A204T + V206L + S323N + K391A + G476K, N54S + V56T + K72R + G109A + Q118R + A174S + G182* + O183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + M105F + G109A + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + K72R + G109A + Q118R + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + A225F + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246A + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y3821 + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + S376H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + I214E + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + I214W + K391A + G476K, N54S + V56T + K72R + G109A + Q172K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + A204T + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + A204S + V206L + O377H + K391A + G476K, N54S + V56T + G109A + T165S + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + T334S + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + A263G + O377H + K391A + G476K,

N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M286F + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y2671 + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + L235V + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + L235A + O377H + K391A + G476K, N54S + V56T + M105L + G109A + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + M1051 + G109A + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + M105V + G109A + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246I + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L250I + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + L250V + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391Y + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391V + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391M + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391E + G476K, N54S + V56T + G109A + T165G + A174S + G182* + 0183* + N195F + V206L + O377H + K391A + G476K, N54R + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54L + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174V + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246F + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246V + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246S + O377H + K391A + G476K, N54S + V56T + G109A + A174R + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174L + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206S + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391D + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391R + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391H + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391W + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K3911 + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250T + D377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250A + D377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250F + D377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250M + D377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391Q + G476K, N54S + V56T + G109A + Q172H + L173A + A174P + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172E + L173A + A174R + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + W167R + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + G184T + N195F + V206L + D377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + G255A + D377H + K391A + G476K, N54S + V56T + G109A + Q118R + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S376T + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S376V + K391A + G476K, N54S + V56T + G109A + Q172E + L173I + A174N + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172D + L173A + A174T + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y267M + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y267H + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V238A + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V238T + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V238G + K391A + G476K, N54S + V56T + M105V + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V264T + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V264I + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L235V + K391A + G476K, N54S + V56T + G109A + W167H + Q172N + L173A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167F + Q172E + L173P + A174K + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172N + L173T + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172K + L173V + A174T + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172E + L173F + A174R + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172E + L173T + A174R + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172N + L173V + A174R + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173V + A174R + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172E + L173M + A174H + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172E + L173F + A174P + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167Y + 0172E + L173V + A174H + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173A + A174T + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + V291A + F328L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246S + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L250V + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L250F + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + L250M + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + A204S + V206L + K391A + G476K, N54S + V56T + G109A + T165G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + T334S + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + A263G + M286F + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + A263G + M286L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y267H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y267L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y267I + K391A + G476K, T51A + N54S + V56T + G109A + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391L + G476K, N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + Q172G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + F113Q + R116Q + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q1720 + A174S + G182* + 0183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + 0183* + N195F + A204G + V206L + G255A + K391A + P473R + G476K, N54S + V56T + K72R + G109A + W167F + A174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + G184T + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + W167F + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + W167F + Q172G + A174S + G182* + 0183* + N195F + V206L + K391A + P473G + G476K, N54S + V56T + G109A + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + R116H + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + M105F + G109A + A174S + G182* + O183* + N195F + V206L + R320A + S323N + K391A + C474V + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + N195F + V206L + G255A + K391A + C474V + G476K, N54S + V56T + G109A + T134E + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + O377H + K391A + C474V + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + A265G + K391A + T4440 + G476K + G477A, A37V + N54S + V56T + G109A + W167F + Q172G + A174S + G182* + 0183* +

N195F + V206L + K391A + G346T + G477A + G476K, N54S + V56T + G109A + R116Q + Q1720 + A174S + G182* + O183* + N195F + V206L + G346T + K391A + T444Q + G477A + G476K, N54S + V56T + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G346T + K391A + T444Q + G477A + G476K, N54S + V56T + G109S + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + R116Q + W167F + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + W167F + Q172G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + Q172G + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + 0183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + Q172N + A174S + G182* + 0183* + N195F + V206L + V264I + K391A + C474V + G476K, N54S + V56T + G109A + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + S280Q + H321Y + K391A + C474V + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + C474V + G476K, N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + F113Q + R116Q + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + R116H + W167F + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + T165G + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116Q + T165G + Q172G + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + F289I + K391A + G476K, N54S + V56T + G109A + F113Q + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + R116Q + Q172M + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + Q172M + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116H + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + T165G + Q172M + A174S + G182* + 0183* + N195F + V206L + K391A + Q395P + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + 0183* + N195F + V206L + A265G + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + T134E + T165G + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116Q + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + A265G + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, T51V + N54S + V56T + G109A + Q172O + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + S280Q + H321Y + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + Q172G + L173V + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + G133Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + Q395P + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172N + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + K72R + G109A + R116H + W167F + Q172R + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109S + R116Q + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109S + F113Q + W167F + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109S + F113Q + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + 0183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + 0183* + N195F + V206L + G255A + K391A + Q395P + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + T134E + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + G255A + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + G255A + K391A + Q395P + A445Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R116H + T134E + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + G255A + K391A + Q395P + G476K, N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + G133Q + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + 0183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172N + L173V + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K, A37H + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + T459N + G476K, N54S + V56T + G109A + F113Q + Q172R + L173V + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + 0183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116H + Q172R + L173V + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + 0183* + N195F + V206 + K391A + G476K, A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + 0183* + N195F + V206 + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172G + L173V + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + 0183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116A + W167F + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + R116A + T165G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + G476K, N54S + V56T + A174S + G182* + O183* + N195F + V206L + G273R + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L, W48G + G109A + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + K391A, N54S + V56T + G109A + F113Q + Q172M + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + 0183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + 0183* + N195F + V206L + G255A + K391A +

P473R + G476K, N54S + V56T + K72R + G109A + R116Q + A174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116A + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + M246F + K391A + G476K, V56A + G109A + A174S + G182* + O183* + N195F + V206L + G476K, N54S + V56T + K72R + G109A + F113Q + R116H + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109S + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + 0183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + 0183* + N195F + V206L + A265G + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + 0183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116H + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K, A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + G109A + R116H + T165G + A174S + G182* + 0183* + N195F + V206L + M246F + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + K391A + G476K, N54S + V56T + G109A + G133S + A174S + G182* + O183* + N195F + K391A + G476H, N54S + V56T + G109A + R116A + Q172N + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116H + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, P45A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + P124A + A174S + G182* + 0183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + P364A + K391A + C474V + G476K, N54S + V56T + G109A + Q118R + T134E + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + G476K, G50A + T51A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V264F + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y382L + K391A + G476K, G7K + N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G255A + Q256A + K391A + G476K, N54S + V56T + G109A + W167Y + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167H + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + T165V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174N + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174Q + G182* + 0183* + N195F + V206L + K391A + G476K, N16H + V17L + N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y382F + K391A + G476K, Q32S + N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Q385L + K391A + G476K, N54S + V56T + R87S + G109A + R171H + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116H + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174* + G182* + 0183* + N195F + V206L + K391A + G476K, K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + 0183* + G184T + N195F + V206L + P473R + G476K, N54S + V56T + G109A + Q169E + Q172G + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + 0183* + G184T + N195F + K391A + P473R + G476K, N54S + V56T + G109A + R116W + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206F + K391A + G476K, V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + N260G + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + Y295N + K391A + G476K, N54S + V56T + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + A186N + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + A288V + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + G255A + K391A + G476K, N54S + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, W48F + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + V291A + K391A + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + R320A + S323N + K391A + G476K, V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + N270T + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + N260G + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172G + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + A1860 + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + A225V + K391A + G476K, G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + G255S + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + K269Q + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G255S + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + W284H + K391A + G476K, N54S + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + S304R + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + S304R + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + 0183* + G184T + N195F + V206L + K391A + P473R, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + M246V + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + K391A, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + N270G + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + Y295F + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174T + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W140Y + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + 0183* + N195F + V206L + F328L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + A288L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + N270G + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K269Q + K391A + G476K, W48F + N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174T + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116W + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V238A + K391A + G476K, N54S + V56T + G109A + A174S + V206L + G182* + 0183* + K391A + G476K, N54S + V56T + 0169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* +

N195F + V206F + K391A + G476K, N54S + V56T + G109A + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174N + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + A186N + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + F289L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + Y295F + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + V291A + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + S244Q + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + A225V + K391A + G476K, N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + P473R + G476K, W48Y + N54S + V56T + G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + Q98A + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206Y + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K, G109A + A174S + G182* + 0183* + N195F + V206L + K391A + G476K, A174S + D183* + G182*, A174S + G182* + O183* + N195F, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K, or N54S + V56T + G109A + Q169E + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K, wherein the positions correspond to SEQ ID NO: 13.

16. The variant of claim 1, which has an IF of >1 when compared to the parent alpha-amylase, and which comprises: TABLE-US-00011 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T + D476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + A 174S + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476G, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A+ I405L + A421 H + A422P + A428T + D476K + G477S, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476K, N54S + V56T + K72R + G109A + W167F + A174S + G182* + O183* + N195F + V206L + 1405L + A421H + A422P + A428T, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + S304N + I405L + A421H + A422P + A428T + 0476K, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476K + G477A, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476G + G477T, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477Q, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477A, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476K + G477Q, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476N, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + 0476Y, N54S + V56T + G109A + A 174S + G182* + 0183* + N195F + V2061 + I405L + A421H + A422P + A428T, N54S + V56T + G109A + Q1720 + A174S + G182* + 0183* + N195F + A204G + V206L + 1405L + A421H + A422P + A428T, N54S + V56T + K72R+G109A + T134E + A 174S + G182* + 0183* + N195F + A204G + V206L + G255A + 1405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + T165G + Q172G + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + Q172N + A 174S + G182* + O183* + N195F + V206L + 1405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + G196R + V206L + G255A + I405L + A421H + A422P + A428T + N475S, N54S + V56T + G109A + T134E + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, T40O + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + N1250 + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + R116V + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + R 116Q + G1330 + T165G + Q172G + A174S + G182* + 0183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421 H + A422P + A428T + G448O, N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + L173V + A174S + G182* + 0183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q 172G + A 174S + G182* + 0183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + R 116H + W167F + Q172G + L173V + A174S + G182* + 0183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + W167F + Q172G + A174S + G182* + 0183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + F116H + W167F + Q172G + L173V + A174S + G182* + 0183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + 0183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + A 174S + M105F + G182* + O183* + N195F + V206L + L2281 + R320A + S323N + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + O183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + V120L + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, or A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, wherein the positions correspond to SEQ ID NO: 14.

17. The variant of claim 1, which comprises: G7K+N54S+V56T+G109A+A174S+G182*+O183*+N195F+V206L+K391A+G476K, I9L+N54S+V56T+G109A+Q125S+V131T+A174*+G182*+O183*+V206L+Y267F+F289H+Q361E+K391 A+G476K, N16H+V17L+N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+D377H+K391 A+G476K, N16H+V17L+N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+K391A+G476K, N16Y+N54S+V56T+G109A+R116H+A174S+G182*+O183*+N195F+V206L+G337E+O377H+K391A+G476K, Q22N+N54S+V56T+G109A+A 174S+G182*+O183*+N195F+V206L+K391 A+G476K, H28N+N29S+N54S+V56T+G109A+A174S+G182*+O183*+N195F+V206L+K391A+G476K, H28N+N54S+V56T+G109A+A 174S+G182*+O183*+N195F+V206L+K391A+G476K, N29S+N54S+V56T+G109A+A 174S+G182*+O183*+N195F+V206L+K391A+G476K, Q32S+N54S+V56T+G109A+A 174S+G182*+O183*+N195F+V206L+K391A+G476K,

T40K+N54S+V56T+G109A+G149H+A174S+G182*+O183*+N195F+V206L+K391A+G476K, G50A+T51A+N54S+V56T+G109A+A174S+G182*+O183*+N195F+V206L+K391A+G476K, G50A+T51A+V56T+G109A+A174S+G182*+O183*+N195F+V206L+G476K, T51G+N54S+V56T+G109A+G11 ON+A174S+G182*+O183*+N195F+V206L+K391A+G476K, T51G+N54S+V56T+G109A+A 174S+G182*+O183*+N195F+Y203G+V206L+K391A+G476K, T51Q+N54S+V56T+G109A+A 174S+G182*+O183*+N195F+Y203F+V206L+K391A+G476K, N54S+V56T+K72R+G109A+A 174S+G182*+O183*+N195F+V206L+G255A+K391A+G476K, N54S+V56T+M105F+G109A+A174S+G182*+O183*+N195F+V206L+M246F+K391A+G476K, N54S+V56T+M105F+G109A+A174S+G182*+O183*+N195F+V206L+R320A+S323N+K391A+G476K, N54S+V56T+G109A+Q125S+A174S+N175G+G182*+O183*+N195F+V206L+K391A+G476K, N54S+V56T+G109A+Q125S+A174S+G182*+O183*+N195F+V206L+K391 A+G476K, N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K, N54S+V56T+G109A+Q169E+Q172K+A174*+N175Q+G182*+D183*+N195F+V206L+K391 A+G476K, N54S+V56T+G109A+Q169E+Q172K+A174*+R176K+G182*+D183*+N195F+V206L+K391 A+G476K, N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, N54S+V56T+G109A+A 174S+N175G+G182*+D183*+N195F+V206L+K391A+G476K, N54S+V56T+G109A+A 174S+R176K+G182*+D183*+N195F+V206L+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+M202V+V206L+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+L235I+K391A+G476K, N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L235V+V238A+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+L235V+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+M246F+A265G+V291A+F328L+Q365S+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+A265G+V291A+F328L+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+A265G+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+K269S+A274K+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+V291A+Y295N+Q299T+K391 A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+R320S+H321N+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+Q365S+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K, N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+K391A+H402R+G476K, or N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+K391A+G476K, wherein the positions correspond to SEQ ID NO: 13, and wherein the IF is determined by use of Model A detergent.

18. The variant of claim 1, which has an IF of at least >1.5 when compared to the parent alpha-amylase, and which comprises: TABLE-US-00012 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + R87S + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, T40G + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, T51K + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72H + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, A37M + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37S + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37Y + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37R + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113W + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113N + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Y + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118N + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116W + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116L + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + G142T + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125P + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S381G + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246T + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + V213T + K391A + G476K, A37L + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377S + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377G + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377R + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377A + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214G + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S381A + K391A + G476K, N54S + V56T + G109A + W1671 + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109A + W167G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G346S + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G346P + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214S + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214T + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420Q + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420S + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420K + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420L + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + K391Y + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + K391R + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + P473A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + P473G + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + P473T + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + Q395P + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + T444S +

G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + T444D + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + T444Y + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + S280W + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + S280L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + S280T + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + S280A + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + G255N + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + T285L + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + T285Q + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + Q449T + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + P473K + G476K, N54S + V56T + K72S + G109A + A 174S + G182* + 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+ K391A + G476K, N54S + V56T + Q98S + G109A + A 174S + G182* + D183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + H324L + K391A + G476K, N54S + V56T + G109A + G149Q + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G346T + K391A + G476K + G477A, N54S + V56T + G109AG255A + + A174S + G182* + O183* + N195F + V206L + K391A + G476K, G50A + N54S + V56T + G109A + F113L + R116L + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173V + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K + G477Q, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Q345O + K391A + G476K + 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+ O183* + N195F + V206L + K391A + T444H + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + T444V + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + T444R + G476K, G50A + N54S + V56T + G109A + L173V + A174S + G182* + O183* + N195F + A204V + V206L + K391A + G476K, G50A + N54S + V56T + G109A + F113L + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, G7K + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + R320A + K391 A + G476K, N54S + V56T + G109A + Q118G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S280K + K391A + G476K, N54S + V56T + G109A + Q1720 + A174S + G182* + O183* + N195F + V206L + G346T + K391A + G476K + G477A, N54S + V56T + G109A + Q118S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + T134E + A 174S + 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K391A + G476K, N54S + V56T + K72R + G109A + Q172M + A174S + G182* + O183* + N195F + V206L + Y295N + K391A + G476K, N54S + V56T + K72Q + G109A + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + Q172T + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72M + G109A + Q172S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72S + G109A + A 174H + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A1740 + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A 1740 + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A 174K + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174G + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A 174Q + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A 174G + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72S + G109A + A 174N + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A 174H + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + K391S + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + T444A + G476K, N54S + V56T + G109A + F113L + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, G50A + N54S + V56T + G109A + A 174E + G182* + O183* + N195F + A204T + V206L + S323N + K391A + G476K, N54S + V56T + K72R + G109A + Q118R + A174S + G182* + O183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + M105F + G109A + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + K72R + G109A + Q118R + A 174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + A225F + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246A + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y3821 + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S376H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214E + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214W + K391A + G476K, N54S + V56T + K72R + G109A + Q172K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + A204T + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + A204S + V206L + O377H + K391A + G476K, N54S + V56T + G109A + T165S + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + T334S + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + A263G + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M286F + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y267I + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + L235V + O377H + K391A + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + L235A + O377H + K391A + G476K, N54S + V56T + M105L + G109A + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + M1051 + G109A + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + M105V + G109A + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246I + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + L250I + O377H + K391A + G476K, 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A 174S + G182* + O183* + N195F + V206L + K391M + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391E + G476K, N54S + V56T + G109A + T165G + A 174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54L + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174V + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246F + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246V + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246S + O377H + K391A + G476K, N54S + V56T + G109A + A 174R + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174L + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206S + K391A + G476K, N54S + V56T + G109A + 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+ T444Q + G477A + G476K, N54S + V56T + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G346T + K391A + T444Q + G477A + G476K, N54S + V56T + G109S + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109S + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + Q172N + A174S + G182* + O183* + N195F + V206L + V264I + K391A + C474V + G476K, N54S + V56T + G109A + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S280Q + H321Y + K391A + C474V + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + C474V + G476K, N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F +

V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + F113Q + R116Q + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + R116H + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + T165G + Q172G + A 174S + G182* + O183* + N195F + V206L + K391 A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116Q + T165G + Q172G + A174S + G182* + O183* + N195F + V206L + K391 A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + F289I + K391A + G476K, N54S + V56T + G109A + F113Q + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + R116Q + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116H + Q172M + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109A + F113Q + R116H + T165G + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + Q395P + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + A265G + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + T134E + T165G + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + A265G + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51V + N54S + V56T + G109A + Q172O + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S280Q + H321Y + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + Q172G + L173V + A 174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116Q + W167F + Q172N + L173V + A 174S + G182* + O183* + N195F + V206L + K391 A + T444Q + G476K, N54S + V56T + G109A + G133Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + Q395P + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172N + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + K72R + G109A + R116H + W167F + Q172R + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R 116H + W167F + Q172G + L173V + A174S + G182* + O183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109S + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109S + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + K391 A + P473R + G476K, N54S + V56T + G109S + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L +K391A + G476K, A37V + N54S + V56T + G109S + F113Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + T134E + W167F + Q172R + A174S + G182* +O183* + N195F + V206L + G255A + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q 172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + A445Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R 116H + T134E + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + G476K, N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + G133Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172N + L173V + A174S + G182* + O183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K, A37H + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, A37H + 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N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116H + Q172R + A174S + G182* + O183* + N195F + A204G

+ V206L + K391A + P473R + G476K, A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + G109A + R116H + T165G + A174S + G182* + O183* + N195F + V206L + M246F + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + K391 A + G476K, N54S + V56T + G109A + G133S + A174S + G182* + O183* + N195F + K391A + G476H, N54S + V56T + G109A + R116A + Q172N + L173V + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116H + T165G + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, P45A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + P124A + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + P364A + K391A + C474V + G476K, N54S + V56T + G109A + Q118R + T134E + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, G50A + T51A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + V264F + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y382L + K391A + G476K, G7K + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G255A + Q256A + K391A + G476K, N54S + V56T + G109A + W167Y + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + T165V + A 174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174N + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174Q + G182* + O183* + N195F + V206L + K391A + G476K, N16H + V17L + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y382F + K391A + G476K, Q32S + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Q385L + K391A + G476K, N54S + V56T + R87S + G109A + R171H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116H + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174* + G182* + O183* + N195F + V206L + K391A + G476K, K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + P473R + G476K, N54S + V56T + G109A + Q169E + Q172G + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + K391A + P473R + G476K, N54S + V56T + G109A + R116W + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + 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+ K391A + G476K, W48F + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + V291A + K391A + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + R320A + S323N + K391A + G476K, V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + N270T + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + K391A + G476K,O183* + N195F + V206L + N260G + N54S + V56T + G109A + F113Q + R116H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + A1860 + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + A225V + K391A + G476K, G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + G255S + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K269Q + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G255S + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + W284H + K391A + G476K, N54S + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + S304R + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246V + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + N270G + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + Y295F + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174T + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W140Y + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + F328L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + A288L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + N270G + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K269Q + K391A + G476K, W48F + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A174T + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116W + A174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + V238A + K391A + G476K, N54S + V56T + G109A + A 174S + V206L + G 182* + D183* + K391A + G476K, N54S + V56T + 0169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206F + K391A + G476K, N54S + V56T + G109A + Q172K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174N + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + A186N + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + F289L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + Y295F + K391 A + G476K, or N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + V291A + K391A + G476K, wherein the positions correspond to SEQ ID NO: 13.

19. The variant of claim 1, which has an IF of at least >1.5 when compared to the parent alpha-amylase, and which comprises: TABLE-US-00013 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476G + G477T, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477Q, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477A, N54S + V56T + G109A + A 174S + G182*

+ D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476N, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476Y, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V2061 + I405L + A421H + A422P + A428T, N54S + V56T + G109A + T134E + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, T40O + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + N125O + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + L173V + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q 172G + A 174S + G182* + O183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + R 116H + W167F + Q172G + L173V + A174S + G182* + O183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + F116H + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + T134E + A 174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + V120L + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, wherein the positions correspond to SEQ ID NO: 14, and wherein the IF is determined by use of Model A detergent.

20. The variant of claim 1, which has an IF of at least >2.0 when compared to the parent alpha-amylase, and which comprises: TABLE-US-00014 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + R87S + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, T40G + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, T51K + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72H + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37M + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37S + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37Y + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, A37R + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113W + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116K + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125P + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S381G + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246T + K391A + G476K, A37L + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q125A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377S + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377A + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S381A + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G346P + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214S + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420Q + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420S + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420K + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + A420L + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + K391Y + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + P473A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + P473G + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391 A + T4440 + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + T444Y + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S280W + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S280L + K391A + G476K, N54S + V56T + K72S + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, T40K + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + G346T + K391A + G476K, G50A + N54S + V56T + G109A + + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K + K484S, H1*K + G50A + N54S + V56T + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173V + A174S + G182* + O183* + N195F + A204T + V206L + K391A + G476K, N54S + V56T + G109A + W140Y + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G346T + K391 A + G476K + G477A, N54S + V56T + G109A + G255A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, G50A + N54S + V56T + G109A + F113L + R116L + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + R320V + K391A + G476Y, N54S + V56T + G109A + R116M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172Y + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + L217V + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + L217H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + P2110 + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + K391Q + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + T444H +

G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + T444V + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + T444R + G476K, G50A + N54S + V56T + G109A + L173V + A174S + G182* + O183* + N195F + A204V + V206L + K391A + G476K, G7K + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + R320A + K391 A + G476K, N54S + V56T + G109A + Q1720 + A174S + G182* + O183* + N195F + V206L + G346T + K391A + G476K + G477A, N54S + V56T + G109A + Q118S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + R320A + K391A + G476K, N54S + V56T + G109A + T134E + A 174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y382M + K391 A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + M105F + G109A + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + M105F + G109A + T134E + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + R320A + S323N + O377H + K391A + G476K, T51E + N54S + V56T + G109A + A174H + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + L173A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Q299V + K391A + G476K, N54S + V56T + G109A + L173G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + T355L + K391 A + G476K, G50A + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + A204T + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A174S + G182* + O183* + N195F +V206L + R320A + S323N + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246L + K391A + G476K, N54S + V56T + K72Q + G109A + W167H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118R + T134E + A 174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + Q118R + A174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + T134E + A 174S + G182* + O183* + G184T + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + Q1720 + A174Q + G182* + O183* + N195F + V206L + K391A + G476K + G477A, N54S + V56T + K72R + G109A + A 174S + G182* + O183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + G184T + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G255A + O377H + K391A + G476K, N54S + V56T + G109A + T134E + A174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + V238T + O377H + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + V291A + O377H + K391 A + G476K, N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + G184T + N195F + V206L + G255A + K391A + G476K, N54S + V56T + G109A + L173H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72N + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72N + G109A + W167L + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q118R + T134E + A 174S + G182* + O183* + N195F + V206L + G255A + O377H + K391 A + G476K, N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + G184T + V206L + K391 A + G476K, N54S + V56T + K72R + G109A + Q118R + A174S + G182* + O183* + N195F + V206L + G255A + O377H + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377H + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377H + K391A, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + O377S + K391A + G476K, N54S + V56T + G109A + R116E + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + T134E + A 174S + G182* + O183* + N195F + V206L + O377H + K391A + G476K, N54S + V56T + G109A + R116S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116| + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51A + N54S + V56T + G109A + Q172S + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51E + N54S + V56T + G109A + Q172T + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51S + N54S + V56T + G109A + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, T51E + N54S + 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G109A + A 1740 + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A 174K + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174G + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A 174Q + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72Q + G109A + A 174G + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72S + G109A + A 174N + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + A 174H + G182* + O183* + N195F + V206L + K391A + G476K, G50A + N54S + V56T + G109A + A 174E + G182* + O183* + N195F + A204T + V206L + S323N + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y3821 + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + 1214W + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + A204S + V206L + 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D183* + N195F + V206L + K3911 + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + L250T + D377H + K391 A + G476K, N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250A + D377H + K391A + G476K, N54S + V56T + G109A + A174S +

G182* + D183* + N195F + V206L + L250F + D377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + L250M + D377H + K391A + G476K, N54S + V56T + G109A + Q172H + L173A + A174P + G182* + D183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + G184T + N195F + V206L + G255A + D377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + S376T + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + S376V + K391A + G476K, N54S + V56T + G109A + Q172E + L1731 + A174N + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172O + L173A + A174T + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y267M + O377H + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + V238G + K391A + G476K, N54S + V56T + M105V + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 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V206L + A265G + K391A + T4440 + G476K + G477A, A37V + N54S + V56T + G109A + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G346T + G477A + G476K, N54S + V56T + G109A + R116Q + Q172O + A174S + G182* + O183* + N195F + V206L + G346T + K391A + T444Q + G477A + G476K, N54S + V56T + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G346T + K391A + T444Q + G477A + G476K, N54S + V56T + G109S + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109S + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + Q172N + A174S + G182* + O183* + N195F + V206L + V264I + K391A + C474V + 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+ N195F + V206L + G255A + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R 116H + T134E + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + G476K, N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + G133Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172N + L173V + A174S + G182* + O183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K, A37H + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + T459N + G476K, N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + O183* + 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+ V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + P473R + G476K, N54S + V56T + G109A + Q169E + Q172G + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + K391A + P473R + G476K, N54S + V56T + G109A + R116W + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206F + K391A + G476K, V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + N260G + K391A + G476K, N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + G184T + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + Y295N + K391A + G476K, N54S + V56T + G109A + W167F + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + A186N + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + A288V + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, W48F + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + V291A + K391A + G476K, N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + R320A + S323N + K391A + G476K, V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + N270T + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174*

+ G182* + O183* + N195F + V206L + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + N260G + K391A + G476K, N54S + V56T + G109A + F113Q + R116H + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + A1860 + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + A225V + K391A + G476K, G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + G255S + K391A + G476K, N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K269Q + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + K391 A + G476K, N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + G255S + K391A + G476K, or N54S + V56T + G109A + Q169E + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K wherein the positions correspond to SEQ ID NO: 13.

21. The variant of claim 1, which has an IF of at least >2.0 when compared to the parent alpha-amylase, and which comprises: TABLE-US-00015 T40D + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + N125D + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + G255A + A265G + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + L173V + A174S + G182* + D183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q 172G + A 174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T, N54S + V56T + K72R + G109A + R116Q + V120L + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T, N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, or A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T, wherein the positions correspond to SEQ ID NO: 14, and wherein the IF is determined by use of Model A detergent.

22. The variant of claim 1, which comprises: N54S+V56T+K72R+G109A+F113Q+R 116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K, N54S+V56T+K72R+G109A+F113Q+R 116Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K, N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+O183*+N195F+V206L+G255A+K391A+Q395P+T444Q+P473R+ N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K, N54S+V56T+K72R+G109A+G133Q+W167F+Q172R+A174S+G182*+O183*+N195F+V206L+K391A+G476K, N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+O183*+N195F+V206L+A265G+K391A+P473R+G476K, N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K, N54S+V56T+G109A+R116Q+W167F+A174S+G182*+O183*+G184T+N195F+V206L+K391 A+G476K, N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, and N54S+V56T+G109A+A 174S+G182*+D183*+N195F+V206L+K391A+G476K, wherein the positions correspond to SEQ ID NO: 13.

23. A composition comprising the variant of claim 1.

24. The composition of claim 23, which further comprises at least one further active component.

25. The composition of claim 24, wherein the further active component is an enzyme selected from the group consisting of a cellulase, a lipase, a mannanase, a pectate lyase and a protease.

26. The composition of claim 23, which is a liquid laundry or liquid dish wash composition, an Automatic Dish Wash (ADW) liquid detergent composition, or a powder laundry, a soap bar, or powder dish wash composition, an ADW unit dose detergent composition and a Hand Dish Wash (HOW) detergent composition.

27. The composition of claim 23, which further comprises a surfactant, builder, bleach, polymer, or which is phosphate-free.